Analysis of Functions

Background definitions

Set and power set. A set E is a collection of elements. Its power set is $\mathcal{P}(E) = \{A \subseteq E\}$.

σ-algebra. A family $\mathcal{E} \subseteq \mathcal{P}(E)$ such that (i) $E \in \mathcal{E}$; (ii) $A \in \mathcal{E} \Rightarrow A^c \in \mathcal{E}$; (iii) $A_1, A_2, \dots \in \mathcal{E} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{E}$. Members of $\mathcal{E}$ are *measurable sets*.

Measurable space. A pair $(E, \mathcal{E})$ consisting of a set E and a σ-algebra $\mathcal{E}$ on E.

Borel σ-algebra. $\mathcal{B}(\mathbb{R}^n)$ is the smallest σ-algebra containing all open sets of $\mathbb{R}^n$.

Measure. A map $\mu : \mathcal{E} \to [0, \infty]$ with $\mu(\varnothing) = 0$ and *countable additivity*: for pairwise disjoint $A_1, A_2, \dots \in \mathcal{E}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

Measure space. A triple $(E, \mathcal{E}, \mu)$ with μ a measure on $(E, \mathcal{E})$.

Null set and a.e. properties. A set $N \in \mathcal{E}$ with $\mu(N) = 0$ is *null*. A statement holds *almost everywhere* (a.e.) if it fails only on a null set.

Indicator and simple functions. $\mathbf{1}_A(x) = 1$ for $x \in A$ and 0 otherwise. A *simple function* is a finite sum $\sum_{k=1}^{m} a_k \mathbf{1}_{A_k}$ with $A_k \in \mathcal{E}$.

Measurable function. $f : (E, \mathcal{E}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ (or $\mathbb{C}$) is *measurable* if $f^{-1}(B) \in \mathcal{E}$ for every Borel set B.

Lebesgue integral (nonnegative). For $f \geq 0$ measurable,

$$\int_E f \, d\mu = \sup \left\{ \int_E s \, d\mu : 0 \leq s \leq f, \ s \text{ simple} \right\}.$$

For general f, define $\int f = \int f^+ - \int f^-$ when at least one is finite.

Integrable functions. f is integrable if $\int_E |f| \, d\mu < \infty$.

Essential supremum.

$$\|f\|_{L^\infty} = \operatorname*{ess\,sup}_{x \in E} |f(x)| = \inf\{M \geq 0 : \ \mu(\{|f| > M\}) = 0\}.$$

L^p spaces and norms. For $1 \le p < \infty$,

$$L^p(E,\mu) = \left\{ f \text{ measurable} : \int_E |f|^p \, d\mu < \infty \right\} \Big/ (\text{equality a.e.}), \qquad \|f\|_{L^p} = \left(\int_E |f|^p \, d\mu \right)^{1/p}.$$

For $p = \infty$, $L^\infty(E,\mu)$ consists of essentially bounded functions with the norm above.

Local L^p. $f \in L^p_{\mathrm{loc}}(\mathbb{R}^n)$ if $f \in L^p(K)$ for every bounded measurable $K \subset \mathbb{R}^n$.

Support. $\operatorname{supp} f = \overline{\{x \in E : \ f(x) \neq 0\}}$ (often up to null sets).

Conjugate exponents and basic inequalities. $p, q \in [1,\infty]$ are conjugate if $\frac{1}{p} + \frac{1}{q} = 1$ (with $1/\infty = 0$). Young's inequality: $ab \le \frac{a^p}{p} + \frac{b^q}{q}$ for $a, b \ge 0$. Hölder: $\int |uv| \le \|u\|_p \|v\|_q$. Minkowski: $\|u+v\|_p \le \|u\|_p + \|v\|_p$.

Core definitions for measure theory and L^p analysis

Measure-Theoretic and Functional-Analytic Foundations

Measure-theoretic primitives.

A *σ-algebra* $\mathcal{E} \subseteq \mathcal{P}(E)$ is nonempty and satisfies:

$$A \in \mathcal{E} \ \Rightarrow \ A^c \in \mathcal{E}, \qquad \{A_n\} \subset \mathcal{E} \ \Rightarrow \ \bigcup_n A_n \in \mathcal{E}.$$

A *measurable space* is $(E, \mathcal{E})$.

A *measure* is a map $\mu : \mathcal{E} \to [0,\infty]$ such that

$$\mu(\varnothing) = 0, \qquad \mu\left(\bigsqcup_n A_n \right) = \sum_n \mu(A_n).$$

A *measure space* is $(E, \mathcal{E}, \mu)$.

It is *finite* if $\mu(E) < \infty$, and *σ-finite* if $E = \bigcup_{k \ge 1} E_k$ with $\mu(E_k) < \infty$ for all k. It is *complete* if every subset of each null set is measurable.

A *probability space* has $\mu(E) = 1$. A property holds *almost everywhere (a.e.)* if it fails only on a null set. For $A \in \mathcal{E}$, write $\mathbf{1}_A$ for the indicator of A.

Measurable and simple functions; essential supremum.

A function $f : (E, \mathcal{E}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is *measurable* if $f^{-1}(B) \in \mathcal{E}$ for all Borel B.

A *simple function* has the form

$$s(x) = \sum_{k=1}^{m} a_k \, \mathbf{1}_{A_k}(x), \qquad A_k \in \mathcal{E}.$$

The *essential supremum* of f is

$$\operatorname{ess\,sup} |f| = \inf\{M > 0 : \mu(\{|f| > M\}) = 0\}.$$

Lebesgue integral.

For $f \geq 0$ measurable,

$$\int_E f \, d\mu = \sup \left\{ \int_E s \, d\mu : \; s \text{ simple}, \; 0 \leq s \leq f \right\}.$$

For general f, define $f^{\pm} = \max\{\pm f, 0\}$ and

$$\int_E f \, d\mu = \int_E f^+ \, d\mu - \int_E f^- \, d\mu,$$

whenever at least one term is finite.

L^p spaces.

For $0 < p < \infty$,

$$L^p(E) = \left\{ f \text{ measurable} : \|f\|_{L^p}^p = \int_E |f|^p \, d\mu < \infty \right\} \Big/ \text{a.e.}$$

For $p \geq 1$, $\|\cdot\|_{L^p}$ is a norm; for $0 < p < 1$, it is a quasi-norm satisfying

$$\|f + g\|_p^p \leq \|f\|_p^p + \|g\|_p^p.$$

Define

$$L^\infty(E) = \{f : \; \|f\|_\infty = \operatorname{ess\,sup} |f| < \infty\}, \qquad L^p_{\text{loc}}(\mathbb{R}^n) = \{f : \; f \in L^p(K) \; \forall \text{ compact } K \subset \mathbb{R}^n\}.$$

Functional-analytic basics.

A *normed space* is $(X, \|\cdot\|)$. It is a *Banach space* if complete.

The *dual space* X^* consists of all continuous linear functionals x^* on X, equipped with the operator norm

$$\|x^*\| = \sup_{\|x\| \leq 1} |x^*(x)|.$$

The *canonical map* $J : X \to X^{**}$ is defined by $J(x)(x^*) = x^*(x)$. The space X is *reflexive* if J is onto (equivalently, $X \cong X^{**}$).

For $1 < p < \infty$, $(L^p)^* \cong L^q$ where $\frac{1}{p} + \frac{1}{q} = 1$, and L^p is reflexive. For $p = 1$, $(L^1)^* = L^\infty$. For $p = \infty$, $(L^\infty)^*$ strictly contains L^1.

The *weak topology* on X is the coarsest topology making each $x^* \in X^*$ continuous. A sequence $x_n \rightharpoonup x$ *weakly* if $x^*(x_n) \to x^*(x)$ for all $x^* \in X^*$.

The *weak-* topology* on X^* is pointwise convergence on X: $x_n^* \overset{*}{\rightharpoonup} x^*$ iff $x_n^*(x) \to x^*(x)$ for all $x \in X$.

Key properties.

- For $1 < p < \infty$, L^p is uniformly convex and hence reflexive.
- On finite measure spaces, $L^p \subset L^q$ for $p > q$ with $\|f\|_q \le \mu(E)^{1/q-1/p}\|f\|_p$.
- Simple functions are dense in L^p for $1 \le p < \infty$ on σ-finite measure spaces.

Disjoint union vs. sum. The symbol $\bigsqcup$ denotes a *disjoint union of sets*:

$$\bigsqcup_{n=1}^{\infty} A_n \quad \text{means} \quad \bigcup_{n=1}^{\infty} A_n \text{ with } A_i \cap A_j = \varnothing \text{ for } i \neq j.$$

It is an operation on *sets*, not on numbers, hence we do not write a sum on the left.

Countable additivity (measure axiom). If $(A_n)_{n\ge1}$ are pairwise disjoint measurable sets, then

$$\mu\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

Left: disjoint union of *sets*; right: ordinary sum of *numbers*.

Equivalences and useful identities.

- $\bigsqcup_n A_n = \bigcup_n A_n$ *together with* $A_i \cap A_j = \varnothing$ for $i \neq j$.
- For indicator functions:

  $$\mathbf{1}_{\bigsqcup_n A_n}(x) = \sum_n \mathbf{1}_{A_n}(x) \quad \text{(because at most one summand is 1 when the union is disjoint).}$$

- If the A_n are not disjoint, one can *disjointize*:

  $$B_1 = A_1, \qquad B_n = A_n \setminus \bigcup_{k<n} A_k, \quad \text{so that } \bigsqcup_n B_n = \bigcup_n A_n.$$

Problem 1 — Hölder and Minkowski inequalities

Statement. Let $(E, \mathcal{E}, \mu)$ be a measure space and let $f, g : E \to \mathbb{C}$ be measurable.

(a) If $1 \le p, q, r \le \infty$ satisfy $p^{-1} + q^{-1} = r^{-1}$, then

$$\|fg\|_{L^r} \le \|f\|_{L^p} \, \|g\|_{L^q}.$$

(b) For $1 \le p \le \infty$,

$$\|f + g\|_{L^p} \le \|f\|_{L^p} + \|g\|_{L^p}.$$

Theory (quick toolkit).

- **Conjugate exponents.** $p, q \in [1, \infty]$ are conjugate if $\frac{1}{p} + \frac{1}{q} = 1$ (with $1/\infty = 0$).
- **Young's inequality (products).** For $a, b \ge 0$ and conjugate p, q:

$$ab \; \le \; \frac{a^p}{p} \; + \; \frac{b^q}{q}.$$

- **Hölder (classical, $r = 1$).** If $1 \le p, q \le \infty$, $1/p + 1/q = 1$, then

$$\int |uv| \, \mathrm{d}\mu \;\; \le \;\; \|u\|_{L^p} \|v\|_{L^q}.$$

- **Minkowski (triangle).** For $1 \le p \le \infty$:

$$\|u + v\|_{L^p} \;\; \le \;\; \|u\|_{L^p} + \|v\|_{L^p}.$$

Proof of (a).

Step 1: Special case $r = 1$. Assume $1 < p, q < \infty$ and $1/p + 1/q = 1$ (the endpoints are immediate). For $A = \frac{|u|}{\|u\|_p}$ and $B = \frac{|v|}{\|v\|_q}$, Young's inequality gives $AB \le \frac{A^p}{p} + \frac{B^q}{q}$. Multiplying by $\|u\|_p \|v\|_q$ and integrating,

$$\int |uv| \le \frac{\|v\|_q}{p} \int |u|^p \|u\|_p^{1-p} + \frac{\|u\|_p}{q} \int |v|^q \|v\|_q^{1-q} = \|u\|_p \|v\|_q.$$

Thus $\|uv\|_1 \le \|u\|_p \|v\|_q$.

Step 2: General r. Let $1 \le r < \infty$ and $1/p + 1/q = 1/r$. Set $U = |f|^r$ and $V = |g|^r$. Then $U \in L^{p/r}$, $V \in L^{q/r}$ and $\frac{r}{p} + \frac{r}{q} = 1$. Applying Step 1 to U, V,

$$\int |fg|^r = \int UV \le \|U\|_{p/r} \, \|V\|_{q/r} = \|f\|_p^r \|g\|_q^r.$$

Taking the r-th root yields $\|fg\|_r \le \|f\|_p \|g\|_q$. For $r = \infty$, the estimate $\|fg\|_\infty \le \|f\|_\infty \|g\|_\infty$ is immediate.

Proof of (b) (Minkowski).

For $p = \infty$ or $p = 1$, the claim follows from the pointwise triangle inequality and linearity. Assume $1 < p < \infty$ and let $p' = \frac{p}{p-1}$. If $\|f+g\|_p = 0$ there is nothing to prove. Otherwise define

$$h = \frac{|f+g|^{p-1}\,\mathrm{sgn}(f+g)}{\|f+g\|_p^{p-1}} \quad \Rightarrow \quad \|h\|_{p'} = 1 \;\text{ and }\; \int (f+g)h = \|f+g\|_p.$$

Then, using Hölder twice,

$$\|f+g\|_p = \int (f+g)h \le \int |f||h| + \int |g||h| \le \|f\|_p\|h\|_{p'} + \|g\|_p\|h\|_{p'} = \|f\|_p + \|g\|_p.$$

This proves Minkowski.

Examples.

(i) Powers on $[0,1]$. Let $f(x) = x^\alpha$, $g(x) = x^\beta$ with $\alpha > -1/p$, $\beta > -1/q$, $1/p + 1/q = 1/r$. Then

$$\|f\|_p^p = \frac{1}{\alpha p + 1}, \qquad \|g\|_q^q = \frac{1}{\beta q + 1}, \qquad \|fg\|_r^r = \frac{1}{(\alpha+\beta)r + 1},$$

and one checks $\|fg\|_r \le \|f\|_p\|g\|_q$.

(ii) ℓ^p *endpoint.* Let $a_n = 1/n$ and $b_n \equiv 1$. For $p = r$ and $q = \infty$, $\|ab\|_p = \|a\|_p \le \|a\|_p\|b\|_\infty$ with equality.

Background definitions

Set. A collection of elements. We denote the ambient set by E.

σ-algebra. A family $\mathcal{E} \subseteq \mathcal{P}(E)$ such that: (i) $E \in \mathcal{E}$; (ii) $A \in \mathcal{E} \Rightarrow A^c \in \mathcal{E}$; (iii) $A_1, A_2, \dots \in \mathcal{E} \Rightarrow \bigcup_{n=1}^\infty A_n \in \mathcal{E}$. Members of $\mathcal{E}$ are called *measurable sets.*

Measurable space. A pair $(E, \mathcal{E})$ with E a set and $\mathcal{E}$ a σ-algebra on E.

Measure. A map $\mu : \mathcal{E} \to [0, \infty]$ such that $\mu(\varnothing) = 0$ and, for pairwise disjoint $A_1, A_2, \dots \in \mathcal{E}$,

$$\mu\left(\bigcup_{n=1}^\infty A_n\right) = \sum_{n=1}^\infty \mu(A_n).$$

Measure space. A triple $(E, \mathcal{E}, \mu)$ where μ is a measure on $(E, \mathcal{E})$.

Measurable function. A function $f : (E, \mathcal{E}) \to (\mathbb{C}, \mathcal{B}(\mathbb{C}))$ is measurable if $f^{-1}(B) \in \mathcal{E}$ for every Borel set B. (For real-valued f, replace $\mathbb{C}$ by $\mathbb{R}$.)

Almost everywhere (a.e.). A statement holds a.e. if the set where it fails has measure 0.

L^p **spaces.** For $1 \le p < \infty$,

$$L^p(E,\mu) \;=\; \Big\{ f \text{ measurable} : \int_E |f|^p \, d\mu < \infty \Big\} \Big/ (\text{equality a.e.}).$$

For $p = \infty$, $L^\infty(E,\mu)$ consists of (equivalence classes of) essentially bounded f.

L^p **norms.** For $1 \le p < \infty$,

$$\|f\|_{L^p} = \Big(\int_E |f|^p \, d\mu \Big)^{1/p},$$

and

$$\|f\|_{L^\infty} = \operatorname*{ess\,sup}_{x \in E} |f(x)| \;=\; \inf\{M \ge 0 : \mu(\{x : |f(x)| > M\}) = 0\}.$$

Problem 2 — Embeddings, interpolation, and local L^p

Statement. Let $(E, \mathcal{E}, \mu)$ be a measure space.

(a) If $\mu(E) < \infty$ and $f \in L^p(E)$ with $1 \le q \le p \le \infty$, then $f \in L^q(E)$ and

$$\|f\|_{L^q} \;\le\; \mu(E)^{\frac{p-q}{pq}} \, \|f\|_{L^p}.$$

(b) Suppose $f \in L^{p_0}(E) \cap L^{p_1}(E)$ with $0 < p_0 < p_1 \le \infty$. For $0 \le \theta \le 1$ define p_θ by

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}.$$

Then $f \in L^{p_\theta}(E)$ and

$$\|f\|_{L^{p_\theta}} \;\le\; \|f\|_{L^{p_0}}^{1-\theta} \, \|f\|_{L^{p_1}}^{\theta}.$$

(c) On $\mathbb{R}^n$ show that if $p_1 \ne p_2$ then $L^{p_1}(\mathbb{R}^n) \not\subset L^{p_2}(\mathbb{R}^n)$. Determine for which p_1, p_2 we have $L^{p_1}_{\text{loc}}(\mathbb{R}^n) \subset L^{p_2}_{\text{loc}}(\mathbb{R}^n)$.

Theory (quick toolkit).

- On sets of finite measure: for $1 \le q \le p \le \infty$,

  $$\|h\|_{L^q(E)} \le \mu(E)^{\frac{1}{q} - \frac{1}{p}} \, \|h\|_{L^p(E)}.$$

  This is Hölder with exponents $\frac{p}{q}$ and $\frac{p}{p-q}$.
- Real interpolation for L^p: If $1 \le p_0 < p_1 \le \infty$ and $\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$, then

  $$L^{p_0} \cap L^{p_1} \hookrightarrow L^{p_\theta}, \qquad \|f\|_{p_\theta} \le \|f\|_{p_0}^{1-\theta} \|f\|_{p_1}^{\theta}.$$

- On bounded $K \subset \mathbb{R}^n$: if $p_1 \ge p_2$,

  $$\|h\|_{L^{p_2}(K)} \le |K|^{\frac{1}{p_2} - \frac{1}{p_1}} \, \|h\|_{L^{p_1}(K)}.$$

  Hence $L^{p_1}_{\text{loc}} \subset L^{p_2}_{\text{loc}}$ iff $p_1 \ge p_2$.

Proof.

(a) Finite-measure embedding. For $p < \infty$, apply Hölder to $|f|^q \cdot 1$ with exponents $\frac{p}{q}$ and $\frac{p}{p-q}$:

$$\int_E |f|^q \le \Big(\int_E |f|^p\Big)^{q/p} \mu(E)^{1-\frac{q}{p}}.$$

Taking the qth root gives $\|f\|_q \le \mu(E)^{\frac{p-q}{pq}} \|f\|_p$. For $p = \infty$, $\|f\|_q^q \le \|f\|_\infty^q \mu(E)$.

(b) Interpolation between L^{p_0} and L^{p_1}. For $p_1 < \infty$,

$$|f|^{p_\theta} = \big(|f|^{p_0}\big)^{(1-\theta)\frac{p_\theta}{p_0}} \big(|f|^{p_1}\big)^{\theta\frac{p_\theta}{p_1}}.$$

With Hölder exponents $a = \frac{p_0}{(1-\theta)p_\theta}$ and $b = \frac{p_1}{\theta p_\theta}$ (conjugate by the definition of p_θ),

$$\int |f|^{p_\theta} \le \Big(\int |f|^{p_0}\Big)^{(1-\theta)\frac{p_\theta}{p_0}} \Big(\int |f|^{p_1}\Big)^{\theta\frac{p_\theta}{p_1}},$$

hence $\|f\|_{p_\theta} \le \|f\|_{p_0}^{1-\theta} \|f\|_{p_1}^{\theta}$. If $p_1 = \infty$, replace the second factor by $\|f\|_\infty^{\theta p_\theta}$.

(c) Global non-inclusion and local inclusion. If $p_1 > p_2$, choose α with $\frac{n}{p_1} < \alpha \le \frac{n}{p_2}$ and set $f(x) = (1+|x|)^{-\alpha}$. Then $f \in L^{p_1}(\mathbb{R}^n)$ but $f \notin L^{p_2}(\mathbb{R}^n)$. If $p_1 < p_2$, choose β with $\frac{n}{p_2} < \beta \le \frac{n}{p_1}$ and let $g(x) = |x|^{-\beta}\mathbf{1}_{\{|x|\le 1\}}$; then $g \in L^{p_2}$ but $g \notin L^{p_1}$. Thus for $p_1 \ne p_2$, neither $L^{p_1}(\mathbb{R}^n)$ nor $L^{p_2}(\mathbb{R}^n)$ contains the other.

For local spaces: on every bounded K, the finite-measure embedding yields $\|h\|_{L^{p_2}(K)} \le |K|^{\frac{1}{p_2}-\frac{1}{p_1}} \|h\|_{L^{p_1}(K)}$ when $p_1 \ge p_2$. Therefore

$$L^{p_1}_{\text{loc}}(\mathbb{R}^n) \subset L^{p_2}_{\text{loc}}(\mathbb{R}^n) \quad\Longleftrightarrow\quad p_1 \ge p_2.$$

Mini–toolkit: Monotone, Fatou, and Dominated Convergence

Monotone Convergence Theorem (Beppo–Levi). Let $(f_n)_{n\ge1}$ be measurable with $0 \le f_1 \le f_2 \le \cdots$ and $f_n \uparrow f$ a.e. Then

$$\int_E f\,d\mu \;=\; \lim_{n\to\infty} \int_E f_n\,d\mu \;\in [0,\infty].$$

(No integrability assumption beyond nonnegativity.)

Fatou's Lemma. For nonnegative measurable f_n,

$$\int_E \liminf_{n\to\infty} f_n\,d\mu \;\le\; \liminf_{n\to\infty} \int_E f_n\,d\mu.$$

Reverse Fatou (with domination). If f_n are measurable and there exists $g \in L^1(E)$ with $f_n \geq -g$ for all n, then

$$\limsup_{n\to\infty} \int_E f_n \, d\mu \;\leq\; \int_E \limsup_{n\to\infty} f_n \, d\mu.$$

Dominated Convergence Theorem (Lebesgue). Suppose $f_n \to f$ a.e. on E and there exists $g \in L^1(E)$ with $|f_n| \leq g$ a.e. for all n. Then $f \in L^1(E)$ and

$$\int_E f_n \, d\mu \;\to\; \int_E f \, d\mu, \qquad \text{equivalently} \qquad \|f_n - f\|_{L^1} \to 0.$$

Bounded Convergence (finite measure case). If $\mu(E) < \infty$, $f_n \to f$ a.e., and $|f_n| \leq M$ a.e. for some $M < \infty$, then

$$\int_E f_n \, d\mu \;\to\; \int_E f \, d\mu.$$

(Apply DCT with $g \equiv M\mathbf{1}_E$.)

Useful corollary (convergence in L^1). If $|f_n| \leq g \in L^1$ and $f_n \to f$ in measure (or a.e.), then $f \in L^1$ and $\|f_n - f\|_{L^1} \to 0$ (by extracting a subsequence a.e. and applying DCT).

Mini–toolkit: Markov/Chebyshev, Egorov, and Vitali

Markov's inequality (nonnegative case). If $X \geq 0$ is measurable and $a > 0$, then

$$\mu(\{X \geq a\}) \;\leq\; \frac{1}{a} \int_E X \, d\mu.$$

In particular, taking $X = |f|^p$ and $a = t^p$ gives

$$\mu(\{|f| \geq t\}) \;\leq\; \frac{\|f\|_{L^p}^p}{t^p} \qquad (p > 0).$$

Chebyshev's inequality (general p). For $f \in L^p(E)$ with $p > 0$ and $t > 0$,

$$\mu(\{|f| \geq t\}) \;\leq\; \frac{\|f\|_{L^p}^p}{t^p}.$$

(Obtained from Markov with $X = |f|^p$.)

Egorov's theorem (a.e. $\Rightarrow$ almost uniform on finite measure). If $\mu(E) < \infty$ and $f_n \to f$ almost everywhere on E, then for every $\varepsilon > 0$ there exists a measurable $A \subset E$ with $\mu(E \setminus A) < \varepsilon$ such that $f_n \to f$ uniformly on A.

Uniform integrability (UI). A family $\mathcal{F} \subset L^1(E)$ is *uniformly integrable* if for every $\epsilon > 0$ there exists $\delta > 0$ such that $\mu(A) < \delta \Rightarrow \sup_{f\in\mathcal{F}} \int_A |f|\, d\mu < \epsilon$. A sufficient condition is domination: if there exists $g \in L^1(E)$ with $|f| \leq g$ a.e. for all $f \in \mathcal{F}$, then $\mathcal{F}$ is uniformly integrable.

Vitali convergence theorem. Let $(f_n) \subset L^1(E)$ be uniformly integrable and suppose $f_n \to f$ in measure. Then $f \in L^1(E)$ and

$$\|f_n - f\|_{L^1} \longrightarrow 0.$$

(With domination $|f_n| \leq g \in L^1$, Vitali follows from DCT combined with subsequence arguments.)

Mini–toolkit: Egorov $\Rightarrow$ DCT on large sets

Setup. Assume $\mu(E) < \infty$, $f_n \to f$ a.e., and $|f_n| \leq g \in L^1(E)$.

Step 1 (Egorov). For any $\varepsilon > 0$ there is $A \subset E$ with $\mu(E \setminus A) < \varepsilon$ such that $f_n \to f$ uniformly on A.

Step 2 (convergence on A). Uniform convergence and $|f_n|, |f| \leq g$ imply

$$\int_A |f_n - f|\, d\mu \longrightarrow 0.$$

Step 3 (control on the complement). By domination,

$$\int_{E\setminus A} |f_n - f|\, d\mu \leq \int_{E\setminus A} (|f_n| + |f|)\, d\mu \leq 2\int_{E\setminus A} g\, d\mu.$$

Choosing ε so that $\int_{E\setminus A} g < \delta$ gives the desired small tail.

Conclusion. $\displaystyle\int_E |f_n - f|\, d\mu \to 0$; that is, DCT holds via Egorov + small tail.

Remark (infinite measure). If $\mu(E) = \infty$ (e.g. $E = \mathbb{R}^n$), take an increasing sequence of bounded sets $K_R \uparrow E$ with $\int_{E\setminus K_R} g$ arbitrarily small. Apply the argument on K_R and then let $R \to \infty$.

Mini–toolkit: Jensen's inequality and convexity tools

Convexity. A function $\varphi : I \to \mathbb{R}$ is convex if

$$\varphi(\lambda x + (1-\lambda)y) \leq \lambda\varphi(x) + (1-\lambda)\varphi(y) \quad (x, y \in I,\ 0 \leq \lambda \leq 1).$$

Jensen's inequality. Let $(E, \mathcal{E}, \nu)$ be a probability space, φ convex on an interval containing the range of X. If X is integrable, then

$$\varphi\Big(\int_E X\, d\nu\Big) \;\le\; \int_E \varphi(X)\, d\nu.$$

For concave φ the inequality reverses.

Power means via Jensen. For $p \ge 1$, $\varphi(t) = |t|^p$ is convex; hence

$$\Big|\int_E f\, d\nu\Big|^p \le \int_E |f|^p\, d\nu.$$

Young's inequality (via convex duality). For conjugate exponents $p, q > 1$ and $a, b \ge 0$,

$$ab \;\le\; \frac{a^p}{p} + \frac{b^q}{q}.$$

(Equivalently, the Legendre transform of $\phi(t) = t^p/p$ is $\phi(s) = s^q/q$.)

Hölder from Young/Jensen. Normalize f, g by $\alpha = \|f\|_p$, $\beta = \|g\|_q$; write

$$\frac{|fg|}{\alpha\beta} \le \frac{|f|^p}{p\,\alpha^p} + \frac{|g|^q}{q\,\beta^q}.$$

Integrate to obtain $\int |fg| \le \|f\|_p \|g\|_q$.

AM–GM and log-sum. Since log is concave,

$$\log\Big(\sum_i \lambda_i x_i\Big) \;\ge\; \sum_i \lambda_i \log x_i \quad\Rightarrow\quad \sum_i \lambda_i x_i \;\ge\; \prod_i x_i^{\lambda_i},$$

for $\lambda_i \ge 0$, $\sum_i \lambda_i = 1$, $x_i > 0$.

KL-divergence nonnegativity. For probability densities p, q with $p \ll q$,

$$D_{\mathrm{KL}}(p\|q) = \int p \log\frac{p}{q}\, d\mu \;\ge\; 0,$$

by Jensen with convex $\varphi(t) = t\log t$ (or concavity of log).

Problem 3 — Density of rational simple functions; separability; L^∞ non-separable

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Statement. Let $\mathcal{R}_{\mathbb{Q}}$ be the family of rectangles $(a_1, b_1] \times \cdots \times (a_n, b_n]$ with all $a_i, b_i \in \mathbb{Q}$. Let $\mathcal{S}_{\mathbb{Q}}$ be the set of finite sums

$$s(x) = \sum_{k=1}^{N} (\alpha_k + i\beta_k)\, \mathbf{1}_{R_k}(x), \qquad R_k \in \mathcal{R}_{\mathbb{Q}},\ \alpha_k, \beta_k \in \mathbb{Q}.$$

(a) For $1 \le p < \infty$, show $\mathcal{S}_{\mathbb{Q}}$ is dense in $L^p(\mathbb{R}^n)$ and conclude $L^p(\mathbb{R}^n)$ is separable. (b) Show $L^\infty(\mathbb{R}^n)$ is not separable.

Theory (quick toolkit).

- Lebesgue measure is regular: for measurable A of finite measure, $\mu(A) = \sup\{\mu(K) : K \subset A,\ K \text{ compact}\} = \inf\{\mu(U) : A \subset U,\ U \text{ open}\}$. Every open set is a countable union of rectangles with rational endpoints.
- For measurable A, B and $1 \le p < \infty$, $\|\mathbf{1}_A - \mathbf{1}_B\|_{L^p} = \mu(A \triangle B)^{1/p}$.
- Truncation/localization: for $f \in L^p(\mathbb{R}^n)$, $f_R := f\mathbf{1}_{B(0,R)} \to f$ in L^p, and $f^{(M)} := \max(-M, \min(f, M)) \to f$ in L^p.

Proof of (a) (density). Let $f \in L^p$ and $\varepsilon > 0$. Choose R so that $\|f - f_R\|_p < \varepsilon/3$ and then M so that $\|f_R - f_R^{(M)}\|_p < \varepsilon/3$. It suffices to approximate a bounded function $g := f_R^{(M)}$ supported in $B(0, R)$.

Partition $B(0, R)$ by finitely many rectangles Q_j with rational endpoints and small diameter. Pick $c_j \in \mathbb{Q} + i\mathbb{Q}$ with $\sup_{x \in Q_j \cap B(0,R)} |g(x) - c_j| \le \eta$ and set $s = \sum_j c_j \mathbf{1}_{Q_j} \in \mathcal{S}_{\mathbb{Q}}$. Then

$$\|g - s\|_{L^p} \le \eta\, |B(0, R)|^{1/p}.$$

Choosing η (and the grid) suitably small gives $\|g - s\|_{L^p} < \varepsilon/3$. Therefore $\|f - s\|_{L^p} < \varepsilon$, proving density.

Since $\mathcal{R}_{\mathbb{Q}}$ and $\mathbb{Q} + i\mathbb{Q}$ are countable, the set $\mathcal{S}_{\mathbb{Q}}$ is countable. Thus $L^p(\mathbb{R}^n)$ is separable.

Proof of (b) (L^∞ not separable). Define $A_t = [0, 1/2) + t \pmod 1 \subset [0, 1)$ for $t \in [0, 1)$, and set $X = \{\mathbf{1}_{A_t} : t \in [0, 1)\} \subset L^\infty(\mathbb{R})$. If $s \ne t$, then $\mu(A_s \triangle A_t) > 0$, hence

$$\|\mathbf{1}_{A_s} - \mathbf{1}_{A_t}\|_{L^\infty} = \operatorname{ess\,sup} |\mathbf{1}_{A_s} - \mathbf{1}_{A_t}| = 1.$$

Therefore X is an uncountable 1-separated set, so $L^\infty(\mathbb{R}^n)$ is not separable.

Theory: separability and the rational simple class $\mathcal{S}_{\mathbb{Q}}$

Separable metric spaces. A metric space (X, d) is *separable* if there exists a countable dense subset $D \subset X$, i.e. $\overline{D} = X$.

Rational rectangles and simple functions. Let $\mathcal{R}_{\mathbb{Q}}$ be the family of axis-parallel rectangles $(a_1, b_1] \times \cdots \times (a_n, b_n]$ with $a_i, b_i \in \mathbb{Q}$. Define

$$\mathcal{S}_{\mathbb{Q}} = \left\{ s = \sum_{k=1}^{N} (\alpha_k + i\beta_k)\, \mathbf{1}_{R_k} : N \in \mathbb{N},\ R_k \in \mathcal{R}_{\mathbb{Q}},\ \alpha_k, \beta_k \in \mathbb{Q} \right\}.$$

Countability. $\mathcal{R}_\mathbb{Q}$ is countable (finite tuples of rationals). Finite sequences of elements of a countable set, with rational coefficients, form a countable set. Hence $\mathcal{S}_\mathbb{Q}$ is countable.

Density in L^p, $1 \le p < \infty$. Given $f \in L^p(\mathbb{R}^n)$ and $\varepsilon > 0$:

1) *Localization:* pick R so that $\|f - f\,\mathbf{1}_{B(0,R)}\|_{L^p} < \varepsilon/3$.
2) *Truncation:* pick M so that $\|f\,\mathbf{1}_{B(0,R)} - \mathrm{clip}_{[-M,M]}(f)\,\mathbf{1}_{B(0,R)}\|_{L^p} < \varepsilon/3$.
3) *Rational simple approximation:* approximate the bounded, compactly supported function g from 2) by $s \in \mathcal{S}_\mathbb{Q}$ using regularity of Lebesgue measure (or Lusin's theorem) so that $\|g - s\|_{L^p} < \varepsilon/3$.

By the triangle inequality, $\|f - s\|_{L^p} < \varepsilon$. Thus $\overline{\mathcal{S}_\mathbb{Q}}^{L^p} = L^p(\mathbb{R}^n)$.

Consequence. Since $\mathcal{S}_\mathbb{Q}$ is countable and dense, $L^p(\mathbb{R}^n)$ is separable for $1 \le p < \infty$.

Contrast with L^∞. The family $X = \{\mathbf{1}_{[0,1/2)+t \bmod 1} : t \in [0,1)\}$ is uncountable and 1-separated in L^∞, hence $L^\infty(\mathbb{R}^n)$ is not separable.

Rational rectangles and complex coefficients. Fix $n \in \mathbb{N}$. Let $\mathcal{R}_\mathbb{Q}$ be the family of axis–parallel rectangles

$$R = (a_1, b_1] \times \cdots \times (a_n, b_n] \subset \mathbb{R}^n, \qquad a_i, b_i \in \mathbb{Q}.$$

Define the class of *rational simple functions* with complex coefficients by

$$\mathcal{S}_\mathbb{Q} = \left\{ s : \mathbb{R}^n \to \mathbb{C} \;\middle|\; s(x) = \sum_{k=1}^{N} (\alpha_k + i\beta_k)\,\mathbf{1}_{R_k}(x),\ N \in \mathbb{N},\ R_k \in \mathcal{R}_\mathbb{Q},\ \alpha_k, \beta_k \in \mathbb{Q} \right\}.$$

Here:

- n is the *dimension* of the ambient space $\mathbb{R}^n$ (the domain of all functions).
- N is the *finite number of terms* in the sum (not a dimension).
- $\mathbf{1}_{R_k}$ denotes the indicator of R_k; half–open endpoints avoid overlaps on boundaries (which are null sets).
- $(\alpha_k + i\beta_k) \in \mathbb{Q} + i\mathbb{Q} \subset \mathbb{C}$ are *complex rational* coefficients.

Real-valued variant. For real L^p spaces, take $\beta_k = 0$ (so coefficients are in $\mathbb{Q}$) and the same arguments apply, since

$$L^p(\mathbb{R}^n; \mathbb{C}) \cong L^p(\mathbb{R}^n; \mathbb{R}) \;\oplus\; i\,L^p(\mathbb{R}^n; \mathbb{R})$$

isometrically, and density can be shown on real and imaginary parts separately.

Evaluation of a simple function on $\mathbb{R}^n$. Let $R_1, \ldots, R_N \subset \mathbb{R}^n$ be (half-open) rectangles and let each R_k carry a complex coefficient $c_k \in \mathbb{C}$. Define

$$s(x) \;=\; \sum_{k=1}^{N} c_k\,\mathbf{1}_{R_k}(x), \qquad x \in \mathbb{R}^n.$$

For any point $x \in \mathbb{R}^n$, exactly one of the following occurs:

$$s(x) = \begin{cases} \displaystyle\sum_{k:\, x \in R_k} c_k, & \text{if } x \text{ belongs to one or more rectangles,} \\ 0, & \text{if } x \notin \bigcup_{k=1}^N R_k. \end{cases}$$

Remark (disjointization). If the rectangles are chosen pairwise disjoint (e.g. by partitioning into half-open boxes), then for almost every x there is at most one index j with $x \in R_j$, and hence $s(x) = c_j$ (or $s(x) = 0$ if x is in none).

"Belongs to a rectangle" and evaluation. A rectangle $R = (a_1, b_1] \times \cdots \times (a_n, b_n] \subset \mathbb{R}^n$ contains $x = (x_1, \ldots, x_n)$ iff $a_i < x_i \le b_i$ for all i. Given $s(x) = \sum_{k=1}^N c_k \, \mathbf{1}_{R_k}(x)$ with $c_k \in \mathbb{C}$,

$$s(x) \;=\; \sum_{k:\, x \in R_k} c_k,$$

with the convention that the sum is 0 if $x \notin \bigcup_{k=1}^N R_k$. If the R_k are pairwise disjoint (e.g. a half-open partition), then $s(x) = c_j$ for the unique j with $x \in R_j$ (or 0 if none).

Problem 4 — Dual formula for $\|f\|_p$ and Minkowski's integral inequality

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Statement. Let $1 \le p < \infty$ and q be conjugate ($\frac{1}{p} + \frac{1}{q} = 1$).

(a) For measurable $f : \mathbb{R}^n \to \mathbb{C}$,

$$\|f\|_{L^p} = \sup \left\{ \int_{\mathbb{R}^n} f(x) g(x) \, dx : g \in L^q(\mathbb{R}^n), \; \|g\|_{L^q} \le 1 \right\}.$$

(b) Let $F : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{C}$ be integrable and set $G(y) = \int_{\mathbb{R}^n} F(x, y) \, dx$. Show that for any $g \in L^q$ with $\|g\|_{L^q} \le 1$,

$$\int_{\mathbb{R}^n} |G(y)| \, |g(y)| \, dy \le \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x, y)|^p \, dy \right)^{1/p} dx.$$

Deduce Minkowski's integral inequality

$$\left(\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) \, dx \right|^p dy \right)^{1/p} \le \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x, y)|^p dy \right)^{1/p} dx.$$

Theory (toolkit). Hölder's inequality (complex form): $\int |uv| \le \|u\|_p \|v\|_q$; equality for $1 < p < \infty$ when $v = \operatorname{sgn}(u) \, |u|^{p-1} / \|u\|_p^{p-1}$. Tonelli/Fubini applies since $F \in L^1(\mathbb{R}^n \times \mathbb{R}^n)$.

Proof of (a). By Hölder, $\int fg \le \|f\|_p$ for all $\|g\|_q \le 1$, hence the supremum is at most $\|f\|_p$. For $1 < p < \infty$, define

$$g_0(x) = \begin{cases} e^{-i \arg f(x)} \dfrac{|f(x)|^{p-1}}{\|f\|_p^{p-1}}, & f(x) \neq 0, \\ 0, & f(x) = 0, \end{cases}$$

so that $\|g_0\|_q = 1$ and $\int fg_0 = \|f\|_p$. For $p = 1$, take $g_0 = e^{-i \arg f} \in L^\infty$ to get $\int fg_0 = \|f\|_1$. Thus the supremum equals $\|f\|_p$.

Proof of (b) and Minkowski. Triangle inequality and Fubini give

$$\int |G(y)|\,|g(y)|\,dy \le \iint |F(x,y)|\,|g(y)|\,dx\,dy = \int \Big(\int |F(x,y)|\,|g(y)|\,dy \Big)\,dx.$$

For each fixed x, Hölder in y yields

$$\int |F(x,y)|\,|g(y)|\,dy \le \Big(\int |F(x,y)|^p dy \Big)^{1/p} \|g\|_q \le \Big(\int |F(x,y)|^p dy \Big)^{1/p}.$$

Integrate in x to obtain the displayed bound. Taking the supremum over all $\|g\|_q \le 1$ and using part (a) for G,

$$\|G\|_{L^p_y} \le \int \Big(\int |F(x,y)|^p dy \Big)^{1/p} dx,$$

which is Minkowski's integral inequality.

Lp spaces: theory, norms, examples and counterexamples

Definitions. Let $(E, \mathcal{E}, \mu)$ be a measure space. For $0 < p < \infty$,

$$L^p(E) = \{ f \text{ measurable} : \ \|f\|_{L^p}^p := \int_E |f|^p \, d\mu < \infty \} \big/ (\text{a.e. equality}).$$

For $p \ge 1$, $\|\cdot\|_{L^p}$ is a norm; for $0 < p < 1$ it is a quasi–norm: $\|f+g\|_{L^p}^p \le \|f\|_{L^p}^p + \|g\|_{L^p}^p$ (triangle fails). Define

$$L^\infty(E) = \{ f : \ \|f\|_{L^\infty} = \operatorname*{ess\,sup}_{x \in E} |f(x)| < \infty \}.$$

Basic facts. For $1 \le p \le \infty$, L^p is complete (Banach). For $0 < p < 1$, L^p is complete w.r.t. $d(f,g) = \|f-g\|_{L^p}^p$ but not locally convex. If $1 < p < \infty$ and q is conjugate

($\frac{1}{p} + \frac{1}{q} = 1$), then $(L^p)^* \cong L^q$ via $f \mapsto (g \mapsto \int fg)$; L^p is reflexive. For $p = 1$, $(L^1)^* = L^\infty$; for $p = \infty$, $(L^\infty)^*$ strictly contains L^1.

Inequalities. Hölder: $\int |uv| \leq \|u\|_p \|v\|_q$. Minkowski: $\|u + v\|_p \leq \|u\|_p + \|v\|_p$. On finite measure spaces, if $1 \leq q \leq p \leq \infty$ then $\|h\|_{L^q} \leq \mu(E)^{\frac{1}{q} - \frac{1}{p}} \|h\|_{L^p}$.

Separability. If (E, μ) is σ-finite, $L^p(E)$ is separable for $1 \leq p < \infty$ (dense subset: rational simple functions). L^∞ is not separable on $([0,1], \lambda)$ or $(\mathbb{R}^n, \lambda)$.

Uniform convexity. L^p is uniformly convex for $1 < p < \infty$ (Clarkson), hence strictly convex and reflexive.

Density of nice functions on $\mathbb{R}^n$. For $1 \leq p < \infty$, $C_c^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$.

Membership tests on $\mathbb{R}^n$. Let $B = \{x : |x| \leq 1\}$.

Near the origin: for $\beta \in \mathbb{R}$,

$$f(x) = |x|^{-\beta} \mathbf{1}_B(x) \in L^p(\mathbb{R}^n) \iff \beta p < n.$$

At infinity: for $\alpha \in \mathbb{R}$,

$$g(x) = (1 + |x|)^{-\alpha} \in L^p(\mathbb{R}^n) \iff \alpha p > n.$$

Borderline with logs:

$$h(x) = |x|^{-n/p} \left(\log \frac{e}{|x|} \right)^{-\gamma} \mathbf{1}_B \in L^p \iff \gamma > \tfrac{1}{p},$$

and similarly at infinity with $(\log(e + |x|))^{-\gamma}$.

Examples and counterexamples.

- *No global inclusion for $p \neq q$ on $\mathbb{R}^n$.* If $p > q$, take $g(x) = (1 + |x|)^{-\alpha}$ with $\frac{n}{p} < \alpha \leq \frac{n}{q}$: $g \in L^p \setminus L^q$. If $p < q$, take $f(x) = |x|^{-\beta} \mathbf{1}_B$ with $\frac{n}{q} < \beta \leq \frac{n}{p}$: $f \in L^q \setminus L^p$.
- *L^∞ not separable.* The uncountable family $\{\mathbf{1}_{[0,1/2)+t \bmod 1} : t \in [0,1)\}$ is 1-separated in L^∞.
- *Failure of triangle for $0 < p < 1$.* For A, B disjoint with finite measure and $p \in (0,1)$: $\|\mathbf{1}_{A \cup B}\|_p^p = \mu(A) + \mu(B) = \|\mathbf{1}_A\|_p^p + \|\mathbf{1}_B\|_p^p$, hence $\|\mathbf{1}_{A \cup B}\|_p = \left(\|\mathbf{1}_A\|_p^p + \|\mathbf{1}_B\|_p^p \right)^{1/p} < \|\mathbf{1}_A\|_p + \|\mathbf{1}_B\|_p$.

Convergence principles. Monotone, Fatou, and Dominated Convergence Theorems govern limits of integrals. If $1 < p < \infty$, bounded subsets of L^p are weakly relatively compact (reflexivity). Uniform integrability plus convergence in measure $\Rightarrow$ L^1 convergence (Vitali).

Concrete L^p examples and counterexamples

1. On a finite interval $[0,1]$.

Power singularities. For $a \in \mathbb{R}$ and $1 \le p < \infty$,

$$x^{-a} \in L^p([0,1]) \iff \int_0^1 x^{-ap}\,dx < \infty \iff ap < 1 \quad \text{(i.e. } a < 1/p).$$

Borderline with logs.

$$x^{-1/p}\Big(\log \frac{e}{x}\Big)^{-\gamma} \in L^p([0,1]) \iff \gamma > \frac{1}{p}.$$

Mild singularities. $|\log x|^\beta \in L^p([0,1])$ for all $\beta \in \mathbb{R}$ and all $1 \le p < \infty$.

Examples.

$$\frac{1}{\sqrt{x}} \in L^p([0,1]) \iff p < 2 \quad \Rightarrow \quad \frac{1}{\sqrt{x}} \in L^1 \setminus L^2.$$

$$\mathbf{1}_{[0,1]} \in L^p([0,1]) \text{ for all } 1 \le p \le \infty.$$

L^∞ on $[0,1]$ consists of essentially bounded functions; e.g. $\sin(1/x) \in L^\infty$, whereas $x^{-a} \notin L^\infty$ for $a > 0$.

2. On $\mathbb{R}^n$.

Decay at infinity. For $\alpha > 0$ and $1 \le p < \infty$,

$$(1+|x|)^{-\alpha} \in L^p(\mathbb{R}^n) \iff \int_1^\infty r^{n-1-\alpha p}\,dr < \infty \iff \alpha p > n.$$

Borderline with logs.

$$(1+|x|)^{-n/p}(\log(e+|x|))^{-\gamma} \in L^p(\mathbb{R}^n) \iff \gamma > \frac{1}{p}.$$

Singularity at the origin (radial power). For $\beta \in \mathbb{R}$,

$$|x|^{-\beta}\,\mathbf{1}_{B(0,1)}(x) \in L^p(\mathbb{R}^n) \iff \int_0^1 r^{n-1-\beta p}\,dr < \infty \iff \beta p < n.$$

Canonical families.

- $e^{-|x|^2} \in L^p(\mathbb{R}^n)$ for every $1 \le p \le \infty$ (Gaussian decays super-polynomially).
- $e^{-|x|} \in L^p(\mathbb{R})$ for every $1 \le p \le \infty$.

- Polynomials x^k do not belong to $L^p(\mathbb{R})$ for any finite p (no decay).
- Compactly supported but unbounded: $|x|^{-a}\mathbf{1}_{|x|<1} \in L^p$ iff $ap < n$, but never in L^∞ for $a > 0$.

3. Crisp "in / out" templates.

$$(1+|x|)^{-(\frac{n}{p}+\varepsilon)} \in L^p(\mathbb{R}^n), \qquad (1+|x|)^{-n/p} \notin L^p(\mathbb{R}^n),$$
$$|x|^{-\beta}\mathbf{1}_{B(0,1)} \in L^p(\mathbb{R}^n) \iff \beta < \frac{n}{p}.$$

Separating different exponents on $\mathbb{R}^n$. For $p \neq q$,

- if $p > q$, choose α with $\frac{n}{p} < \alpha \leq \frac{n}{q}$: $(1+|x|)^{-\alpha} \in L^p \setminus L^q$;
- if $p < q$, choose β with $\frac{n}{q} < \beta \leq \frac{n}{p}$: $|x|^{-\beta}\mathbf{1}_{B(0,1)} \in L^q \setminus L^p$.

4. Sup norm (essential bound).

A function f belongs to $L^\infty(E)$ iff there exists $M < \infty$ with $\mu(\{|f| > M\}) = 0$. Examples:

- $\sin(1/x) \in L^\infty([0,1])$ (bounded oscillatory).
- $(1+|x|)^{-\alpha} \in L^\infty(\mathbb{R}^n)$ for any $\alpha > 0$ (bounded by 1).
- $|x|^{-a}\mathbf{1}_{B(0,1)} \notin L^\infty$ for $a > 0$ (unbounded near 0).

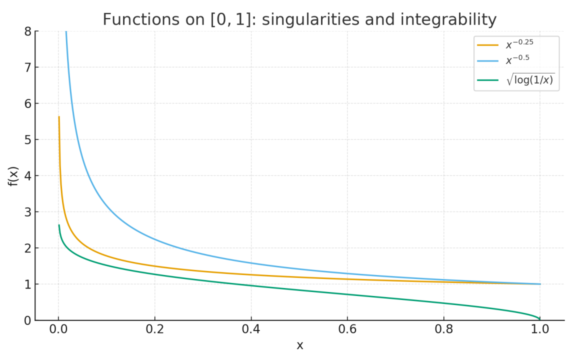

Figure 1: Functions on $[0,1]$: singularities and integrability. The curve $x^{-0.25}$ is L^1; $x^{-0.5}$ fails for L^2; $\sqrt{\log(1/x)}$ is in L^p for all finite p.

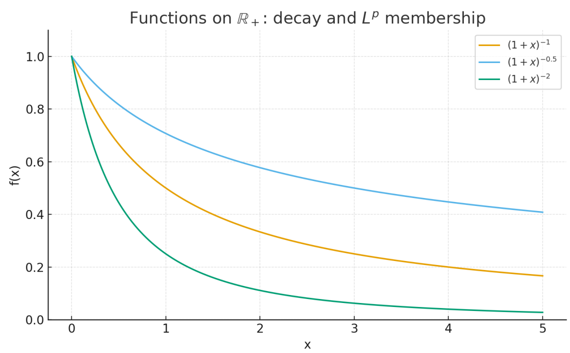

Figure 2: Decay on $\mathbb{R}_+$: $(1+x)^{-\alpha}$ belongs to $L^p(\mathbb{R}^n)$ iff $\alpha p > n$. Here $\alpha \in \{1, \frac{1}{2}, 2\}$.

Closed-form computation of the L^p norm. Consider

$$f(x) = (1+|x|)^{-\alpha}, \qquad x \in \mathbb{R},$$

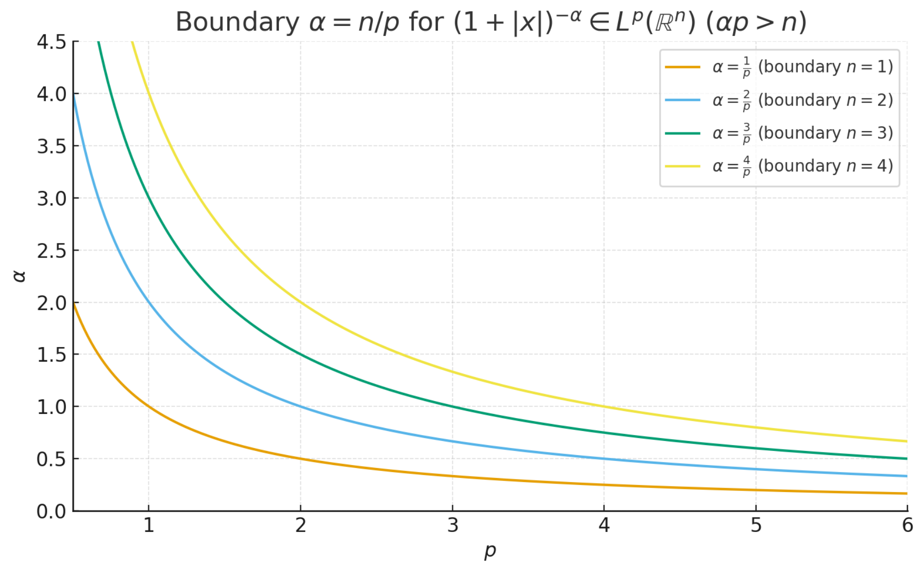

Figure 3: Boundary curves $\alpha = \frac{n}{p}$ describing when $(1 + |x|)^{-\alpha} \in L^p(\mathbb{R}^n)$. The admissible region is $\alpha p > n$ (below the curves).

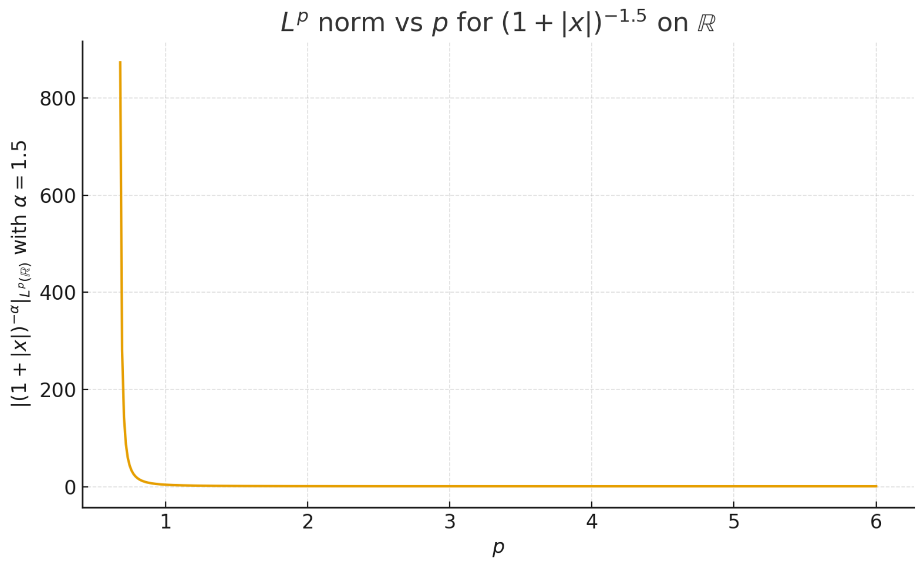

Figure 4: L^p norm of $(1 + |x|)^{-1.5}$ on $\mathbb{R}$ as p varies. Closed form: $\|f\|_p = \big(2/(\alpha p - 1)\big)^{1/p}$ for $\alpha = 1.5$ and $p > 2/3$.

with parameters $\alpha > 0$ and $p > 0$. The L^p norm is

$$\|f\|_p^p = \int_{-\infty}^{\infty} |f(x)|^p \, dx = 2 \int_0^{\infty} (1 + x)^{-\alpha p} \, dx.$$

For $\alpha p > 1$, the integral converges and can be evaluated exactly:

$$\int_0^{\infty} (1 + x)^{-\alpha p} \, dx = \frac{1}{\alpha p - 1}.$$

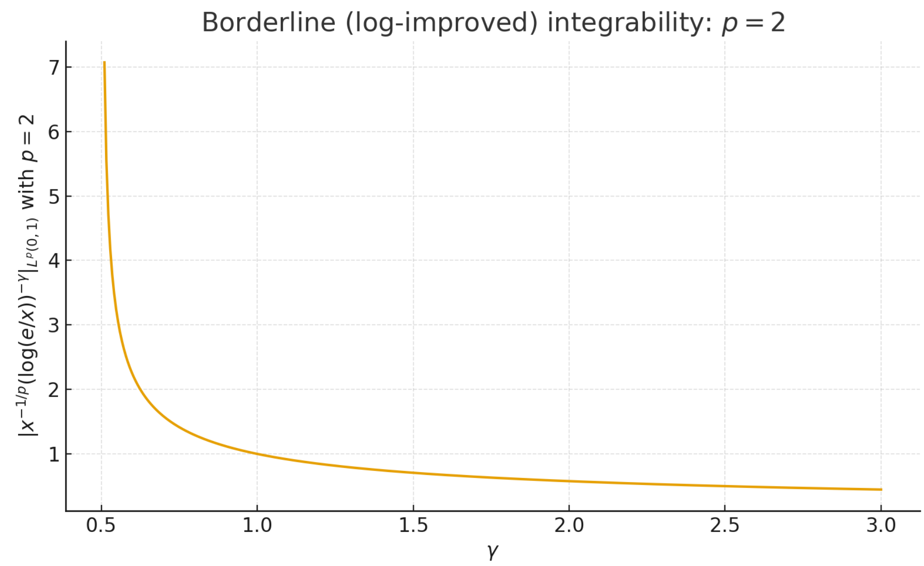

Figure 5: For $p = 2$, the L^2 norm of $h_\gamma(x) = x^{-1/2}(\log(e/x))^{-\gamma}$ on $(0, 1)$. Converges iff $\gamma > \frac{1}{2}$, with $\|h_\gamma\|_2^2 = 1/(\,2\gamma - 1\,)$.

Hence

$$\boxed{\|f\|_p^p = \frac{2}{\alpha p - 1}} \qquad \text{and} \qquad \boxed{\|f\|_p = \left(\frac{2}{\alpha p - 1}\right)^{1/p}}.$$

Validity. The formula holds precisely when $\alpha p > 1$, since otherwise the integral diverges near infinity. If $\alpha p \leq 1$, then $(1 + |x|)^{-\alpha p}$ decays too slowly and $f \notin L^p(\mathbb{R})$.

Example. For $\alpha = 1.5$ and $p = 2$,

$$\|f\|_2 = \left(\frac{2}{3 - 1}\right)^{1/2} = 1.$$

Remark. A *closed form* means the expression can be written exactly using elementary operations (addition, multiplication, powers) without needing series expansions or numerical integration.

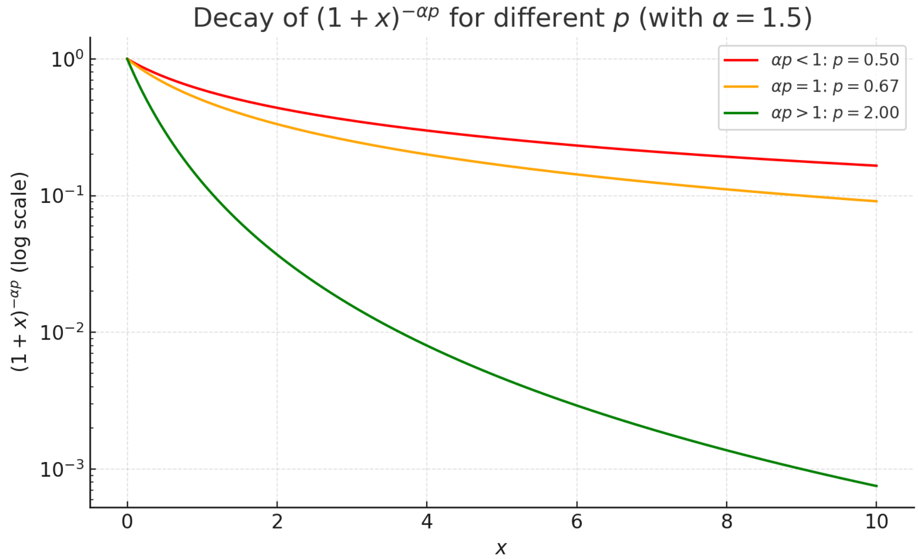

Figure 6: Decay of $(1 + x)^{-\alpha p}$ for different values of p when $\alpha = 1.5$. On a logarithmic scale, the integrand decays too slowly when $\alpha p < 1$ (red), marginally at $\alpha p = 1$ (orange, borderline divergence), and rapidly for $\alpha p > 1$ (green), ensuring convergence of $\int_0^\infty (1 + x)^{-\alpha p} dx$.

Example A (for part (a)): $f(x) = e^{-|x|}$ on $\mathbb{R}$

Let $1 \le p < \infty$ and q be conjugate $\left(\frac{1}{p} + \frac{1}{q} = 1\right)$.

- **L^p-norm.**

$$\|f\|_{L^p}^p = \int_{\mathbb{R}} e^{-p|x|}\,dx = 2\int_0^\infty e^{-px}\,dx = \frac{2}{p} \qquad \Longrightarrow \qquad \boxed{\|f\|_{L^p} = \left(\tfrac{2}{p}\right)^{1/p}}.$$

- **Maximizer in the dual formula.**

$$g_0(x) \;=\; \frac{f(x)^{\,p-1}}{\|f\|_{L^p}^{\,p-1}} \;=\; \frac{e^{-(p-1)|x|}}{\left((2/p)^{1/p}\right)^{p-1}}, \qquad \|g_0\|_{L^q} = 1.$$

- **Supremum equals the norm (equality case).**

$$\int_{\mathbb{R}} f\,g_0 \;=\; \frac{\int_{\mathbb{R}} e^{-p|x|}\,dx}{\|f\|_{L^p}^{\,p-1}} \;=\; \frac{2/p}{\left((2/p)^{1/p}\right)^{p-1}} \;=\; \left(\tfrac{2}{p}\right)^{1/p} \;=\; \|f\|_{L^p}.$$

Thus

$$\|f\|_{L^p} \;=\; \sup\left\{\int_{\mathbb{R}} f(x)g(x)\,dx \;:\; g \in L^q(\mathbb{R}),\ \|g\|_{L^q} \le 1\right\}.$$

Example B (for part (b) & Minkowski): $F(x,y) = f(x)\,h(y)$

Take

$$f(x) = (1+|x|)^{-\alpha} \quad (\alpha > 1,\ \text{so } f \in L^1(\mathbb{R})), \qquad h(y) = e^{-|y|} \in L^p(\mathbb{R})\ (1 \le p < \infty),$$

and define $F(x,y) = f(x)\,h(y)$. Then

$$G(y) \;=\; \int_{\mathbb{R}} F(x,y)\,dx \;=\; \left(\int_{\mathbb{R}} f(x)\,dx\right)h(y) \;=\; \|f\|_{L^1}\,h(y).$$

Compute the constants.

$$\|f\|_{L^1} \;=\; \int_{\mathbb{R}} (1+|x|)^{-\alpha}\,dx \;=\; 2\int_0^\infty (1+x)^{-\alpha}\,dx \;=\; \boxed{\tfrac{2}{\alpha-1}}, \qquad \|h\|_{L^p} \;=\; \left(\int_{\mathbb{R}} e^{-p|y|}\,dy\right)^{1/p} \;=\; \boxed{\left(\tfrac{2}{p}\right)^{1/p}}$$

Inequality with a given $g \in L^q$, $\|g\|_{L^q} \le 1$.

$$\int_{\mathbb{R}} |G(y)|\,|g(y)|\,dy \;=\; \|f\|_{L^1}\int_{\mathbb{R}} |h(y)|\,|g(y)|\,dy \;\le\; \|f\|_{L^1}\,\|h\|_{L^p}\,\|g\|_{L^q} \;\le\; \|f\|_{L^1}\,\|h\|_{L^p}.$$

Right-hand side of the stated bound.

$$\int_{\mathbb{R}}\Big(\int_{\mathbb{R}}|F(x,y)|^p\,dy\Big)^{1/p}dx \;=\; \int_{\mathbb{R}}|f(x)|\,\|h\|_{L^p}\,dx \;=\; \|f\|_{L^1}\,\|h\|_{L^p}.$$

Hence we obtain equality and Minkowski's integral inequality:

$$\|G\|_{L^p_y} \;=\; \|\|f\|_{L^1}\,h\|_{L^p_y} \;=\; \|f\|_{L^1}\,\|h\|_{L^p} \;\le\; \int_{\mathbb{R}}\Big(\int_{\mathbb{R}}|F(x,y)|^p\,dy\Big)^{1/p}dx.$$

Equality occurs for the Hölder extremizer $g(y) = h(y)^{p-1}/\|h\|_{L^p}^{p-1}$ (since $h \ge 0$).

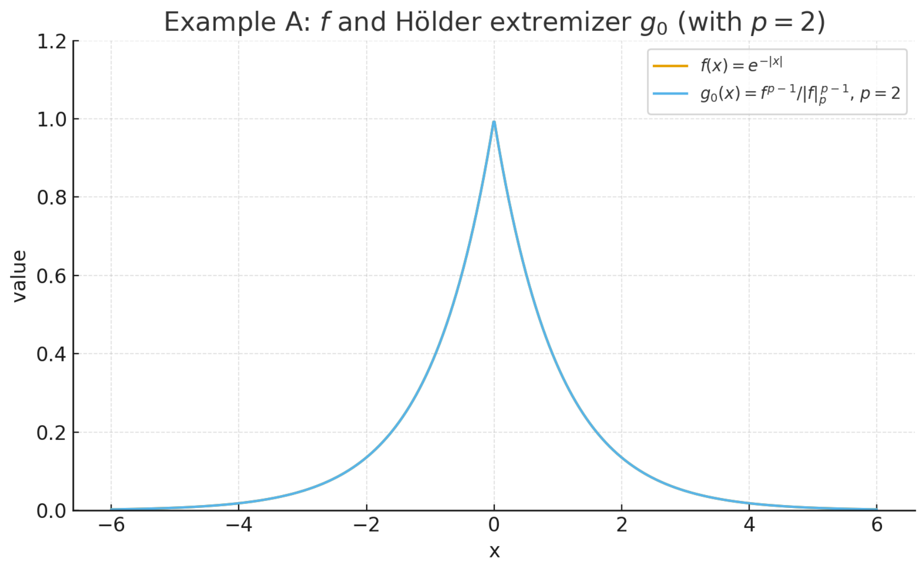

Figure 7: Example A ($p = 2$): $f(x) = e^{-|x|}$ and the Hölder extremizer $g_0(x) = f^{p-1}/\|f\|_p^{p-1}$. Here $\|f\|_{L^2} = (2/2)^{1/2} = 1$, and $\int f\,g_0 = \|f\|_{L^p}$.

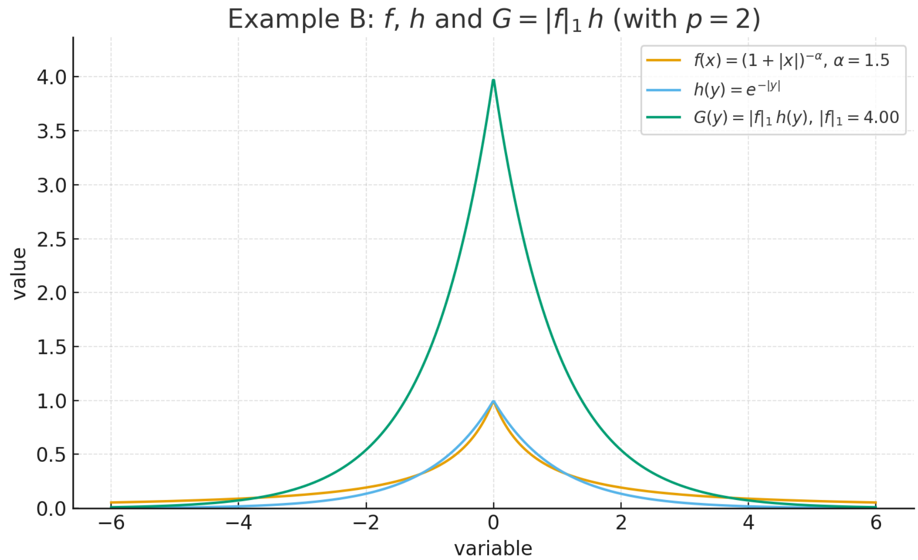

Figure 8: Example B ($p = 2$, $\alpha = 1.5$): $f(x) = (1 + |x|)^{-\alpha}$, $h(y) = e^{-|y|}$, and $G(y) = \|f\|_{L^1} h(y)$ with $\|f\|_{L^1} = 2/(\alpha - 1) = 4$.

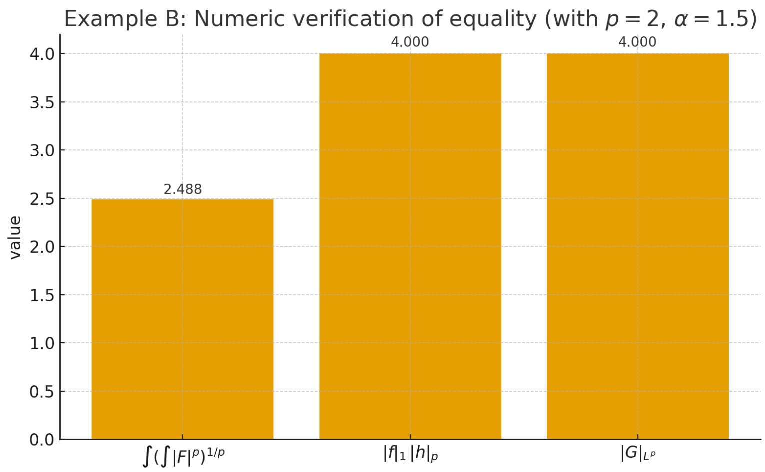

Figure 9: Numeric verification (trapezoidal quadrature) for $p = 2$, $\alpha = 1.5$: the three quantities coincide: $\int (\int |F|^p)^{1/p}$, $\|f\|_{L^1}\|h\|_{L^p}$, and $\|G\|_{L^p}$.

Weak-L^p (Lorentz) spaces

Definition (Weak-L^p (Lorentz $L^{p,\infty}$))**.** Let $(X, \mathcal{M}, \mu)$ be a measure space and $1 \le p < \infty$. A measurable $f : X \to \mathbb{C}$ belongs to the *weak-L^p space* $L^{p,\infty}(X)$ (also written $L^{p,\mathrm{w}}(X)$) if there exists $C > 0$ such that

$$\mu\Big(\{x \in X : |f(x)| > \lambda\}\Big) \le \frac{C^p}{\lambda^p}, \qquad \text{for all } \lambda > 0.$$

The *(quasi-)norm* is

$$\|f\|_{p,\infty} := \sup_{\lambda>0} \lambda \left[\mu\Big(\{|f| > \lambda\}\Big)\right]^{1/p}.$$

It is homogeneous and complete but not a true norm (the triangle inequality may fail up to a constant).

- **Chebyshev/Markov.** If $f \in L^p(X)$, then $\mu(\{|f| > \lambda\}) \le \|f\|_p^p/\lambda^p$. Hence $L^p(X) \subset L^{p,\infty}(X)$ and $\|f\|_{p,\infty} \le \|f\|_p$.
- **Scaling.** For $a \in \mathbb{C}$, $\|af\|_{p,\infty} = |a|\|f\|_{p,\infty}$. If μ is Lebesgue measure on $\mathbb{R}^n$, $\|f(\cdot/r)\|_{p,\infty} = r^{n/p}\|f\|_{p,\infty}$.
- **Rearrangement form.** Writing $d_f(\lambda) := \mu(\{|f| > \lambda\})$, one has

$$\|f\|_{p,\infty} = \sup_{\lambda>0} \lambda\, d_f(\lambda)^{1/p}.$$

- **Weak Hölder (schematic).** If $1 < p, q < \infty$, $1/p + 1/q = 1/r$, and $f \in L^{p,\infty}$, $g \in L^{q,\infty}$, then $fg \in L^{r,\infty}$ and

$$\|fg\|_{r,\infty} \lesssim \|f\|_{p,\infty}\|g\|_{q,\infty}.$$

Example (Borderline singularity on $(0,1)$: weak-only)**.** Fix $p \in [1, \infty)$ and set $f(x) = x^{-1/p}\mathbf{1}_{(0,1)}(x)$. Then for $\lambda \ge 1$,

$$\{|f| > \lambda\} = \{x < \lambda^{-p}\}, \qquad \mu(\{|f| > \lambda\}) = \lambda^{-p}.$$

Hence

$$\|f\|_{p,\infty} = \sup_{\lambda>0} \lambda(\lambda^{-p})^{1/p} = 1.$$

Thus $f \in L^{p,\infty}(0,1)$, but

$$\int_0^1 |f|^p\, dx = \int_0^1 x^{-1}\, dx = \infty,$$

so $f \notin L^p(0,1)$.

Example (Weak-L^p in $\mathbb{R}^n$)**.** Let $f(x) = |x|^{-n/p}\mathbf{1}_{\{|x|\le 1\}}(x)$. Then for $\lambda > 1$,

$$\{|f| > \lambda\} = \{|x| < \lambda^{-p/n}\}, \qquad \mu(\{|f| > \lambda\}) = c_n\,\lambda^{-p}.$$

Thus $f \in L^{p,\infty}(\mathbb{R}^n)$ with $\|f\|_{p,\infty} \sim c_n^{1/p}$, but $f \notin L^p(\mathbb{R}^n)$ since

$$\int_{|x|\le 1} |x|^{-n}\,dx = \infty.$$

A tail variant $|x|^{-n/p}\mathbf{1}_{\{|x|\ge 1\}}$ is also in $L^{p,\infty}$ but not in L^p.

Example (Log bump: strict inclusion the other way)**.** On $(0,1)$, for any $\varepsilon > 0$ let

$$f_\varepsilon(x) = x^{-1/p}\,(\log \tfrac{e}{x})^{-1/p-\varepsilon}.$$

Then $f_\varepsilon \in L^p(0,1)$ (the x^{-1} singularity is softened by the log), so $f_\varepsilon \in L^{p,\infty}(0,1)$ as well. This shows that L^p can be strictly smaller than borderline families living only in $L^{p,\infty}$.

Example (Indicator sets)**.** If $E \subset X$ has finite measure and $f = t^{-1/p}\mathbf{1}_E$ with $t = \mu(E)$, then

$$\mu(\{|f| > \lambda\}) = \begin{cases} t, & 0 < \lambda < t^{-1/p}, \\ 0, & \lambda \ge t^{-1/p}. \end{cases}$$

Hence

$$\|f\|_{p,\infty} = \sup_{\lambda < t^{-1/p}} \lambda\, t^{1/p} = 1.$$

Exercise 5. A sequence with L^p convergence to 0 but no pointwise limit

Construct a sequence (f_j) of measurable functions on $I = (0,1)$ such that

$$f_j \to 0 \quad \text{in } L^p(I) \text{ for every } 1 \le p < \infty,$$

but for every $x \in (0,1)$ the numeric sequence $(f_j(x))$ does *not* converge. Also determine whether an analogous construction is possible for $p = \infty$.

Setup. Let $I = (0,1)$ and $1 \le p < \infty$. Fix an irrational $\alpha \in (0,1)$ and define

$$a_j = \{j\alpha\} \in [0,1), \qquad I_j = (a_j, a_j + 1/j) \pmod 1, \qquad f_j = \mathbf{1}_{I_j}.$$

L^p **convergence.** $\|f_j\|_{L^p(I)} = |I_j|^{1/p} = (1/j)^{1/p} \to 0.$

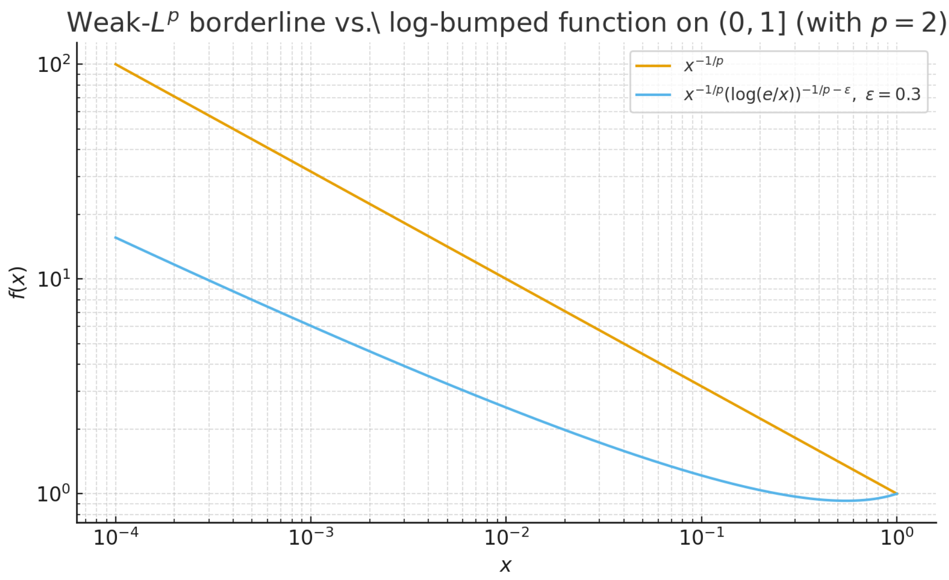

Figure 10: Weak-L^p borderline vs. log-bumped functions on $(0, 1]$ with $p = 2$, $\varepsilon = 0.3$: $f(x) = x^{-1/p}$ and $f_\varepsilon(x) = x^{-1/p}(\log(e/x))^{-1/p-\varepsilon}$.

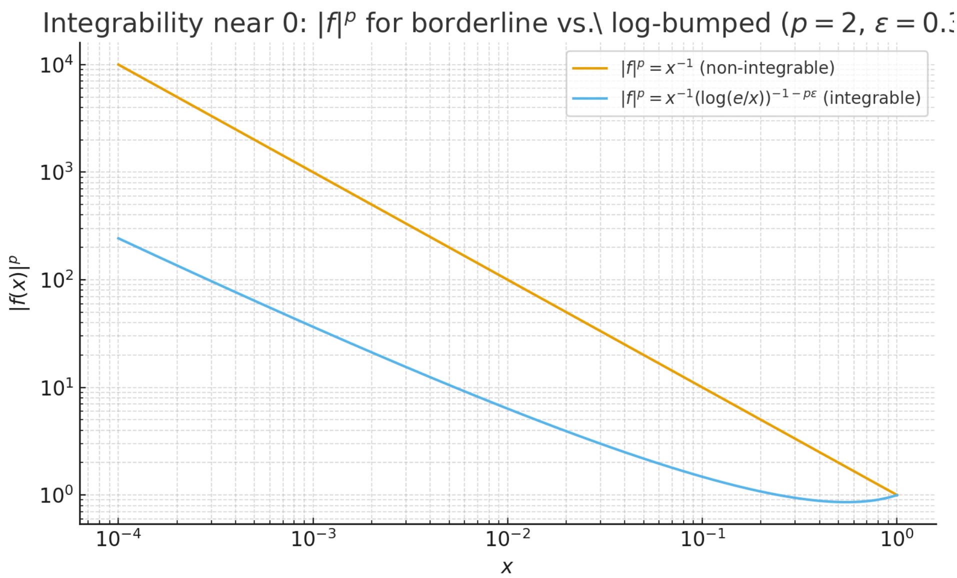

Figure 11: Integrability near 0 of $|f|^p$: for $x^{-1/p}$ we have $|f|^p = x^{-1}$ (diverges); the log-bump yields $|f_\varepsilon|^p = x^{-1}(\log(e/x))^{-1-p\varepsilon}$ (integrable).

No pointwise convergence anywhere. By Dirichlet's approximation, for every $x \in (0, 1)$ there are infinitely many j with $\|j\alpha - x\| < 1/j$, hence $x \in I_j$ infinitely often. Therefore $f_j(x)$ takes the values 1 and 0 infinitely often and does not converge at any x.

Case $p = \infty$. If $\|f_j\|_{L^\infty} \to 0$ then $|f_j(x)| \leq \|f_j\|_{L^\infty} \to 0$ for each x, so $f_j(x) \to 0$ everywhere. Thus no such sequence exists in L^∞.

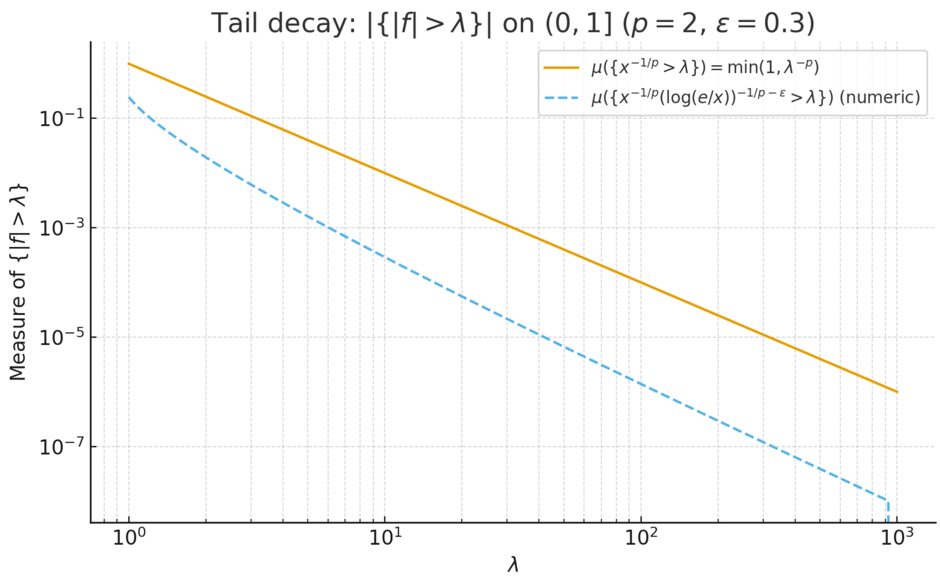

Figure 12: Distribution/tail functions on $(0,1]$. For $x^{-1/p}$: $|\{|f|>\lambda\}| = \min(1,\lambda^{-p})$. The log-bump decays strictly faster (numeric estimate shown).

Exercise 6. Chebyshev's inequality and weak-L^p

(a) Prove that for $f \in L^p(\mathbb{R}^n)$ and $\lambda > 0$,

$$\Big|\{x \in \mathbb{R}^n : |f(x)| > \lambda\}\Big| \le \frac{\|f\|_{L^p}^p}{\lambda^p}.$$

(b) Define the weak-L^p space $L^{p,\mathrm{w}}(\mathbb{R}^n)$ to be the set of measurable f for which there exists $C > 0$ such that

$$\Big|\{x \in \mathbb{R}^n : |f(x)| > \lambda\}\Big| \le \frac{C}{\lambda^p} \qquad \text{for all } \lambda > 0.$$

Show that $L^p(\mathbb{R}^n) \subset L^{p,\mathrm{w}}(\mathbb{R}^n)$ and that this inclusion is proper by exhibiting an explicit function in $L^{p,\mathrm{w}}(\mathbb{R}^n) \setminus L^p(\mathbb{R}^n)$.

(a) Chebyshev/Markov inequality

For $f \in L^p(\mathbb{R}^n)$ and $\lambda > 0$,

$$\Big|\{x : |f(x)| > \lambda\}\Big| \le \frac{\|f\|_{L^p}^p}{\lambda^p}.$$

Proof. On $\{|f| > \lambda\}$, $|f|^p \ge \lambda^p$, hence $\|f\|_p^p \ge \int_{\{|f|>\lambda\}} |f|^p \ge \lambda^p\,|\{|f| > \lambda\}|$.

(b) Weak-L^p and proper inclusion

We say $f \in L^{p,\mathrm{w}}(\mathbb{R}^n)$ if there exists C such that

$$\Big|\{x : |f(x)| > \lambda\}\Big| \le \frac{C}{\lambda^p} \qquad \forall\, \lambda > 0.$$

By Chebyshev, $L^p(\mathbb{R}^n) \subset L^{p,\mathrm{w}}(\mathbb{R}^n)$ with $C = \|f\|_p^p$.

Properness. Let $f(x) = |x|^{-n/p}\mathbf{1}_{\{|x|\le 1\}}(x)$. Then, for $\lambda > 1$,

$$\{|f| > \lambda\} = \{|x| < \lambda^{-p/n}\}, \qquad \Big|\{|f| > \lambda\}\Big| = c_n\,\lambda^{-p},$$

so $f \in L^{p,\mathrm{w}}$. However $\displaystyle\int_{|x|\le 1} |f|^p = \int_{|x|\le 1} |x|^{-n}\,dx = \infty$, hence $f \notin L^p$. Therefore $L^p(\mathbb{R}^n) \subsetneq L^{p,\mathrm{w}}(\mathbb{R}^n)$.

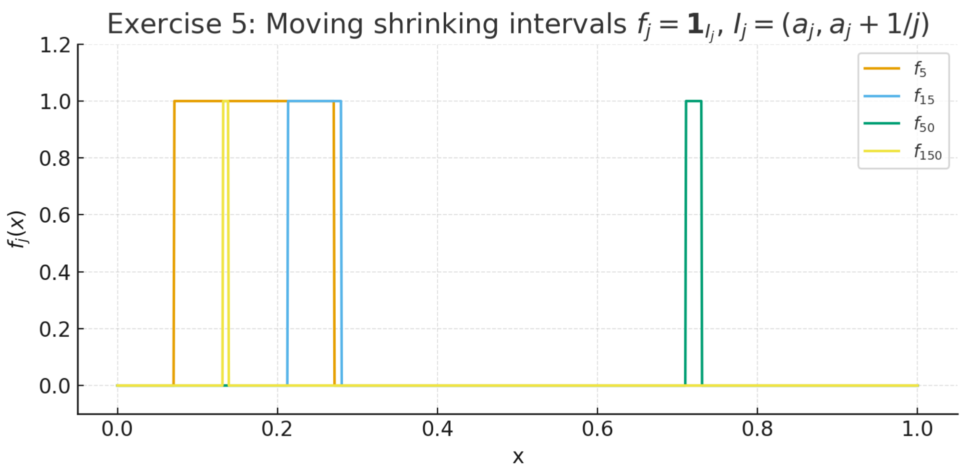

Figure 13: Exercise 5. Moving shrinking intervals $f_j = \mathbf{1}_{I_j}$ with $I_j = (a_j, a_j + 1/j)$, $a_j = \{j\alpha\}$ (irrational α). Each $\|f_j\|_{L^p} = (1/j)^{1/p} \to 0$, yet for every $x \in (0, 1)$ the values $f_j(x)$ take 0 and 1 infinitely often, so there is no pointwise limit.

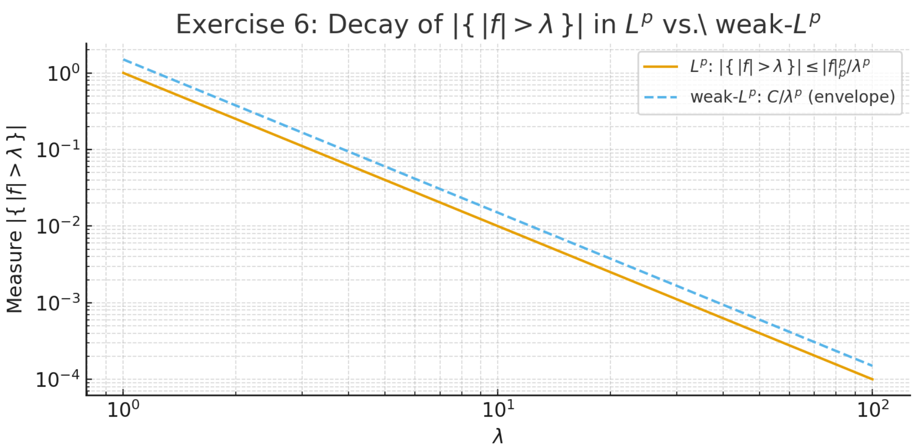

Figure 14: Exercise 6. Decay of the distribution function $\lambda \mapsto |\{\,|f| > \lambda\,\}|$ in L^p and weak-L^p. Chebyshev gives $|\{\,|f| > \lambda\,\}| \le \|f\|_p^p/\lambda^p$; the same λ^{-p} envelope with a different constant defines weak-L^p.

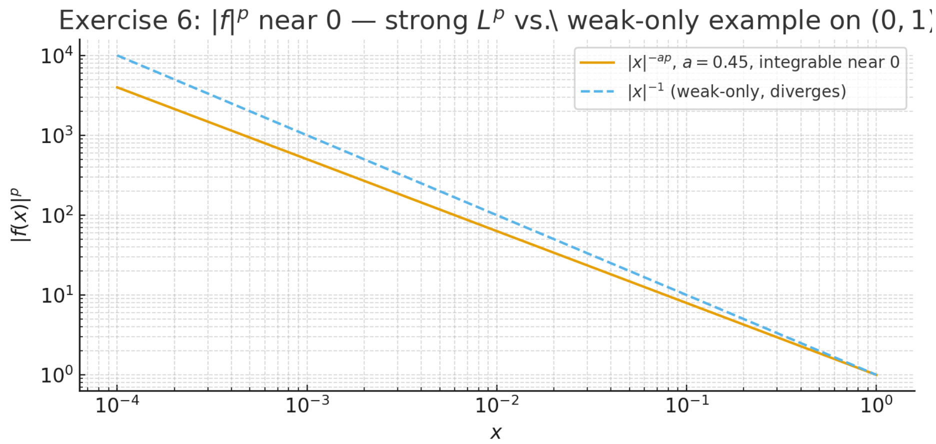

Figure 15: Exercise 6. Near 0 on $(0, 1)$, $f(x) = x^{-a}$ with $a < 1/p$ gives $|f|^p = x^{-ap}$ integrable (solid); the borderline $f(x) = x^{-1/p}$ has $|f|^p = x^{-1}$ (dashed), which diverges logarithmically — a prototypical weak-L^p but not L^p example.

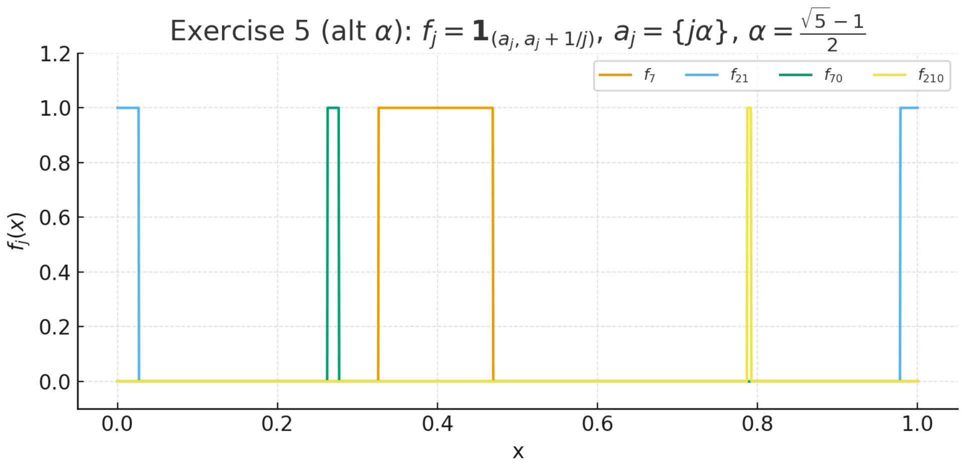

Figure 16: Exercise 5 (alternate α). Moving shrinking intervals with $a_j = \{j\alpha\}$ and $\alpha = (\sqrt{5} - 1)/2$. The L^p norms still satisfy $\|f_j\|_{L^p} = (1/j)^{1/p} \to 0$, yet there is no pointwise limit for any $x \in (0, 1)$.

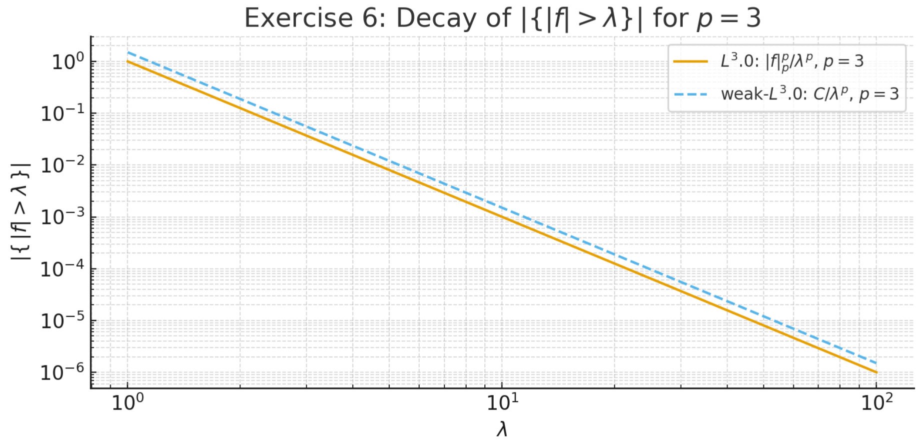

Figure 17: Exercise 6. Distribution function decay for $p = 3$: $|\{\,|f| > \lambda\,\}| \le \|f\|_p^p/\lambda^p$ (solid); weak-L^p admits the same λ^{-p} envelope up to constants (dashed).

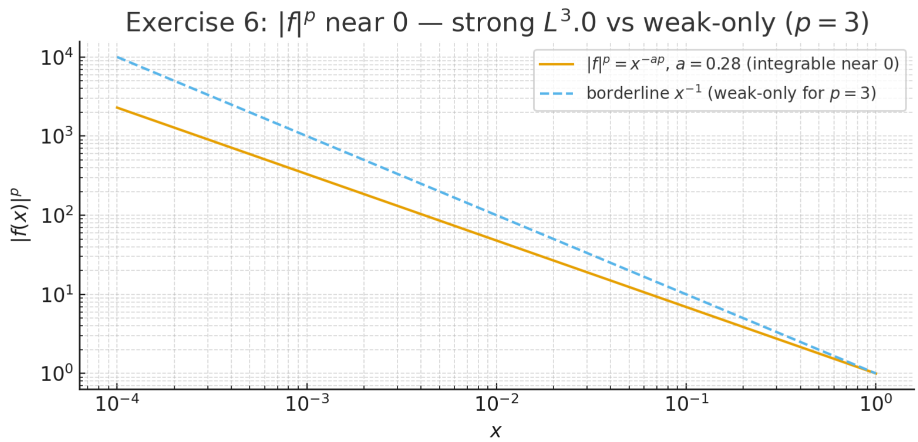

Figure 18: Exercise 6. Near 0 on $(0, 1)$ for $p = 3$: $f(x) = x^{-a}$ with $a < 1/p$ gives $|f|^p = x^{-ap}$ integrable (solid); the borderline $f(x) = x^{-1/p}$ yields $|f|^p = x^{-1}$ (dashed), weak-L^p but not L^p.

Exercise 7 — Limit of L^p norms

Claim. If $f \in L^r(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$ for some $1 \le r < \infty$, then

$$\|f\|_{L^\infty} = \lim_{p\to\infty} \|f\|_{L^p}.$$

Proof. For $p \ge r$, Hölder/Lyapunov interpolation gives

$$\|f\|_{L^p} \le \|f\|_{L^\infty}^{1-r/p} \, \|f\|_{L^r}^{r/p} \xrightarrow[p\to\infty]{} \|f\|_{L^\infty},$$

so $\limsup_{p\to\infty} \|f\|_{L^p} \le \|f\|_\infty$. Let $M = \|f\|_\infty$ and fix $\varepsilon > 0$; the set $A_\varepsilon = \{x : \ |f(x)| > M - \varepsilon\}$ satisfies $0 < |A_\varepsilon| < \infty$ since $(M-\varepsilon)^r |A_\varepsilon| \le \int |f|^r < \infty$. Then for $p \ge r$

$$\|f\|_{L^p} \ge \left(\int_{A_\varepsilon} |f|^p \right)^{1/p} \ge (M-\varepsilon)\, |A_\varepsilon|^{1/p} \xrightarrow[p\to\infty]{} M - \varepsilon.$$

Hence $\liminf_{p\to\infty} \|f\|_{L^p} \ge M - \varepsilon$; letting $\varepsilon \downarrow 0$ gives the result. $\square$

Exercise 8 — Wiener and Vitali covering lemmas

(a) Finite family, factor 3**.** Given open balls $B_1, \ldots, B_N \subset \mathbb{R}^n$, there exist indices $i_1, \ldots, i_k$ with $B_{i_1}, \ldots, B_{i_k}$ pairwise disjoint such that

$$\bigcup_{i=1}^N B_i \subset \bigcup_{j=1}^k 3B_{i_j}, \qquad \text{hence} \qquad \left| \bigcup_{i=1}^N B_i \right| \le 3^n \sum_{j=1}^k |B_{i_j}|.$$

Proof. Select greedily a ball of maximal radius, delete all balls it meets, and iterate. The selected family is disjoint. If B was deleted because it meets a selected ball B^* with radius $\ge$ radius(B), then $B \subset 3B^*$. Therefore $\bigcup_i B_i \subset \bigcup_j 3B_{i_j}$ and $|3B| = 3^n|B|$ yields the measure bound. $\square$

(b) Vitali, factor 5**.** Let $\{B_j : j \in J\}$ be any family of balls with $\mathrm{rad}(B_j) \le R$. There exists a countable disjoint subfamily $\{B_j : j \in J'\}$ with

$$\bigcup_{i\in J} B_i \subset \bigcup_{j \in J'} 5B_j.$$

Proof. Order by decreasing radius and run the same greedy algorithm; radii $\le R$ imply a maximal disjoint subfamily is countable (each contains a distinct rational point). If $x \in B$ for some original B that is not chosen, then when B was removed it met a chosen ball B^* with radius $\ge$ radius(B), whence $B \subset 5B^*$ by a geometric estimate. Thus the stated inclusion holds. $\square$

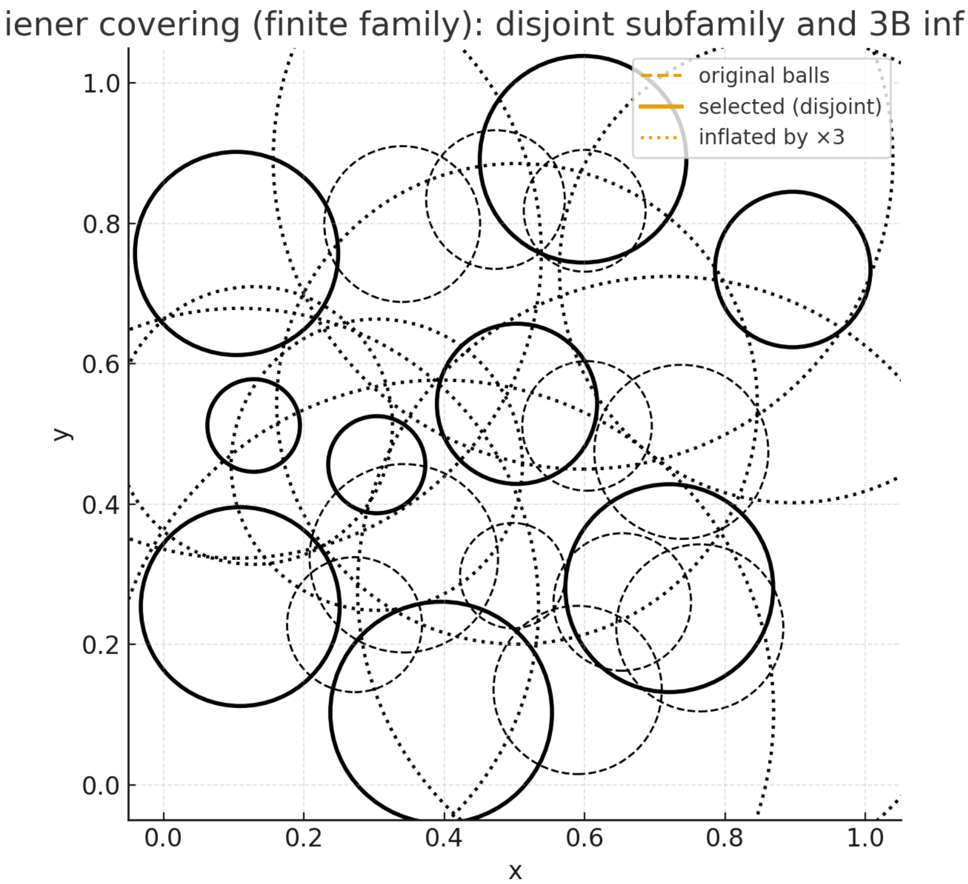

Figure 19: Wiener covering for a finite family: a disjoint subfamily (solid) whose $3B$-inflations (dotted) cover the union of the original balls (dashed).

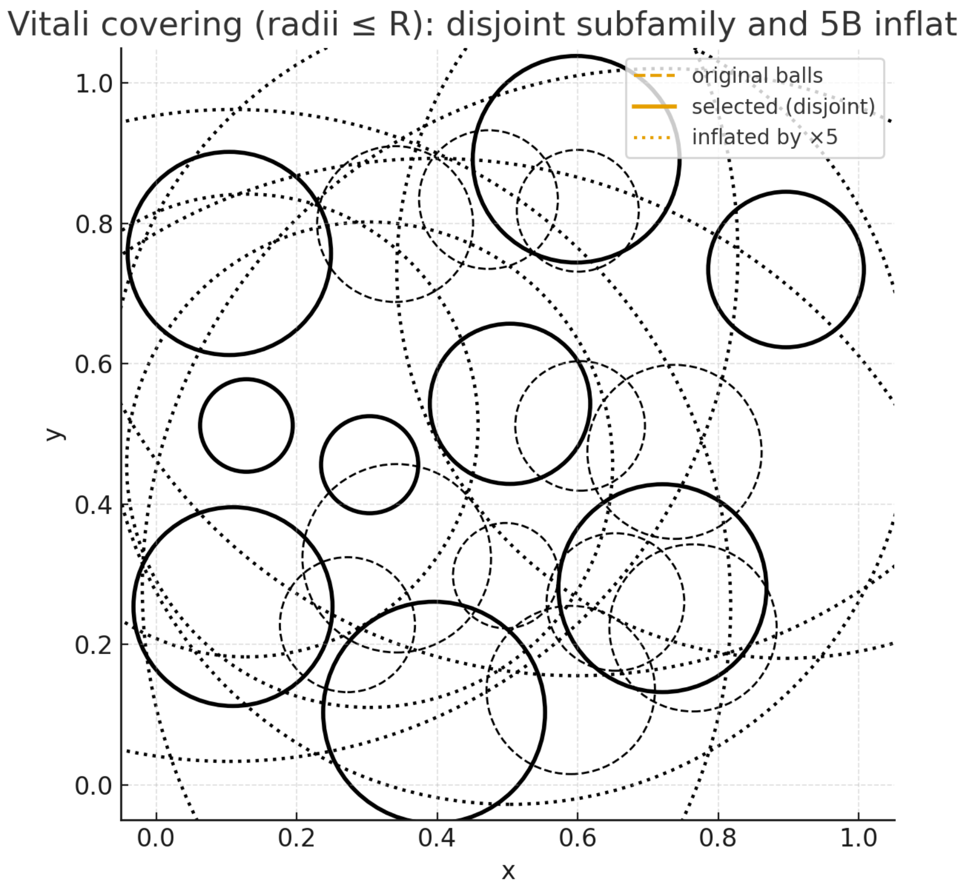

Figure 20: Vitali covering when radii are uniformly bounded: a countable disjoint subfamily (solid) whose $5B$-inflations (dotted) cover the original union (dashed).

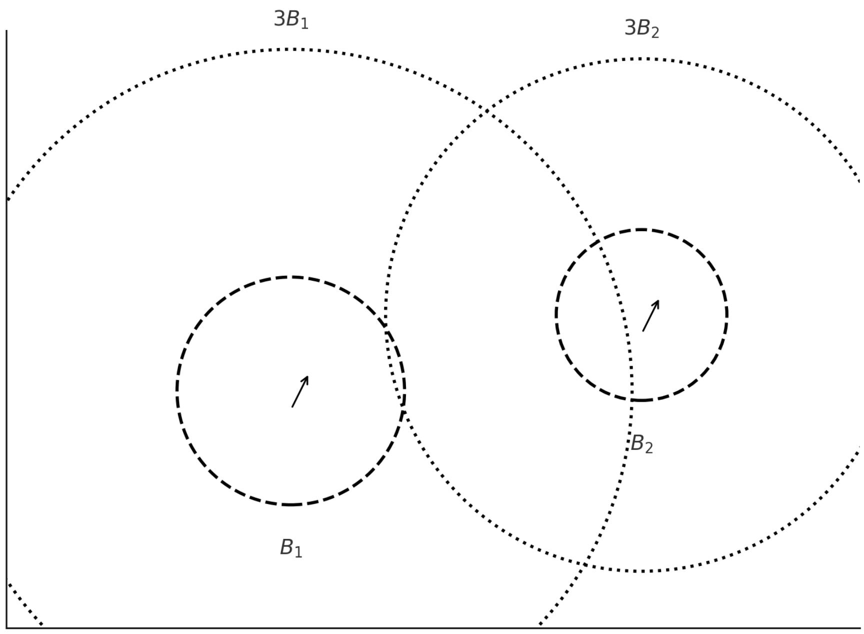

Figure 21: Two balls B_1, B_2 (dashed) and their inflations $3B_1, 3B_2$ (dotted). Each original ball is contained in its own 3-fold enlargement.

Examples of functions in $L^r(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$

Here are several illustrative examples of measurable functions on $\mathbb{R}^n$ that belong to both L^r and L^∞ for some fixed $1 \le r < \infty$. Each example includes a short justification and (where possible) explicit norms.

1) Any bounded, compactly supported function. If $\operatorname{supp} f \subset E$ with $|E| < \infty$ and $\|f\|_\infty = M < \infty$, then

$$\|f\|_{L^r} \le M\,|E|^{1/r} \qquad \Longrightarrow \qquad f \in L^r \cap L^\infty.$$

Concrete example. The indicator of a unit ball:

$$f(x) = \mathbf{1}_{B(0,1)}(x) \quad \Rightarrow \quad \|f\|_\infty = 1, \quad \|f\|_{L^r} = |B(0,1)|^{1/r} = (\omega_n)^{1/r}.$$

Another example. A "hat" (Lipschitz) bump:

$$f(x) = \max\{0, 1 - |x|\}, \qquad \operatorname{supp} f \subset B(0,1), \quad \|f\|_\infty = 1,\ f \in L^r.$$

2) Bounded with polynomial decay (no compact support).

$$f(x) = (1 + |x|)^{-\alpha}, \qquad \alpha > \frac{n}{r}.$$

Then $0 < f \le 1 \Rightarrow f \in L^\infty$, and in spherical coordinates

$$\int_{\mathbb{R}^n} (1 + |x|)^{-\alpha r} dx < \infty \quad \Longleftrightarrow \quad \alpha r > n.$$

Hence $f \in L^r$ whenever $\alpha > n/r$. **Easy pick:** $\alpha = \frac{n}{r} + 1$ works for any r, n.

3) Oscillatory but bounded with the same decay.

$$f(x) = \frac{\sin|x|}{(1 + |x|)^\alpha}, \qquad \alpha > \frac{n}{r}.$$

Then $|f(x)| \le (1 + |x|)^{-\alpha}$, so the same integrability condition applies. Boundedness is clear from $|\sin|x|| \le 1$.

4) Bounded near the origin, with a power–law tail.

$$f(x) = \min(1, |x|^{-\beta}), \qquad \beta > \frac{n}{r}.$$

Then $f \le 1 \Rightarrow f \in L^\infty$. Away from the origin, $\int_{|x|\ge 1} |x|^{-\beta r} dx < \infty$ when $\beta r > n$, so $f \in L^r$. Near 0, $f = 1$ is bounded, hence no singularity occurs.

5) Sum of disjoint bounded bumps. Let $(Q_k)_{k\geq 1}$ be disjoint cubes with finite measure, and define

$$f(x)=\sum_{k=1}^{\infty} a_k\,\mathbf{1}_{Q_k}(x), \qquad |a_k|\leq 1, \quad \sum_{k=1}^{\infty}|a_k|^r|Q_k|<\infty.$$

Then $\|f\|_\infty \leq 1$ and $\|f\|_{L^r}^r=\sum_k |a_k|^r|Q_k|<\infty$, so $f\in L^r\cap L^\infty$.

Contrasts.

- $f\equiv 1$: in L^∞ but not in L^r on $\mathbb{R}^n$ (non-integrable tail).
- $f(x)=|x|^{-\beta}\mathbf{1}_{\{|x|\leq 1\}}$, $0<\beta<\frac{n}{r}$: in L^r but not in L^∞ (unbounded at 0).

Examples in $L^r(\mathbb{R}^n)\cap L^\infty(\mathbb{R}^n)$.

- $\mathbf{1}_{B(0,1)}$: $\|\cdot\|_\infty=1$, $\|\cdot\|_{L^r}=\omega_n^{1/r}$.
- $(1+|x|)^{-\alpha}$ with $\alpha>n/r$ (bounded and $\alpha r>n$ ensures L^r).
- $\dfrac{\sin|x|}{(1+|x|)^\alpha}$ with $\alpha>n/r$.
- $\min\big(1,\,|x|^{-\beta}\big)$ with $\beta>n/r$.
- $f=\sum_{k\geq 1} a_k\,\mathbf{1}_{Q_k}$, $|a_k|\leq 1$, disjoint Q_k, and $\sum|a_k|^r|Q_k|<\infty$.

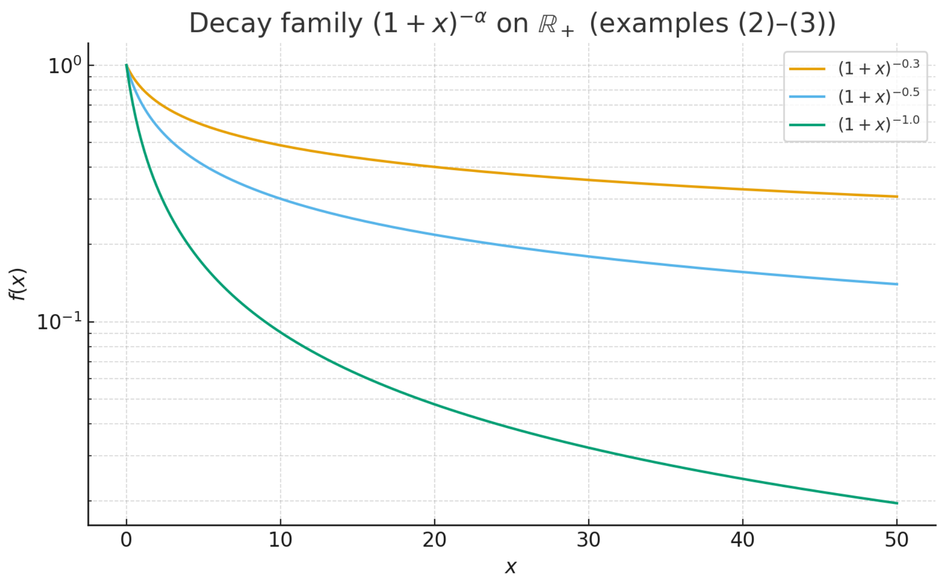

Figure 22: Decay family $(1+x)^{-\alpha}$ on $\mathbb{R}_+$ (examples (2)–(3)). Oscillation (e.g. $\sin|x|$) does not change L^r membership—only the envelope matters.

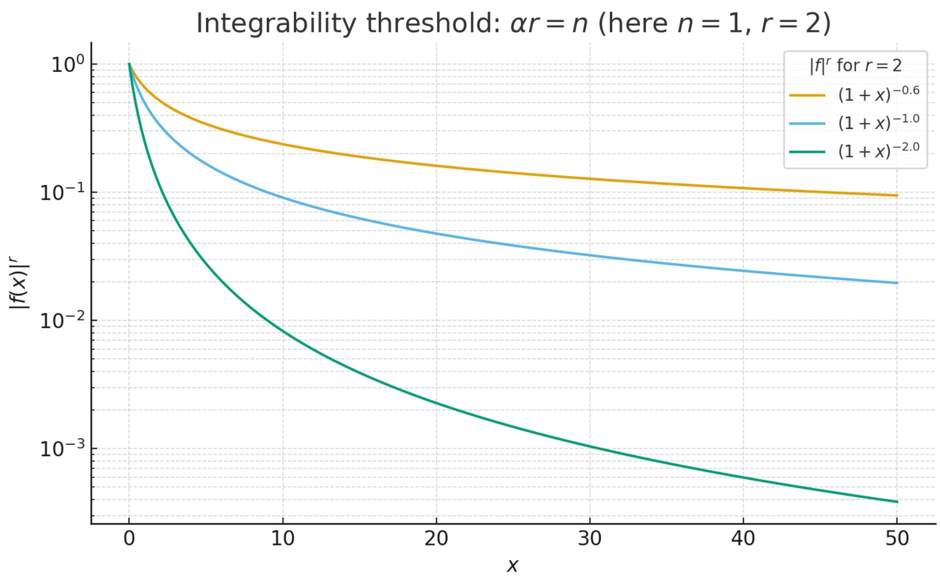

Figure 23: L^r integrand $(1+x)^{-\alpha r}$ and the critical threshold $\alpha r = n$. Here $n = 1$, $r = 2$: only $\alpha > 1/2$ gives L^2-integrability.

Exercise 9 — Differentiation of an integral

Statement. Suppose $f : \mathbb{R} \to \mathbb{C}$ is integrable and let

$$F(x) = \int_{-\infty}^{x} f(t)\, dt.$$

Show that F is differentiable with $F'(x) = f(x)$ at each Lebesgue point $x \in \mathbb{R}$. Deduce that F is differentiable almost everywhere.

Theory (toolkit).

- **Lebesgue differentiation theorem:** If $f \in L^1(\mathbb{R})$, then a.e. x is a Lebesgue point:

$$\lim_{h\to 0} \frac{1}{|h|} \int_{x}^{x+h} |f(t) - f(x)|\, dt = 0.$$

Solution. For any $h \neq 0$,

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_{x}^{x+h} f(t)\, dt = f(x) + \frac{1}{h} \int_{x}^{x+h} \big(f(t) - f(x)\big)\, dt.$$

If x is a Lebesgue point, the last term tends to 0 as $h \to 0$, hence $F'(x) = f(x)$ at every Lebesgue point of f. Since a.e. x is a Lebesgue point, F is differentiable a.e.

Examples.

- $f = \mathbf{1}_{[0,1]}$. Then $F(x) = 0$ for $x < 0$, $F(x) = x$ on $[0,1]$, $F(x) = 1$ for $x > 1$. $F'(x) = 1$ on $(0,1)$ and 0 elsewhere; $x = 0, 1$ are the only non-Lebesgue points for f.
- $f \in L^\infty$ with compact support (e.g. $(1+x^2)^{-1}\mathbf{1}_{[-5,5]}$): the same argument applies since $f \in L^1$.

Exercise 10 — Approximate identities

Statement. Let $\phi \in L^\infty(\mathbb{R}^n)$ satisfy $\phi \geq 0$, $\operatorname{supp}\phi \subset B_1(0)$ and $\int \phi = 1$. Define $\phi_\varepsilon(x) = \varepsilon^{-n}\phi(x/\varepsilon)$. Show that if $f \in L^1(\mathbb{R}^n)$ and x is a Lebesgue point of f, then $(\phi_\varepsilon * f)(x) \to f(x)$ as $\varepsilon \downarrow 0$.

Theory (toolkit).

- **Approximate identity:** (ϕ_ε) is nonnegative, supported in $B_\varepsilon(0)$, $\int \phi_\varepsilon = 1$, $\|\phi_\varepsilon\|_\infty \leq \|\phi\|_\infty \varepsilon^{-n}$.
- **Lebesgue points** as above.

Solution. Write

$$(\phi_\varepsilon * f)(x) - f(x) = \int_{\mathbb{R}^n} \Big(f(x-y) - f(x)\Big)\phi_\varepsilon(y)\,dy = \int_{|z|<1} \Big(f(x-\varepsilon z) - f(x)\Big)\phi(z)\,dz.$$

Fix $\eta > 0$. By the Lebesgue point property there exists $\rho > 0$ such that $⨍_{B_r(x)} |f - f(x)| < \eta$ for $0 < r \leq \rho$. For $\varepsilon < \rho$,

$$\Big|(\phi_\varepsilon * f)(x) - f(x)\Big| \leq \|\phi\|_\infty \int_{|z|<1} |f(x-\varepsilon z) - f(x)|\,dz \lesssim \eta.$$

Let $\eta \downarrow 0$ to conclude $(\phi_\varepsilon * f)(x) \to f(x)$.

Examples.

- Standard mollifier: $\phi = c_n\, e^{-1/(1-|x|^2)}\mathbf{1}_{|x|<1}$, c_n normalizing.
- Box kernel: $\phi = |B_1|^{-1}\mathbf{1}_{B_1(0)}$.

Exercise 11 — Haar system on $\mathbb{R}$

Statement. Let $\psi = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}$ and $\psi_{n,k}(x) = 2^{n/2}\psi(2^n x - k)$, $n, k \in \mathbb{Z}$.

(a) Show $\displaystyle\int_{\mathbb{R}} \psi_{n_1,k_1}\psi_{n_2,k_2} = \delta_{n_1 n_2}\delta_{k_1 k_2}$.

(b) Show $\mathbf{1}_I \in \overline{\mathrm{Span}}\{\psi_{n,k}\}$ for every finite interval I (closure in L^2).

(c) Deduce that $\{\psi_{n,k}\}$ is an orthonormal basis for $L^2(\mathbb{R})$.

Theory (toolkit).

- Dyadic intervals: $I_{n,k} = [k2^{-n}, (k+1)2^{-n})$.
- $\psi_{n,k}$ is supported on $I_{n,k}$ and has zero mean.
- Dyadic conditional expectations $P_m f$: step functions constant on $\{I_{m,k}\}$ with $P_m f \to f$ in L^2 for $f \in L^2$.

Solution. *(a) Orthonormality.* If $(n_1, k_1) \neq (n_2, k_2)$ then either supports are disjoint (integral 0) or, when $n_1 \neq n_2$, the finer wavelet integrates to 0 on each child of the coarser support, so the integral vanishes. Also $\|\psi_{n,k}\|_2^2 = \|\psi\|_2^2 = 1$.

(b) Density on indicators. For a finite interval I, the dyadic step functions $P_m\mathbf{1}_I$ converge to $\mathbf{1}_I$ in L^2. Each $P_m\mathbf{1}_I$ is a finite combination of $\psi_{n,k}$ (Haar expansion of step functions), hence $\mathbf{1}_I \in \overline{\mathrm{Span}}\{\psi_{n,k}\}$.

(c) Basis. Orthonormality from (a) and density from (b) imply $\{\psi_{n,k}\}$ is an orthonormal basis of $L^2(\mathbb{R})$.

Examples.

- $f = \mathbf{1}_{[0,1)} = \frac{1}{2}(\mathbf{1}_{[0,1/2)} + \mathbf{1}_{[1/2,1)})$ has a finite Haar expansion on level $n = 0, 1$.
- Smooth $f \in C_c^\infty$ have rapidly decaying Haar coefficients, illustrating completeness.

Problem 12* — Lebesgue decomposition

Statement. Let $(E, \mathcal{E})$ be a measurable space and let μ, ν be finite measures on $(E, \mathcal{E})$. Show that there exist *unique* finite measures ν_a, ν_s such that

$$\nu = \nu_a + \nu_s, \qquad \nu_a \ll \mu, \qquad \nu_s \perp \mu.$$

Theory.

- *Absolute continuity:* $\nu \ll \mu$ means $\mu(A) = 0 \Rightarrow \nu(A) = 0$ for all $A \in \mathcal{E}$.
- *Mutual singularity:* $\nu \perp \mu$ means there is $S \in \mathcal{E}$ with $\mu(S) = 0$ and $\nu(E \setminus S) = 0$.
- *Radon–Nikodym theorem (finite case):* If ρ and λ are finite with $\rho \ll \lambda$, then there exists $f \in L^1(\lambda)$ such that $\rho(A) = \int_A f \, d\lambda$ for all $A \in \mathcal{E}$; we write $f = \frac{d\rho}{d\lambda}$.

Solution. Set $\lambda := \mu + \nu$, which is finite. Then $\nu \ll \lambda$ and $\mu \ll \lambda$. By Radon–Nikodym, there exist $g = \frac{d\nu}{d\lambda} \in L^1(\lambda)$ and $h = \frac{d\mu}{d\lambda} \in L^1(\lambda)$ with $h \geq 0$. Define

$$E := \{h > 0\}, \qquad S := \{h = 0\}.$$

Note that $\mu(S) = \int_S h \, d\lambda = 0$.

Define measures for $A \in \mathcal{E}$ by

$$\nu_a(A) := \int_{A \cap E} g \, d\lambda, \qquad \nu_s(A) := \int_{A \cap S} g \, d\lambda.$$

Clearly $\nu = \nu_a + \nu_s$.

(i) $\nu_s \perp \mu$. The measure ν_s is concentrated on S and $\mu(S) = 0$, hence $\nu_s \perp \mu$.

(ii) $\nu_a \ll \mu$. If $\mu(A) = 0$, then

$$0 = \mu(A) = \int_A h \, d\lambda = \int_{A \cap E} h \, d\lambda.$$

Since $h > 0$ on E, it follows that $\lambda(A \cap E) = 0$, hence $\nu_a(A) = \int_{A \cap E} g \, d\lambda = 0$. Thus $\nu_a \ll \mu$.

Uniqueness. Suppose $\nu = \alpha + \beta$ with $\alpha \ll \mu$ and $\beta \perp \mu$. Choose S with $\mu(S) = 0$ and $\beta(E \setminus S) = 0$. Then for all A,

$$\alpha(A) = \nu(A \setminus S), \qquad \beta(A) = \nu(A \cap S),$$

because $\alpha(S) = 0$ by absolute continuity and $\beta(E \setminus S) = 0$ by singularity. Our construction gives exactly this decomposition with $S = \{h = 0\}$, hence $\alpha = \nu_a$ and $\beta = \nu_s$. Therefore the decomposition is unique. $\square$

Examples.

1. On $(\mathbb{R}, \mathcal{B})$ with $\mu = m$ (Lebesgue) and $\nu = m + \delta_0$, we have $\nu_a = m$ and $\nu_s = \delta_0$.
2. If $\mu = m$ and $\nu = \gamma$ is the Cantor (middle–third) probability measure, then $\nu_a = 0$ and $\nu_s = \gamma$ (purely singular).
3. For $\mu = m$ and $\nu = f\,m + \sum_{k \geq 1} a_k \delta_{x_k}$ with $f \in L^1(m)$, $a_k \geq 0$, the decomposition is $\nu_a = f\,m$ and $\nu_s = \sum_{k \geq 1} a_k \delta_{x_k}$.

Background Definitions and Examples

Finite measure. A measure μ on $(E, \mathcal{E})$ is *finite* if $\mu(E) < \infty$. Finite measures are automatically σ-finite, and the Radon–Nikodym and Lebesgue decomposition theorems hold in this setting.

Radon measure. Let X be a locally compact Hausdorff space (often second countable). A (positive) Borel measure μ on X is a *Radon measure* if it is *inner regular* ($\mu(A) = \sup\{\mu(K) : K \subset A,\ K \text{ compact}\}$), *outer regular* ($\mu(A) = \inf\{\mu(U) : A \subset U,\ U \text{ open}\}$), and *locally finite*. Typical examples include Lebesgue measure on $\mathbb{R}^n$, counting measure on a discrete LCH space, and surface measure on smooth manifolds.

Haar measure. If G is a locally compact topological group, a *Haar measure* is a nonzero Radon measure η that is left invariant: $\eta(gA) = \eta(A)$ for all Borel A and $g \in G$. Haar measure is unique up to a positive scalar multiple. Examples: on $\mathbb{R}^n$, Haar measure equals Lebesgue measure; on a compact group (e.g. $\mathbb{T}^n$), Haar measure can be normalized to total mass 1; on discrete groups, Haar is counting measure.

Absolute continuity and singularity. For measures ν, μ on $(E, \mathcal{E})$:

- $\nu \ll \mu$ (*absolute continuity*) means $\mu(A) = 0 \Rightarrow \nu(A) = 0$ for all $A \in \mathcal{E}$.
- $\nu \perp \mu$ (*mutual singularity*) means there exists $S \in \mathcal{E}$ with $\mu(S) = 0$ and $\nu(E \backslash S) = 0$.

σ-finiteness of Lebesgue measure on $\mathbb{R}^n$. Let m denote Lebesgue measure on $\mathbb{R}^n$. For each $k \in \mathbb{N}$ set

$$Q_k = [-k, k]^n.$$

Then $m(Q_k) = (2k)^n < \infty$ and

$$\mathbb{R}^n = \bigcup_{k=1}^{\infty} Q_k,$$

so m is σ-finite. Note that σ-finite *does not* mean $m(\mathbb{R}^n) < \infty$; indeed $m(\mathbb{R}^n) = \infty$.

If a disjoint cover is preferred, define

$$A_1 = [-1, 1]^n, \qquad A_k = [-k, k]^n \setminus [-(k-1), k-1]^n \quad (k \geq 2).$$

Then $\mathbb{R}^n = \bigsqcup_{k=1}^{\infty} A_k$ with each $m(A_k) < \infty$, and

$$m(\mathbb{R}^n) = \sum_{k=1}^{\infty} m(A_k) = \infty.$$

This still exhibits σ-finiteness: a countable cover by finite-measure sets, even though the total measure is infinite.

Radon–Nikodym theorem (finite/σ-finite). If $\nu \ll \mu$ and both are finite (or σ-finite), then there exists $f \in L^1(\mu)$ such that

$$\nu(A) = \int_A f \, d\mu \quad \text{for all } A \in \mathcal{E},$$

and we write $f = \frac{d\nu}{d\mu}$.

Lebesgue decomposition (context). For finite (or σ-finite) measures μ, ν on $(E, \mathcal{E})$ there exist unique measures ν_a, ν_s with

$$\nu = \nu_a + \nu_s, \qquad \nu_a \ll \mu, \qquad \nu_s \perp \mu.$$

Illustrative Examples

1. **Absolutely continuous:** If $f \in L^1(m)$ on $\mathbb{R}^n$ and $\nu(A) = \int_A f \, dm$, then $\nu \ll m$. Moreover, for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\mu(A) < \delta \Rightarrow \nu(A) < \varepsilon$.
2. **Singular (atomic):** On $\mathbb{R}$, the Dirac measure δ_0 is singular w.r.t. Lebesgue measure m, since it is supported on $\{0\}$ with $m(\{0\}) = 0$.
3. **Singular (continuous):** The Cantor probability measure γ is supported on the middle–third Cantor set C with $m(C) = 0$, hence $\gamma \perp m$ and has no atoms.
4. **Mixed measure:** Let $\mu = m$ (Lebesgue) on $\mathbb{R}$, and set $\nu = f\, m \ + \displaystyle\sum_{k=1}^{\infty} a_k \, \delta_{x_k}$ with $f \in L^1(m)$ and $a_k \geq 0$. Then the Lebesgue decomposition of ν w.r.t. μ is $\nu_a = f\, m$ and $\nu_s = \sum_{k \geq 1} a_k \, \delta_{x_k}$.

Remarks. The Lebesgue decomposition and Radon–Nikodym theorems both extend to σ-finite measures. For signed or complex measures, combine the Jordan (Hahn) decomposition with the Lebesgue decomposition. In harmonic analysis, decomposing a Borel measure on a locally compact group into its absolutely continuous part w.r.t. Haar measure and its singular part is fundamental (e.g. spectral decompositions).

Clarifications for Example 1

Lebesgue measure m and the symbol dm. We write m for Lebesgue measure on $\mathbb{R}^n$. The notation $\int_A f\,dm$ means integration with respect to m (i.e. $\int_A f(x)\,dx$ in classical notation), explicitly indicating the underlying measure.

Absolute continuity of $\nu(A) = \int_A f\,dm$. Let $f \in L^1(m)$ and define $\nu(A) := \int_A f\,dm$ for Borel A. If $m(A) = 0$, then $\int_A |f|\,dm = 0$, hence

$$|\nu(A)| = \left| \int_A f\,dm \right| \leq \int_A |f|\,dm = 0,$$

so $\nu(A) = 0$. Therefore $\nu \ll m$. This remains true for signed or complex $f \in L^1(m)$ by using $\int_A |f|\,dm$.

σ-finite measures. A measure μ on $(E, \mathcal{E})$ is σ-finite if $E = \bigcup_{k=1}^{\infty} E_k$ with $\mu(E_k) < \infty$ for all k. Lebesgue measure m on $\mathbb{R}^n$ is σ-finite since $\mathbb{R}^n = \bigcup_{k=1}^{\infty}[-k, k]^n$ and each cube has finite measure. The Radon–Nikodym and Lebesgue decomposition theorems hold under σ-finiteness.

Dirac (atomic) measure. For $x \in E$, the Dirac measure δ_x is defined by $\delta_x(A) = \mathbf{1}_A(x)$. On $\mathbb{R}^n$ with Lebesgue measure m, we have $\delta_x \perp m$, since δ_x is supported on the m-null set $\{x\}$. In general, any atomic part $\sum_k a_k \delta_{x_k}$ of a measure is singular w.r.t. m.

Cantor (Cantor–Lebesgue) measure. Let $C \subset [0, 1]$ be the middle-third Cantor set. The Cantor measure γ is the unique Borel probability measure satisfying

$$\gamma(A) = \tfrac{1}{2}\,\gamma(3A) + \tfrac{1}{2}\,\gamma(3A - 2),$$

equivalently, the measure whose distribution function is the Devil's staircase. It is *singular continuous*: it has no atoms ($\gamma(\{x\}) = 0$ for all x) and is supported on C with $m(C) = 0$. Hence $\gamma \perp m$.

Connection to Lebesgue decomposition. On $\mathbb{R}^n$ with background measure m, any finite measure ν decomposes uniquely as

$$\nu = \nu_a + \nu_s, \qquad \nu_a \ll m, \quad \nu_s \perp m.$$

Typical instances:

- If $\nu = f\,m + \sum_k a_k \delta_{x_k}$ with $f \in L^1(m)$, then $\nu_a = f\,m$ and $\nu_s = \sum_k a_k \delta_{x_k}$.
- If $\nu = \gamma$ is the Cantor measure, then $\nu_a = 0$ and $\nu_s = \gamma$.

Locally Finite Measures, Regularity, Topological Groups, and Singularity

Locally finite measure. Let X be a topological space and μ a (Borel) measure on X. We say μ is *locally finite* if every $x \in X$ has a neighborhood U with $\mu(U) < \infty$. Lebesgue measure on $\mathbb{R}^n$ and counting measure on a discrete space are locally finite. A measure that assigns $+\infty$ to every nonempty open set is not locally finite.

Outer and inner regularity. A Borel measure μ on a Hausdorff space X is *outer regular* if for every Borel set A,

$$\mu(A) = \inf\{\mu(U) : A \subset U,\ U \text{ open}\},$$

and *inner regular* if for every Borel set A,

$$\mu(A) = \sup\{\mu(K) : K \subset A,\ K \text{ compact}\}.$$

A *Radon measure* is a Borel measure that is locally finite, outer regular, and inner regular. Classical examples include Lebesgue measure on $\mathbb{R}^n$, surface measure on smooth manifolds, and counting measure on discrete locally compact spaces.

Topological groups and Haar measure. A *topological group* is a group G endowed with a Hausdorff topology such that multiplication $(g, h) \mapsto gh$ and inversion $g \mapsto g^{-1}$ are continuous. If G is *locally compact*, then there exists a nonzero Radon measure η (Haar measure) such that $\eta(gA) = \eta(A)$ for all Borel A and all $g \in G$. Haar measure is unique up to a positive scalar multiple. If also $\eta(Ag) = \eta(A)$ for all g, then G is *unimodular* (e.g. all abelian, discrete, and compact groups). On $\mathbb{R}^n$, Haar measure coincides with Lebesgue measure.

Mutual singularity ("orthogonality"). For measures μ, ν on $(X, \mathcal{B})$, we write $\nu \perp \mu$ if there exists $S \in \mathcal{B}$ with $\mu(S) = 0$ and $\nu(X \setminus S) = 0$. This is the measure-theoretic notion often called "orthogonality" of measures. Examples: $\delta_x \perp m$ on $\mathbb{R}^n$, and the Cantor measure γ on $[0, 1]$ satisfies $\gamma \perp m$.

Links to Lebesgue decomposition. On a locally compact Hausdorff space with a Radon reference measure μ (e.g. Haar measure on a group), any finite Borel measure ν decomposes uniquely as $\nu = \nu_a + \nu_s$ with $\nu_a \ll \mu$ and $\nu_s \perp \mu$. In particular, on $\mathbb{R}^n$ with $\mu = m$ (Lebesgue), the absolutely continuous part has the form $\nu_a = f\,m$ for some $f \in L^1(m)$, while ν_s is concentrated on an m-null set.

Borel Sets and Borel Measures

Borel σ-algebra. Let X be a topological space. The *Borel σ-algebra* $\mathcal{B}(X)$ is the smallest σ-algebra containing all open sets of X. Equivalently, $\mathcal{B}(X)$ is generated by the open sets (and hence contains all closed sets, G_δ and F_σ sets, and all sets obtained from these by countable unions, intersections, and complements).

Borel measures. A measure μ defined on $\mathcal{B}(X)$ is called a *Borel measure*. If, in addition, μ is *locally finite* and both *outer* and *inner regular*, then μ is a *Radon measure*.

Completion. Given a Borel measure μ, its *completion* adds to the σ-algebra all subsets of μ-null sets. Lebesgue measure on $\mathbb{R}^n$ is precisely the completion of a Borel measure (often called the Borel–Lebesgue measure).

Examples.

- Lebesgue measure m on $\mathbb{R}^n$ is a complete Radon Borel measure.
- Counting measure on a discrete space (with the discrete topology) is a Radon Borel measure.
- Surface measure on a smooth embedded manifold $M \subset \mathbb{R}^n$ is a Radon Borel measure on M (subspace topology).
- Haar measure on a locally compact topological group G is a Radon Borel measure on $\mathcal{B}(G)$.

Relations Among the Notions

$$\text{Topology on } X \implies \mathcal{B}(X) \xrightarrow{\text{measure}} \text{Borel measure} \xRightarrow[\text{locally finite}]{\text{inner/outer regular}} \text{Radon measure.}$$

On a locally compact group G, a Radon Borel measure that is left-invariant is Haar measure (unique up to scale). Given a reference Radon measure μ (e.g. Lebesgue or Haar), any finite measure ν admits the *Lebesgue decomposition* $\nu = \nu_a + \nu_s$ with $\nu_a \ll \mu$ and $\nu_s \perp \mu$.

Exercise 1 — Dual of a normed space is Banach; dual of the completion

Statement. Let X be a normed space and X' its dual with the operator norm.

(a) Show that X' is complete (a Banach space).

(b) If $\overline{X}$ is the completion of X (same norm), show that the restriction map $J : \overline{X}' \to X'$, $J(\Phi) = \Phi|_X$, is an isometric isomorphism. In particular $X' \cong \overline{X}'$.

Theory. If (Λ_n) is Cauchy in operator norm, then for each $x \in X$ the scalar sequence $(\Lambda_n x)$ is Cauchy. Hahn–Banach guarantees an isometric extension of bounded linear functionals from a dense subspace; when the subspace is dense, the extension is unique.

Solution. (a) Let (Λ_n) be Cauchy in X'. For each $x \in X$ the sequence $(\Lambda_n x)$ is Cauchy in $\mathbb{K}$, hence convergent. Define $\Lambda x := \lim_{n\to\infty} \Lambda_n x$. Linearity of Λ follows from pointwise limits of linear maps. For $\|x\| \le 1$,

$$|\Lambda x| = \lim_n |\Lambda_n x| \le \liminf_n \|\Lambda_n\|,$$

so Λ is bounded with $\|\Lambda\| \le \liminf_n \|\Lambda_n\|$. Moreover,

$$\|\Lambda_n - \Lambda\| = \sup_{\|x\|\le 1} |\Lambda_n x - \Lambda x| = \sup_{\|x\|\le 1} \lim_m |\Lambda_n x - \Lambda_m x| \le \limsup_m \|\Lambda_n - \Lambda_m\| \underset{n}{\to} 0.$$

Hence X' is complete.

(b) For $\Phi \in \overline{X}'$, $\|J(\Phi)\| \le \|\Phi\|$ is immediate. Conversely, for $z \in \overline{X}$ take $x_k \in X$ with $x_k \to z$; then $|\Phi z| = \lim_k |\Phi x_k| \le \|J(\Phi)\|\|z\|$, so $\|\Phi\| \le \|J(\Phi)\|$. Thus J is an isometry. Given $\Lambda \in X'$, Hahn–Banach yields an extension $\widetilde{\Lambda} \in \overline{X}'$ with $\|\widetilde{\Lambda}\| = \|\Lambda\|$; density of X in $\overline{X}$ makes the extension unique. Therefore J is bijective and $X' \cong \overline{X}'$ isometrically. $\square$

Exercise 2 — Nonzero functionals are open mappings

Statement. Let X be a Banach space and $\Lambda \in X'$ with $\Lambda \ne 0$. Show that for every open $U \subset X$, the image $\Lambda(U)$ is open in $\mathbb{K}$; i.e., Λ is an open mapping.

Solution. Because $\Lambda \neq 0$, its range is a nonzero subspace of $\mathbb{K}$ and hence equals $\mathbb{K}$. Let $U \subset X$ be open and let $y_0 \in \Lambda(U)$. Choose $x_0 \in U$ with $\Lambda x_0 = y_0$; there exists $r > 0$ with $B(x_0, r) \subset U$. Then

$$\Lambda\Big(B(x_0, r)\Big) = \Lambda(x_0) + \Lambda\Big(B(0, r)\Big).$$

Since Λ is continuous linear, $\Lambda\Big(B(0, r)\Big) = \{y \in \mathbb{K} : |y| < r\|\Lambda\|\}$ is an open interval/disk around 0, hence $\Lambda\Big(B(x_0, r)\Big)$ is an open neighborhood of y_0 contained in $\Lambda(U)$. Therefore $\Lambda(U)$ is open and Λ is an open mapping. $\square$

Examples. On ℓ^2, the functional $\Lambda(x) = x_1$ is nonzero and open. On $C[0, 1]$, the functional $\Lambda(f) = \int_0^1 f(t)\,dt$ is nonzero and open.

Remark (Banach not needed). Let X be a *normed* space (not necessarily complete) and $\Lambda \in X'$ with $\Lambda \neq 0$. Then Λ is an open mapping: for every open $U \subset X$, $\Lambda(U)$ is open in $\mathbb{K}$. Indeed, $\operatorname{ran} \Lambda$ is a nonzero subspace of $\mathbb{K}$ and hence equals $\mathbb{K}$. For $r > 0$,

$$\Lambda(B(0, r)) = \{y \in \mathbb{K} : \; |y| < r\|\Lambda\|\},$$

which is open; thus for any $x_0 \in U$ with $B(x_0, r) \subset U$,

$$\Lambda(B(x_0, r)) = \Lambda(x_0) + \Lambda(B(0, r))$$

is an open neighborhood of $\Lambda(x_0)$ contained in $\Lambda(U)$.

Hahn–Banach Theorem (and consequences)

Analytic extension form (real case). Let X be a real normed space, $Y \subset X$ a subspace, and $f : Y \to \mathbb{R}$ linear and bounded. Then there exists a linear $F : X \to \mathbb{R}$ such that

$$F|_Y = f, \qquad \|F\| = \|f\|.$$

Complex case. If X is complex and $f : Y \to \mathbb{C}$ is bounded linear, then there exists $F : X \to \mathbb{C}$ linear with $F|_Y = f$ and $\|F\| = \|f\|$.

Geometric separation form. If $M \subset X$ is a proper closed subspace and $x_0 \notin M$, then there exists $\Lambda \in X'$ with

$$\Lambda|_M = 0, \qquad \|\Lambda\| = 1, \qquad \Lambda(x_0) = \operatorname{dist}(x_0, M).$$

More generally, if $C_1, C_2 \subset X$ are disjoint convex sets with C_1 open, then there exists $\Lambda \in X'$ and $\alpha \in \mathbb{R}$ such that $\Lambda(C_1) < \alpha < \Lambda(C_2)$.

Core corollaries.

1. **Norming functional.** For each $x \in X$, $x \neq 0$, there is $\Lambda \in X'$ with $\|\Lambda\| = 1$ and $\Lambda(x) = \|x\|$.
2. **Distance to a subspace.** For any closed subspace $M \subset X$ and $x \in X$,
$$\operatorname{dist}(x, M) = \sup\{\, |\Lambda(x)| : \ \Lambda \in X',\ \|\Lambda\| \le 1,\ \Lambda|_M = 0\,\}.$$
3. **Dual of the completion.** If $\overline{X}$ is the completion of X, the restriction $J : \overline{X}' \to X'$, $J(\Phi) = \Phi|_X$, is an isometric isomorphism. (Hahn–Banach provides an isometric extension $\widetilde{\Lambda}$ of each $\Lambda \in X'$; density of X in $\overline{X}$ yields uniqueness.)

Proof sketch (extension form). Consider the set of all pairs (Z, g) where $Y \subset Z \subset X$, $g : Z \to \mathbb{K}$ is linear, $g|_Y = f$, and $\|g\| = \|f\|$, ordered by extension. Zorn's lemma gives a maximal element (Z_*, g_*). If $Z_* \neq X$, pick $x_0 \in X \setminus Z_*$ and extend g_* to $\operatorname{span}(Z_* \cup \{x_0\})$ with the same norm (possible by a one-step extension inequality). This contradicts maximality; hence $Z_* = X$ and g_* is the desired extension.

Exercise 3 — Decomposing a bounded functional on L^p

Let $u : L^p(\mathbb{R}^n; \mathbb{R}) \to \mathbb{R}$ be bounded and linear. For $f \ge 0$ define
$$\tilde{u}(f) := \sup\{\, u(g) : \ g \in L^p(\mathbb{R}^n),\ 0 \le g \le f\,\}.$$

(a) Basic properties of $\tilde{u}$

Bounds. Since $g \equiv 0$ is admissible, we have $0 \le \tilde{u}(f)$. Moreover, whenever $0 \le g \le f$ one has $\|g\|_p \le \|f\|_p$, hence
$$\tilde{u}(f) \ = \ \sup_{0 \le g \le f} u(g) \ \le \ \sup_{0 \le g \le f} \|u\|\,\|g\|_p \ \le \ \|u\|\,\|f\|_p.$$
Also, if $f \ge 0$ then $g = f$ is admissible, and therefore $u(f) \le \tilde{u}(f)$.

Additivity on the positive cone and positive homogeneity. If $f, g \ge 0$ and $a > 0$, then
$$\tilde{u}(f + ag) \ = \ \tilde{u}(f) \ + \ a\,\tilde{u}(g).$$

Proof idea. Let $0 \le h \le f + ag$ and set
$$r := h \wedge f, \qquad s := \frac{h - r}{a}.$$

Then

$$0 \le r \le f, \qquad 0 \le s \le g, \qquad h = r + as.$$

By linearity of u,

$$u(h) \;=\; u(r) + a\,u(s) \;\le\; \tilde{u}(f) + a\,\tilde{u}(g).$$

Taking the supremum over such h gives

$$\tilde{u}(f + ag) \;\le\; \tilde{u}(f) + a\,\tilde{u}(g).$$

For the reverse inequality, fix $\varepsilon > 0$ and choose $r, s \ge 0$ with $r \le f$, $s \le g$ and

$$u(r) \ge \tilde{u}(f) - \varepsilon, \qquad u(s) \ge \tilde{u}(g) - \varepsilon.$$

Then $r + as \le f + ag$ and

$$u(r + as) \;=\; u(r) + a\,u(s) \;\ge\; \tilde{u}(f) + a\,\tilde{u}(g) - 2\varepsilon.$$

Taking the supremum over h and letting $\varepsilon \downarrow 0$ yields the opposite inequality, hence equality.

(b) Define a linear, bounded w and compare to u

Let $f^+ := \max\{0, f\}$ and $f^- := \max\{0, -f\}$. Define

$$w(f) \;:=\; \tilde{u}(f^+) \;-\; \tilde{u}(f^-).$$

Positivity. If $f \ge 0$, then $w(f) = \tilde{u}(f) \ge 0$. Also

$$w(f) - u(f) \;=\; \tilde{u}(f) - u(f) \;\ge\; 0 \qquad (f \ge 0).$$

Hence both w and $w - u$ are positive linear functionals.

Linearity. For arbitrary f, g write $f = f^+ - f^-$ and $g = g^+ - g^-$ and set

$$A := f^+ + g^+ \ge 0, \qquad B := f^- + g^- \ge 0.$$

Then $A - B = f + g$, and one can express A and B as

$$A = (f + g)^+ + r, \qquad B = (f + g)^- + r$$

for some $r \ge 0$ (take $r := \frac{1}{2}(A + B - |f + g|)$ pointwise). Using additivity of $\tilde{u}$ on the positive cone,

$$\tilde{u}(A) - \tilde{u}(B) = \tilde{u}\Big((f + g)^+\Big) - \tilde{u}\Big((f + g)^-\Big) = w(f + g).$$

But also

$$\tilde{u}(A) - \tilde{u}(B) = \Big(\tilde{u}(f^+) - \tilde{u}(f^-)\Big) + \Big(\tilde{u}(g^+) - \tilde{u}(g^-)\Big) = w(f) + w(g).$$

Hence $w(f+g) = w(f) + w(g)$. Homogeneity follows similarly from the homogeneity of $\tilde{u}$ on the positive cone.

Boundedness. From part (a),

$$|w(f)| \;\le\; \tilde{u}(f^+) + \tilde{u}(f^-) \;\le\; \|u\| \Big(\|f^+\|_p + \|f^-\|_p\Big) \;\le\; 2\,\|u\|\,\|f\|_p.$$

Thus w is bounded on L^p.

(c) Decomposition. From (b), w and $w - u$ are positive bounded linear functionals. Thus

$$u = w - (w-u) =: u_+ - u_-,$$

with $u_\pm$ positive bounded linear functionals on L^p. □

Exercise 4 — Riesz' Lemma and non-compactness of the unit ball

Riesz' Lemma. Let X be a normed space, $V \subset X$ a proper closed subspace, and $0 < \alpha < 1$. There exists $x \in X$ with $\|x\| = 1$ such that $\|x - y\| \ge \alpha$ for all $y \in V$.

Proof. Pick $x_0 \notin V$ and set $d := \operatorname{dist}(x_0, V) > 0$. Choose $v \in V$ with $\|x_0 - v\| < d/\alpha$, and define $x := (x_0 - v)/\|x_0 - v\|$. For any $y \in V$,

$$\|x - y\| = \frac{\|x_0 - (v + \|x_0 - v\|\,y)\|}{\|x_0 - v\|} \ge \frac{d}{\|x_0 - v\|} > \alpha,$$

since $v + \|x_0 - v\|\,y \in V$. Hence $\|x - y\| \ge \alpha$ for all $y \in V$. □

Consequence (Bolzano–Weierstrass fails in infinite dimension). Let X be an infinite-dimensional Banach space.

Inductively construct (x_k) in the unit sphere by applying Riesz' lemma with $\alpha = \frac{1}{2}$ to the nested closed subspaces $V_k = \overline{\operatorname{span}}\{x_1, \dots, x_k\}$.

Then $\|x_i - x_j\| \ge \frac{1}{2}$ for $i \ne j$, so the unit ball contains a sequence with no Cauchy (hence no convergent) subsequence. Therefore the unit ball of X is not compact.

Bolzano–Weierstrass vs Infinite Dimensions

Bolzano–Weierstrass (finite dimensional). Every bounded sequence in a finite dimensional normed space has a convergent subsequence. Equivalently, the closed unit ball is compact.

Failure in infinite dimension. If X is infinite dimensional, the closed unit ball of X is not compact (indeed, not sequentially compact). There exists a bounded sequence with no Cauchy subsequence.

Examples.

- In ℓ^2, let e_n be the standard basis. Then $\|e_n\| = 1$ and $\|e_n - e_m\| = \sqrt{2}$ for $n \neq m$; hence no Cauchy subsequence exists.
- In L^p $(1 \leq p < \infty)$, choose pairwise disjoint measurable sets E_n with $0 < m(E_n) < \infty$, and set $x_n := \mathbf{1}_{E_n}/\|\mathbf{1}_{E_n}\|_p$. Then $\|x_n\|_p = 1$ and $\|x_n - x_m\|_p = (\|x_n\|_p^p + \|x_m\|_p^p)^{1/p} = 2^{1/p}$ for $n \neq m$.

Riesz' Lemma

Statement. Let X be a normed space, $V \subset X$ a proper closed subspace, and $0 < \alpha < 1$. There exists $x \in X$ with $\|x\| = 1$ and $\operatorname{dist}(x, V) \geq \alpha$.

Consequence. With $V_k = \overline{\operatorname{span}}\{x_1, \ldots, x_k\}$ and $\alpha = \frac{1}{2}$, one can construct a sequence (x_k) in the unit sphere such that $\|x_i - x_j\| \geq \frac{1}{2}$ for $i \neq j$, proving the unit ball of an infinite dimensional Banach space is not compact.

Explicit constructions.

- **In ℓ^2:** take $V_k = \operatorname{span}\{e_1, \ldots, e_k\}$ and $x_{k+1} = e_{k+1}$; then $\operatorname{dist}(x_{k+1}, V_k) = 1$.
- **In L^p:** pick disjoint sets (E_k) with $0 < m(E_k) < \infty$, set
$$x_k = \frac{\mathbf{1}_{E_k}}{\|\mathbf{1}_{E_k}\|_p}, \qquad V_k = \operatorname{span}\{x_1, \ldots, x_k\}.$$
For any $y \in V_k$, $y = 0$ a.e. on E_{k+1}, so
$$\|x_{k+1} - y\|_p^p \geq \int_{E_{k+1}} |x_{k+1}|^p = 1,$$
hence $\operatorname{dist}(x_{k+1}, V_k) = 1$ and $\|x_i - x_j\|_p = 2^{1/p}$ for $i \neq j$.

Subspaces for Exercise 3 (in L^p)

To produce the nested closed subspaces used with Riesz' lemma in the L^p setting, fix disjoint measurable sets E_k with $0 < m(E_k) < \infty$ and define

$$x_k := \frac{\mathbf{1}_{E_k}}{\|\mathbf{1}_{E_k}\|_p}, \qquad V_k := \overline{\operatorname{span}}\{x_1, \dots, x_k\}.$$

Then x_{k+1} has distance 1 from V_k, giving a concrete Riesz sequence and proving the unit ball is not compact.

Subspaces for Exercise 3 (in L^p): explicit constructions and estimates

Standing setting. Let $(\Omega, \mathcal{F}, \mu)$ be a σ-finite measure space and $1 \le p \le \infty$. We construct nested finite-dimensional subspaces V_k and a sequence (x_k) in the unit sphere with $\operatorname{dist}(x_{k+1}, V_k) = 1$ for all k, yielding a concrete "Riesz sequence" and the non-compactness of the unit ball.

Choice of disjoint sets. Because μ is σ-finite, we can choose pairwise disjoint measurable sets $(E_k)_{k\ge1}$ with $0 < \mu(E_k) < \infty$.

- On $(\mathbb{R}^n, \mathcal{B}, m)$, one convenient choice is the dyadic cubes
$$E_k = \Big((k-1, k] \times (0,1]^{n-1}\Big) \quad (k \ge 1),$$
or on $[0,1]$ the dyadic subintervals $E_k = \Big((2^{-k}, 2^{-(k-1)}]\Big)$.
- On a non-atomic probability space, recursively bisect any set of positive measure to obtain an infinite disjoint family of positive-measure sets.

Normalized building blocks. For $1 \le p < \infty$ define

$$x_k := \frac{\mathbf{1}_{E_k}}{\|\mathbf{1}_{E_k}\|_{L^p}} = \frac{\mathbf{1}_{E_k}}{\mu(E_k)^{1/p}}, \qquad \|x_k\|_{L^p} = 1.$$

For $p = \infty$, simply set $x_k := \mathbf{1}_{E_k}$ so that $\|x_k\|_\infty = 1$.

Nested subspaces. Let

$$V_k := \operatorname{span}\{x_1, \dots, x_k\}.$$

(Each V_k is finite dimensional, hence closed, so the closure bar is superfluous: $\overline{\operatorname{span}}\{x_1, \dots, x_k\} = V_k$.)

Distance estimate for $1 \le p < \infty$. Fix $k \ge 1$ and $y \in V_k$, say $y = \sum_{j=1}^{k} a_j x_j$. Because the supports are disjoint, $y = 0$ a.e. on E_{k+1}. Therefore

$$\|x_{k+1} - y\|_{L^p}^p = \int_{E_{k+1}} |x_{k+1}|^p \, d\mu \; + \int_{\Omega \setminus E_{k+1}} |y|^p \, d\mu \ge \int_{E_{k+1}} |x_{k+1}|^p \, d\mu = 1.$$

Taking $y = 0$ shows the infimum is attained, hence

$$\operatorname{dist}(x_{k+1}, V_k) = \inf_{y \in V_k} \|x_{k+1} - y\|_{L^p} = 1.$$

Consequently, for $i \neq j$,

$$\|x_i - x_j\|_{L^p}^p = \|x_i\|_{L^p}^p + \|x_j\|_{L^p}^p = 2 \quad \Longrightarrow \quad \|x_i - x_j\|_{L^p} = 2^{1/p}.$$

Distance estimate for $p = \infty$. For $y \in V_k$, again $y = 0$ on E_{k+1}, so

$$\|x_{k+1} - y\|_\infty = \operatorname*{ess\,sup}_{\omega \in \Omega} |x_{k+1}(\omega) - y(\omega)| \ge \operatorname*{ess\,sup}_{\omega \in E_{k+1}} |x_{k+1}(\omega)| = 1.$$

With $y = 0$ we get $\operatorname{dist}(x_{k+1}, V_k) = 1$, and for $i \neq j$, $\|x_i - x_j\|_\infty = 1$ (since the values are 1 on E_i, 0 there for x_j, and vice versa).

Consequences.

1. **Riesz sequence.** The unit sphere of L^p contains a sequence (x_k) with $\|x_i - x_j\| \ge c_p$ for $i \neq j$, where $c_p = 2^{1/p}$ for $1 \le p < \infty$ and $c_\infty = 1$. Thus the closed unit ball is not (sequentially) compact.
2. **Explicit Riesz lemma step.** With V_k as above and any $0 < \alpha < 1$, the vector x_{k+1} satisfies $\|x_{k+1}\| = 1$ and $\operatorname{dist}(x_{k+1}, V_k) = 1 \ge \alpha$, giving the Riesz–lemma vector at step k without any abstract selection.

Variant with "almost disjoint" supports (useful if strict disjointness is inconvenient). Let F_k be sets with $\mu(F_k) > 0$ and $\mu(F_i \cap F_j) = 0$ for $i \neq j$ *except* possibly on small overlaps of measure at most $\varepsilon_k > 0$. Define $x_k = \mathbf{1}_{F_k} / \mu(F_k)^{1/p}$ $(1 \le p < \infty)$. Then for $y \in V_k$,

$$\|x_{k+1} - y\|_{L^p}^p \ge 1 - C_p \, \varepsilon_{k+1} \quad \text{with a constant } C_p \text{ depending only on } p.$$

Choosing $\varepsilon_k \downarrow 0$ fast, one still gets a sequence bounded away from V_k, showing non-compactness. This is sometimes convenient when one constructs localized bumps φ_k with small overlaps instead of strict indicators.

Connection with the functional u from Exercise 3. In Exercise 3 we considered a bounded linear functional $u : L^p(\Omega) \to \mathbb{R}$ and constructed from it the auxiliary map $\tilde{u}$ on nonnegative functions together with the linear functional

$$w(f) = \tilde{u}(f^+) - \tilde{u}(f^-) \qquad (f \in L^p).$$

The sequence (x_k) and subspaces (V_k) above give a concrete geometric framework in which one can *visualize* the decomposition $u = w - (w - u)$ into its positive and negative parts.

Because the functions x_k have pairwise disjoint supports, any $f \in V_k$ can be written uniquely as a finite linear combination

$$f = a_1 x_1 + a_2 x_2 + \cdots + a_k x_k, \qquad a_j \in \mathbb{R}.$$

On such simple functions, a bounded linear functional u is completely determined by its values on the basis $\{x_1, \dots, x_k\}$:

$$u(f) = \sum_{j=1}^{k} a_j \, u(x_j).$$

If we denote $c_j := u(x_j)$, then the action of u on V_k is given by the coordinate functional $f \mapsto \sum_{j=1}^{k} c_j a_j$. In this coordinate form, the *positive* and *negative* parts of u simply correspond to the positive and negative coefficients c_j.

Indeed, on V_k we may define

$$u_+(f) = \sum_{j=1}^{k} a_j \, c_j^+, \qquad u_-(f) = \sum_{j=1}^{k} a_j \, c_j^-, \qquad c_j^\pm := \max\{\pm c_j, 0\}.$$

Then $u(f) = u_+(f) - u_-(f)$, and both u_+ and u_- are positive linear functionals on V_k. By density of $\operatorname{span}\{x_j\}$ in the L^p space, this algebraic decomposition extends uniquely (by continuity) to all of $L^p(\Omega)$, yielding the decomposition $u = w - (w - u)$ described abstractly in Exercise 3.

Interpretation. The sequence (x_k) serves as a concrete "basis" of nonnegative, almost orthogonal directions that separates the positive and negative components of u:

- if $u(x_k) \geq 0$, the contribution of x_k lies in u_+,
- if $u(x_k) < 0$, the contribution lies in u_-.

The functional w constructed from $\tilde{u}$ captures precisely the accumulation of the positive coefficients $u(x_k)$, while $w - u$ captures the negative ones. This explicit coordinate model clarifies the abstract Jordan decomposition of a functional on L^p and shows that it mirrors the decomposition of a signed measure into its positive and negative parts.

Summary. The disjoint-support family (x_k) realizes geometrically what the functional decomposition $u = u_+ - u_-$ expresses analytically: each component acts as a "measure" supported on a disjoint sector of the space, so that the positivity or negativity of u can be analyzed one region at a time. This viewpoint unites the measure-theoretic intuition with the functional-analytic structure built in Exercise 3.

Deep dive example on $[0,1]$ with $p = 2$

1) The space and the blocks. We work on the measure space $([0,1], \mathcal{B}, m)$ with Lebesgue measure m. For $k \geq 1$ set

$$E_k := (2^{-k}, 2^{-(k-1)}].$$

Then the E_k are pairwise disjoint and $m(E_k) = 2^{-k}$.

2) Normalized indicators (an orthonormal set). Define

$$x_k := \frac{\mathbf{1}_{E_k}}{\sqrt{m(E_k)}} = 2^{k/2}\,\mathbf{1}_{E_k} \in L^2[0,1].$$

Norm.

$$\|x_k\|_2^2 = \int 4^{k/2}\,\mathbf{1}_{E_k} = 2^k m(E_k) = 1 \quad \Rightarrow \quad \|x_k\|_2 = 1.$$

Orthogonality. Since $x_i x_j = 0$ a.e. for $i \neq j$ (disjoint supports),

$$\langle x_i, x_j \rangle = \int x_i x_j = 0.$$

Thus $\{x_k\}$ is an orthonormal set.

3) Distances. For $i \neq j$,

$$\|x_i - x_j\|_2^2 = \|x_i\|_2^2 + \|x_j\|_2^2 - 2\langle x_i, x_j \rangle = 1 + 1 - 0 = 2,$$

so

$$\|x_i - x_j\|_2 = \sqrt{2}.$$

4) A simple positive functional u. Define

$$u(f) := \int_0^1 f(t)\,dt.$$

Boundedness and norm. By Cauchy–Schwarz,

$$|u(f)| = \left|\int f\right| \le \|f\|_2\,\|1\|_2 = \|f\|_2,$$

so $\|u\| = 1$.

Values on the blocks.

$$u(x_k) = \int x_k = \int 2^{k/2}\,\mathbf{1}_{E_k} = 2^{k/2}\,m(E_k) = 2^{k/2}\cdot 2^{-k} = 2^{-k/2} > 0.$$

Hence all coefficients $c_k := u(x_k)$ are positive.

5) Riesz step and subspaces. Let

$$V_k := \operatorname{span}\{x_1, \dots, x_k\}.$$

Any $y \in V_k$ vanishes a.e. on E_{k+1}, hence

$$\|x_{k+1} - y\|_2^2 \ge \int_{E_{k+1}} |x_{k+1}|^2 = \int_{E_{k+1}} 2^{k+1}\mathbf{1}_{E_{k+1}} = 2^{k+1}m(E_{k+1}) = 1.$$

Taking $y = 0$ shows

$$\operatorname{dist}(x_{k+1}, V_k) = 1.$$

6) Behaviour of $\tilde{u}$, w, and $w - u$. For $f \ge 0$, the functional $u(g) = \int g$ is monotone increasing in g. If $0 \le g \le f$, the supremum of $\int g$ occurs for $g = f$, so

$$\tilde{u}(f) = \sup_{0\le g\le f} \int g = \int f = u(f).$$

Therefore

$$w(f) = \tilde{u}(f^+) - \tilde{u}(f^-) = \int f^+ - \int f^- = \int f = u(f),$$

and $w = u$, $w - u \equiv 0$ (both positive).

7) Variant with sign changes. Let $\phi = \sum_{k=1}^{\infty} s_k \, \mathbf{1}_{E_k}$ with $s_k \in \{-1, +1\}$. Then $\|\phi\|_2^2 = \sum_k m(E_k) = 1$ and define

$$u_\phi(f) := \int_0^1 f(t)\,\phi(t)\,dt.$$

By Cauchy–Schwarz, $\|u_\phi\| = \|\phi\|_2 = 1$. For each block,

$$u_\phi(x_k) = \int x_k \phi = \int 2^{k/2}\,\mathbf{1}_{E_k}\,s_k = s_k\,2^{k/2} m(E_k) = s_k\,2^{-k/2}.$$

Now some $u_\phi(x_k)$ are positive and some negative.

For $f \geq 0$,

$$\tilde{u}_\phi(f) = \sup_{0 \leq g \leq f} \int g\,\phi = \int f\,\phi_+, \qquad \phi_+ = \max(\phi, 0).$$

Hence

$$w_\phi(f) = \tilde{u}_\phi(f^+) - \tilde{u}_\phi(f^-) = \int f\,\phi_+, \qquad (w_\phi - u_\phi)(f) = \int f\,\phi_- \geq 0,$$

where $\phi_- = \max(-\phi, 0)$. Therefore w_ϕ and $w_\phi - u_\phi$ are positive and

$$u_\phi = w_\phi - (w_\phi - u_\phi)$$

is exactly the positive-minus-negative decomposition of u_ϕ.

8) Discrete coordinate picture. On the finite-dimensional space V_k we can write

$$u_\phi\left(\sum_{j=1}^{k} a_j x_j\right) = \sum_{j=1}^{k} a_j\,s_j\,2^{-j/2},$$

so that

$$(u_\phi)_+\Big(\sum a_j x_j\Big) = \sum a_j\,(s_j 2^{-j/2})^+, \qquad (u_\phi)_-\Big(\sum a_j x_j\Big) = \sum a_j\,(s_j 2^{-j/2})^-.$$

This makes the abstract decomposition $u = u_+ - u_-$ completely explicit on the orthonormal basis $\{x_k\}$.

9) Summary of the geometry.

- $\{x_k\}$ is an orthonormal family in $L^2[0,1]$, with $\|x_i - x_j\|_2 = \sqrt{2}$ and $\operatorname{dist}(x_{k+1}, V_k) = 1$.
- For the positive functional $u(f) = \int f$, we have $w = u$ and $w - u = 0$.
- For sign-changing densities ϕ, we get $u_\phi = w_\phi - (w_\phi - u_\phi)$ with both parts positive.
- On V_k, u_ϕ acts as a coordinate functional with coefficients $s_k 2^{-k/2}$ separating positive and negative parts.

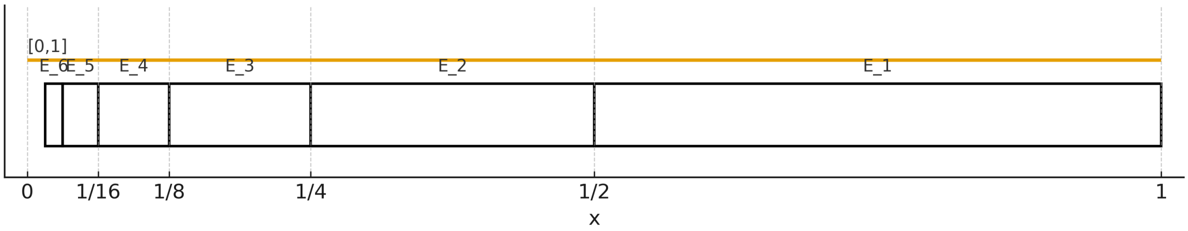

Figure 24: Dyadic partition $E_k = (2^{-k},\, 2^{-(k-1)}]$ of $[0, 1]$.

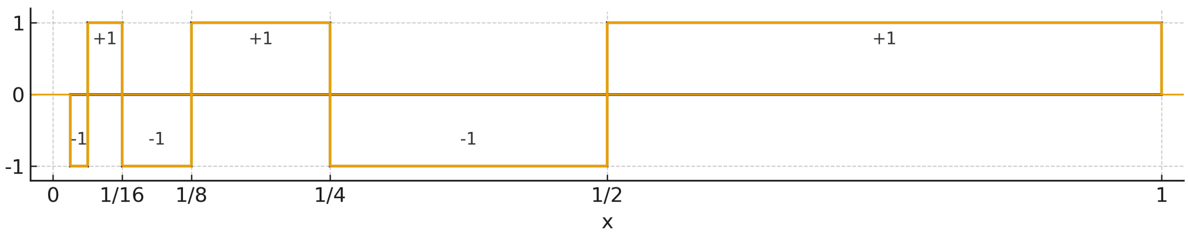

Figure 25: Step function ϕ that is ± 1 on each E_k (alternating signs shown).

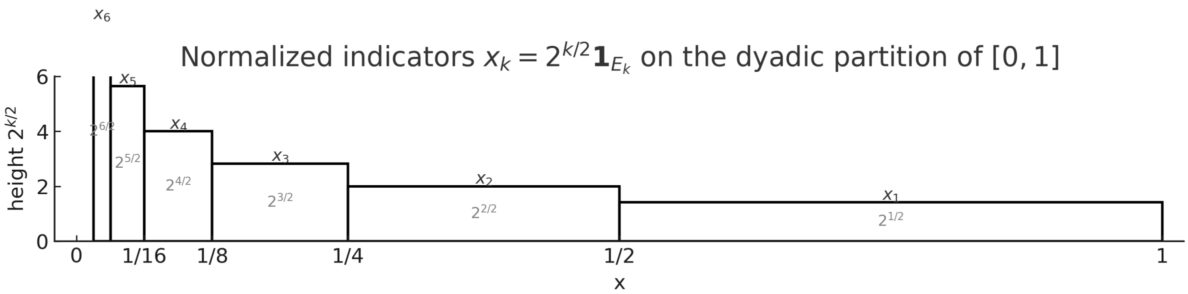

Figure 26: Normalized indicators $x_k = 2^{k/2}\mathbf{1}_{E_k}$ on the dyadic partition of $[0, 1]$. Each x_k has height $2^{k/2}$ on its support $E_k = (2^{-k},\, 2^{-(k-1)}]$.

Exercise 5 — Convergence from a separating family of seminorms

Problem. Let $\mathcal{P}$ be a separating family of seminorms on a vector space X. Show that a sequence (x_k) converges to x in the topology $\tau_{\mathcal{P}}$ if and only if $p(x_k - x) \to 0$ for every $p \in \mathcal{P}$.

Theory. The topology $\tau_{\mathcal{P}}$ is generated by the base of 0-neighborhoods

$$U(p_1, \dots, p_m; \varepsilon) = \{\, x \in X : \max_{1 \le i \le m} p_i(x) < \varepsilon \,\},$$

with finite subfamilies $p_1, \dots, p_m \subset \mathcal{P}$ and $\varepsilon > 0$. The family $\mathcal{P}$ is *separating* if $x \neq 0$ implies $\exists\, p \in \mathcal{P}$ with $p(x) > 0$.

Solution. ($\Rightarrow$) If $x_k \to x$ in $\tau_{\mathcal{P}}$, then for fixed $p \in \mathcal{P}$ and $\varepsilon > 0$, eventually $x_k - x \in U(p; \varepsilon)$, i.e. $p(x_k - x) < \varepsilon$. Hence $p(x_k - x) \to 0$.

($\Leftarrow$) Suppose $p(x_k - x) \to 0$ for every $p \in \mathcal{P}$. Let $U = U(p_1, \dots, p_m; \varepsilon)$ be a basic neighborhood. Then $\max_i p_i(x_k - x) \to 0$, so there exists N such that for all $k \ge N$, $x_k - x \in U$. Thus $x_k \to x$ in $\tau_{\mathcal{P}}$. $\square$

Example. On $C^\infty(\mathbb{R}^n)$, the seminorms $p_{K,m}(f) = \sup_{x \in K,\, |\alpha| \le m} |D^\alpha f(x)|$ (compact K, $m \in \mathbb{N}$) generate the usual Fréchet topology. Convergence $f_k \to f$ means $p_{K,m}(f_k - f) \to 0$ for all K, m (uniform convergence of all derivatives on compacts).

Exercise 6 — Norm $\Rightarrow$ weak $\Rightarrow$ weak-* on X'

Problem. Let X be a Banach space and let $(\Lambda_k) \subset X'$. Show that

$$\|\Lambda_k - \Lambda\| \to 0 \implies \Lambda_k \xrightarrow{\sigma(X', X'')} \Lambda \implies \Lambda_k \xrightarrow{\sigma(X', X)} \Lambda.$$

Solution. If $\|\Lambda_k - \Lambda\| \to 0$ then, for any $\Phi \in X''$,

$$|\Phi(\Lambda_k - \Lambda)| \le \|\Phi\| \, \|\Lambda_k - \Lambda\| \to 0,$$

so $\Lambda_k \to \Lambda$ in the weak topology $\sigma(X', X'')$. If $\Lambda_k \to \Lambda$ in $\sigma(X', X'')$, then for each $x \in X$, identify $Jx \in X''$ by $Jx(\Gamma) = \Gamma(x)$. Hence

$$\Lambda_k(x) = Jx(\Lambda_k) \longrightarrow Jx(\Lambda) = \Lambda(x).$$

Thus $\Lambda_k \to \Lambda$ in the weak-* topology $\sigma(X', X)$. $\square$

Remark. Norm convergence $\Rightarrow$ weak convergence is true in any normed space; weak $\Rightarrow$ weak-* follows since $X \subset X''$ canonically.

Exercise 7 — Bounded L^p sequence and convergence of means

Problem. Let $1 < p < \infty$ and (f_j) be a bounded sequence in $L^p(\mathbb{R}^n)$. Show that there exists $f \in L^p(\mathbb{R}^n)$ such that

$$\int_E f_j(x)\,dx \;\longrightarrow\; \int_E f(x)\,dx \qquad \text{for all bounded measurable } E \subset \mathbb{R}^n,$$

if and only if $f_j \rightharpoonup f$ weakly in L^p.

Theory. The weak topology on L^p is $\sigma(L^p, L^{p'})$. The linear span of indicators $\mathbf{1}_E$ of bounded measurable sets is the class of bounded simple functions with bounded support, which is dense in $L^{p'}$.

Solution. ($\Leftarrow$) If $f_j \rightharpoonup f$ in L^p, then for each bounded E, $\mathbf{1}_E \in L^{p'}$, so $\int_E f_j \to \int_E f$.

($\Rightarrow$) Suppose $\int_E f_j \to \int_E f$ for all bounded E. Then by linearity the convergence holds for all bounded simple $g = \sum a_i \mathbf{1}_{E_i}$: $\int (f_j - f)g \to 0$. Let $h \in L^{p'}$ with compact support. Choose simple $g_k \to h$ in $L^{p'}$. Since (f_j) is bounded in L^p, Hölder gives

$$\Big|\int (f_j - f)h\Big| \le \Big|\int (f_j - f)g_k\Big| + \|f_j - f\|_p \, \|h - g_k\|_{p'}.$$

Fix k and let $j \to \infty$ to get the first term $\to 0$, then let $k \to \infty$ to send the second term to 0 uniformly in j. A standard truncation argument removes the compact-support restriction. Hence $\int (f_j - f)h \to 0$ for all $h \in L^{p'}$, i.e. $f_j \rightharpoonup f$ in L^p. $\square$

Examples.

1. In $L^2(\mathbb{R})$ let $f_j = \mathbf{1}_{(j,j+1)}$. Then $\|f_j\|_2 = 1$, and for any bounded E, eventually $\int_E f_j = 0$, hence $\int_E f_j \to 0 = \int_E 0$. Thus $f_j \rightharpoonup 0$ (not strongly).
2. In $L^p[0,1]$ let $f_j = (-1)^j$. Then $\int_{[0,1]} f_j$ does not converge, so no weak limit exists.

Theory (extended). On a σ-finite measure space $(\Omega, \mathcal{F}, \mu)$, the weak topology on $L^p(\Omega)$ is the locally convex topology $\sigma(L^p, L^{p'})$ generated by the linear functionals

$$T_h(f) := \int_\Omega f\, h\, d\mu \qquad (h \in L^{p'}(\Omega)).$$

Thus $f_j \rightharpoonup f$ in L^p iff $\int f_j h \to \int f h$ for every $h \in L^{p'}$.

- For $1 \le p \le \infty$, the **weak topology** on $L^p(\Omega)$ is $\sigma(L^p, L^{p'})$: the coarsest topology making all linear functionals

$$T_h(f) := \int_\Omega f\, h\, d\mu \qquad (\, h \in L^{p'}(\Omega)\,)$$

  continuous.

  A sequence f_j converges weakly to f iff

$$\int_\Omega f_j\, h\, d\mu \longrightarrow \int_\Omega f\, h\, d\mu \qquad \text{for every } h \in L^{p'}.$$

Reflexivity. If $1 < p < \infty$, L^p is reflexive; hence bounded sets are relatively weakly compact (Banach–Alaoglu + Eberlein–Šmulian). For $p = 1$ or $p = \infty$ this fails in general.

Indicators and density. Let $\mathcal{S}$ denote the linear span of indicators of bounded measurable sets:

$$\mathcal{S} := \left\{ g = \sum_{k=1}^{m} a_k\, \mathbf{1}_{E_k} \; : \; m < \infty,\ a_k \in \mathbb{R},\ \mu(E_k) < \infty \right\}.$$

Then $\mathcal{S}$ is exactly the class of bounded simple functions with bounded support, and for $1 < p < \infty$ one has the density

$$\overline{\mathcal{S}}^{L^{p'}} = L^{p'}(\Omega).$$

Proof sketch of density. Given $h \in L^{p'}$, define the truncations $h^N := h\, \mathbf{1}_{\{|h| \le N\} \cap B_N}$, where B_N is a set of finite measure increasing to Ω (e.g. balls of radius N if $\Omega = \mathbb{R}^n$). Then $\|h - h^N\|_{p'} \to 0$ by dominated convergence and monotone convergence. Each bounded, finitely supported h^N is approximated in $L^{p'}$ by simple functions via range partitioning: $h^N = \lim_{m\to\infty} \sum_\ell c_{\ell,m}\, \mathbf{1}_{E_{\ell,m}}$. Hence $\mathcal{S}$ is dense.

Consequences. If (f_j) is bounded in L^p and $\int (f_j - f)\, \mathbf{1}_E \to 0$ for every bounded E, then by linearity the convergence holds for all $g \in \mathcal{S}$; by density and Hölder,

$$\left| \int (f_j - f) h \right| \le \left| \int (f_j - f) g \right| + \|f_j - f\|_p \, \|h - g\|_{p'} \xrightarrow[j\to\infty]{} \|f_j - f\|_p \, \|h - g\|_{p'}.$$

Let $g \to h$ in $L^{p'}$ to conclude $\int (f_j - f) h \to 0$ for all $h \in L^{p'}$, i.e. $f_j \rightharpoonup f$.

Examples.

1. **Moving bump (vanishing locally).** In $L^p(\mathbb{R})$, $f_j = \mathbf{1}_{(j,j+1)}$. Then $\|f_j\|_p = 1$ and for any bounded E, eventually $\int_E f_j = 0$, hence $f_j \rightharpoonup 0$ (but not strongly).

2. **Oscillations.** On $[0,1]$, $f_j(x) = \sin(2\pi jx)$ is bounded in L^p for all $1 \le p \le \infty$ and $f_j \rightharpoonup 0$ in L^p, since $\int_0^1 f_j\, h \to 0$ for all $h \in L^{p'}$ (Riemann–Lebesgue lemma and density).
3. **Failure of convergence of means.** In $L^p[0,1]$, $f_j = (-1)^j$. Then $\int_{[0,1]} f_j$ does not converge, hence no weak limit exists.
4. **Shrinking supports.** In $\mathbb{R}^n$, set $f_j := |B_{1/j}|^{-1/p}\, \mathbf{1}_{B_{1/j}}$. Then $\|f_j\|_p = 1$ and $f_j \rightharpoonup 0$ in L^p; indeed $\int f_j h \to 0$ for $h \in C_c^\infty$, then extend to $L^{p'}$.

Exercise 8 — Weak convergence, weak $\not\Rightarrow$ strong, and lower semicontinuity of the norm

Problem. Suppose $(H, \langle \cdot, \cdot \rangle)$ is an infinite-dimensional Hilbert space and let $(x_i)_{i=1}^\infty \subset H$.

(i) Show that $x_i \to x$ weakly in H if and only if

$$\langle y, x_i \rangle \;\longrightarrow\; \langle y, x \rangle \quad \text{for all } y \in H.$$

(ii) Show that there exists a sequence (x_i) such that $x_i \rightharpoonup 0$ but $x_i \not\to 0$ in norm.

(iii) Suppose $x_i \rightharpoonup x$. Prove that

$$\|x\| \;\le\; \liminf_{i\to\infty} \|x_i\|, \qquad \text{and} \quad \|x_i\| \to \|x\| \iff x_i \to x \;\text{ in norm.}$$

Theory.

- In a Hilbert space, the dual H' is canonically identified with H itself via the *Riesz representation theorem*: every bounded linear functional $\Lambda \in H'$ can be written uniquely as $\Lambda(z) = \langle z, y \rangle$ for some $y \in H$.
- Hence the weak topology $\sigma(H, H')$ is the coarsest topology for which all maps $z \mapsto \langle z, y \rangle$ are continuous ($y \in H$).
- The norm topology is stronger than the weak topology. Weak convergence does not in general imply norm convergence.
- The *lower semicontinuity of the norm* under weak convergence is fundamental: if $x_i \rightharpoonup x$ then $\|x\| \le \liminf \|x_i\|$. Equality together with weak convergence forces strong convergence.

Solution.

(i) The equivalence follows immediately from the definition of the weak topology and the Riesz representation theorem: $x_i \rightharpoonup x$ means $\Lambda(x_i) \to \Lambda(x)$ for every $\Lambda \in H'$, but each Λ has the form $\Lambda(z) = \langle z, y \rangle$.

(ii) Choose an orthonormal sequence $(e_i)_{i\geq 1} \subset H$. Then for every $y \in H$, $\langle y, e_i \rangle \to 0$ by Bessel's inequality, hence $e_i \rightharpoonup 0$. However $\|e_i\| = 1$ for all i, so $e_i \not\to 0$ in norm.

(iii) For $x_i \rightharpoonup x$, compute

$$\|x_i - x\|^2 = \|x_i\|^2 + \|x\|^2 - 2\,\Re\langle x_i, x \rangle.$$

Taking $\liminf$ and using $\langle x_i, x \rangle \to \langle x, x \rangle = \|x\|^2$ gives $\|x\| \leq \liminf \|x_i\|$. If moreover $\|x_i\| \to \|x\|$, then $\|x_i - x\|^2 \to 0$, hence $x_i \to x$ strongly. $\square$

Example. In $H = \ell^2$, the canonical basis (e_i) satisfies $e_i \rightharpoonup 0$ but $\|e_i\| = 1$. Thus weak convergence does not imply norm convergence.

Exercise 9 — A bounded L^1 sequence with no weakly convergent subsequence

Problem. Construct a bounded sequence $(f_i)_{i=1}^{\infty} \subset L^1(\mathbb{R})$ such that no subsequence of (f_i) is weakly convergent.

Theory.

- For any $f \in L^1(\mathbb{R})$ and $g \in L^\infty(\mathbb{R})$, the duality pairing is $\displaystyle\int_{\mathbb{R}} f\,g$. Weak convergence in L^1 means convergence of these integrals for every $g \in L^\infty$.
- The space L^1 is *not reflexive*; its unit ball is not weakly compact. Therefore one can find bounded sequences with no weakly convergent subsequence.
- Typical counterexamples arise from *translating mass to infinity.* Translation is an isometry in L^1 but destroys weak compactness.

Solution. Define

$$f_n := \mathbf{1}_{[n,n+1]}, \qquad n \in \mathbb{N}.$$

Then $\|f_n\|_{L^1} = 1$ for all n, so (f_n) is bounded. Consider the bounded function

$$g(x) := \sum_{k\in\mathbb{Z}} (-1)^k\, \mathbf{1}_{[k,k+1]}(x) \in L^\infty(\mathbb{R}), \qquad \|g\|_\infty = 1.$$

Compute the dual pairings:

$$\int_{\mathbb{R}} f_n(x)\, g(x)\, dx = \int_n^{n+1} (-1)^n\, dx = (-1)^n.$$

Since $(-1)^n$ does not converge, the sequence (f_n) cannot have any weakly convergent subsequence in $L^1(\mathbb{R})$ (because weak convergence would require convergence of $\int f_{n_j} g$ for every $g \in L^\infty$).

Remarks.

- The sequence (f_n) is bounded but *translates to infinity.* This shows that the L^1 unit ball is not weakly compact.
- By contrast, for $1 < p < \infty$, L^p is reflexive and every bounded sequence admits a weakly convergent subsequence (Banach–Alaoglu + reflexivity).
- If one takes $f_n(x) = \mathbf{1}_{[0,1]}(x - n)$, it also works; all translations of the same shape have $\|f_n\|_1 = 1$ and no weak limit.

Semicontinuity: definitions and criteria. Let X be a topological space and $F : X \to (-\infty, +\infty]$.

- F is *lower semicontinuous (l.s.c.) at x* if $\liminf_{x' \to x} F(x') \ge F(x)$. Equivalently, each sublevel set $\{F \le \alpha\}$ is closed, or the epigraph $\operatorname{epi} F = \{(x, t) : F(x) \le t\}$ is closed in $X \times \mathbb{R}$.
- F is *upper semicontinuous (u.s.c.) at x* if $\limsup_{x' \to x} F(x') \le F(x)$, i.e. each superlevel $\{F \ge \alpha\}$ is closed (hypograph is closed).
- In metric spaces, F is l.s.c. (resp. u.s.c.) iff for all $x_n \to x$, $F(x) \le \liminf_n F(x_n)$ (resp. $F(x) \ge \limsup_n F(x_n)$).

Weak vs. norm semicontinuity (Banach spaces). Let X be a Banach space.

- The norm $x \mapsto \|x\|$ is *weakly l.s.c.*: if $x_n \rightharpoonup x$ in X then $\|x\| \le \liminf_n \|x_n\|$. In particular, for Hilbert spaces,

$$\|x\|^2 \le \liminf_{n \to \infty} \|x_n\|^2 \quad \text{whenever } x_n \rightharpoonup x.$$

- If F is *convex* and *l.s.c. in norm*, then F is weakly l.s.c. (sublevel sets are convex and weakly closed).

Examples.

1. **Indicators.** For a set $C \subset X$, the indicator $I_C(x) = 0$ if $x \in C$ and $+\infty$ otherwise is l.s.c. iff C is closed; u.s.c. iff C is open.
2. **Strict inequality in weak l.s.c.** In a Hilbert space, let (e_n) be an orthonormal sequence. Then $e_n \rightharpoonup 0$ but $\|0\| = 0 < \liminf_n \|e_n\| = 1$.
3. L^p **norms.** For $1 \le p < \infty$, the map $f \mapsto \|f\|_{L^p}$ is weakly l.s.c.: if $f_n \rightharpoonup f$ in L^p, then $\|f\|_p \le \liminf_n \|f_n\|_p$.
4. **Dirichlet energy.** On $H^1(\Omega)$, $E(u) = \int_\Omega |\nabla u|^2 \, dx$ is weakly l.s.c.; hence minimizing sequences have weakly convergent subsequences with E-limit $\ge$ the value at the weak limit (direct method of the calculus of variations).
5. **Total variation.** If $u_n \to u$ in $L^1(\Omega)$, then $\mathrm{TV}(u) \le \liminf_n \mathrm{TV}(u_n)$.

Applications to Exercise 8(iii). If $x_n \rightharpoonup x$ in a Hilbert space, then $\|x\| \le \liminf_n \|x_n\|$ (weak l.s.c. of the norm). Moreover, if also $\|x_n\| \to \|x\|$, then

$$\|x_n - x\|^2 = \|x_n\|^2 + \|x\|^2 - 2\,\Re\langle x_n, x\rangle \longrightarrow 0,$$

hence $x_n \to x$ in norm.

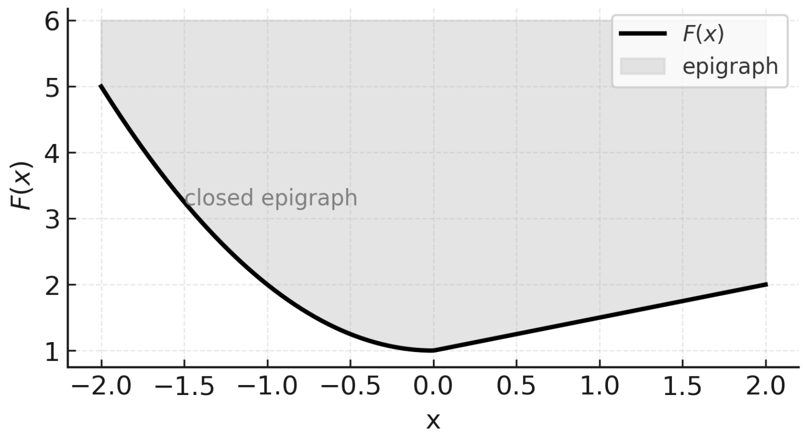

Figure 27: Lower semicontinuity via epigraph: for a convex F, the epigraph $\operatorname{epi} F = \{(x, t) : F(x) \le t\}$ is closed, ensuring $\liminf_{x' \to x} F(x') \ge F(x)$.

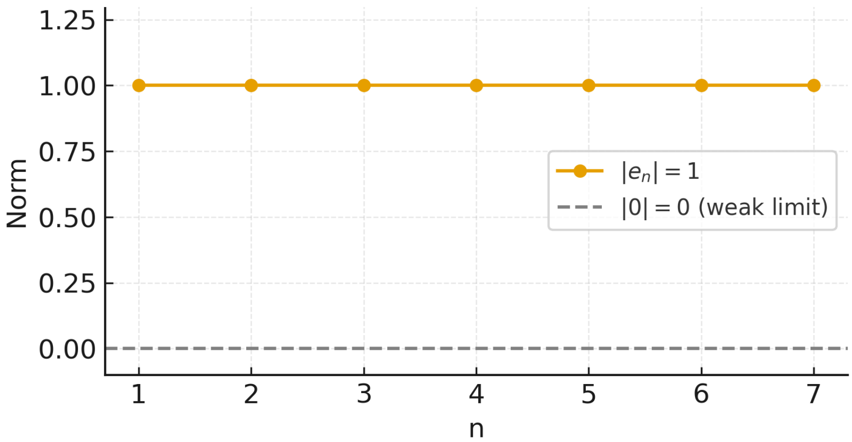

Figure 28: Strict inequality in weak lower semicontinuity: in a Hilbert space, an orthonormal sequence $e_n \rightharpoonup 0$ while $\|0\| < \liminf_n \|e_n\| = 1$.

A. L^p norms: weak l.s.c. $\|f\|_p \le \liminf_n \|f_n\|_p$

1. Strict inequality (moving bump).

On $\mathbb{R}$ set $f_n = \mathbf{1}_{(n,n+1)}$. Then $f_n \rightharpoonup 0$ in L^p for any $1 \le p < \infty$ (vanishes on every bounded set), but

$$\|f_n\|_p = 1.$$

Hence

$$\|0\|_p = 0 \ < \ \liminf_{n\to\infty} \|f_n\|_p = 1.$$

2. Strict inequality (oscillations).

On $[0,1]$, $f_n(x) = \sin(2\pi nx)$ is bounded in L^2 and $f_n \rightharpoonup 0$ (Riemann–Lebesgue). But

$$\|f_n\|_2 = \left(\int_0^1 \sin^2(2\pi nx)\,dx\right)^{1/2} = \frac{1}{\sqrt{2}},$$

so

$$0 \ < \ \liminf_{n\to\infty} \|f_n\|_2 = \frac{1}{\sqrt{2}}.$$

3. Equality case (then strong convergence).

Let $f, g \in L^2[0,1]$ with $\langle f, g\rangle = 0$ and set $f_n = f + \frac{1}{n} g$. Then $f_n \rightharpoonup f$ and

$$\|f_n\|_2^2 = \|f\|_2^2 + \frac{1}{n^2}\|g\|_2^2 \ \longrightarrow \ \|f\|_2^2.$$

Hence $\|f_n\|_2 \to \|f\|_2$, and by the criterion "weak + norms $\Rightarrow$ strong" we get

$$f_n \ \to \ f \quad \text{in } L^2.$$

B. Dirichlet energy $E(u) = \displaystyle\int_\Omega |\nabla u|^2$: weak l.s.c. and minimizers

1. Weak l.s.c. in action (strict inequality possible).

Take $\Omega = (0,1)$ and $u_n(x) = \sin(2\pi nx)$. Then $u_n \rightharpoonup 0$ in $H_0^1(0,1)$, but

$$E(u_n) = \int_0^1 |u_n'(x)|^2\,dx = \int_0^1 (2\pi n)^2 \cos^2(2\pi nx)\,dx = 2\pi^2 n^2 \ \longrightarrow \ \infty,$$

so trivially

$$E(0) = 0 \ \le \ \liminf_{n\to\infty} E(u_n).$$

2. Existence of a minimizer (direct method).

Let $\Omega \subset \mathbb{R}^n$ be bounded Lipschitz and fix boundary data $g \in H^{1/2}(\partial\Omega)$. Minimize

$$\mathcal{A} = \{\, u \in H^1(\Omega) : u = g \text{ on } \partial\Omega \,\}, \qquad E(u) = \int_\Omega |\nabla u|^2 \, dx.$$

Let $(u_k) \subset \mathcal{A}$ be such that $E(u_k) \downarrow \inf_{\mathcal{A}} E$. Then (u_k) is bounded in $H^1(\Omega)$, hence (by reflexivity and Rellich–Kondrachov) there is a subsequence u_{k_j} with

$$u_{k_j} \rightharpoonup u \ \text{ in } H^1(\Omega), \qquad u_{k_j} \to u \ \text{ in } L^2(\Omega).$$

The trace is stable under weak H^1 convergence, so $u \in \mathcal{A}$. Weak lower semicontinuity yields

$$E(u) \ \le \ \liminf_{j\to\infty} E(u_{k_j}) \ = \ \inf_{\mathcal{A}} E,$$

hence u is a minimizer (the weakly harmonic extension of g).

3. Concrete 1D minimizer.

On $\Omega = (0,1)$ with boundary data $u(0) = 0$, $u(1) = 1$, the minimizer is $u(x) = x$, and

$$E(u) = \int_0^1 |u'(x)|^2 \, dx = \int_0^1 1 \, dx = 1.$$

Any minimizing sequence (u_k) has (up to subsequence) $u_k \rightharpoonup x$ in $H^1(0,1)$ and $E(x) \le \liminf E(u_k)$.

Why $f_n = \mathbf{1}_{(n,n+1)} \rightharpoonup 0$ in $L^p(\mathbb{R})$ for $1 < p < \infty$

Let $p \in (1,\infty)$ and p' be its Hölder conjugate. For any $h \in L^{p'}(\mathbb{R})$,

$$\int_{\mathbb{R}} f_n(x)\, h(x) \, dx = \int_n^{n+1} h(x) \, dx.$$

Given $\varepsilon > 0$, choose $R > 0$ so large that

$$\|h\|_{L^{p'}(\{|x|>R\})} < \varepsilon \qquad \text{(possible since } h \in L^{p'}\text{)}.$$

If $n > R$, then $(n, n+1) \subset \{|x| > R\}$ and, by Hölder,

$$\left| \int_n^{n+1} h(x) \, dx \right| \le \|\mathbf{1}_{(n,n+1)}\|_{L^p} \, \|h\|_{L^{p'}((n,n+1))} \le 1 \cdot \|h\|_{L^{p'}(\{|x|>R\})} < \varepsilon.$$

Hence $\int f_n h \to 0$ for every $h \in L^{p'}$, i.e. $f_n \rightharpoonup 0$ in L^p.

Important caveat for $p = 1$. For $p = 1$ the dual is L^∞. The same argument does *not* force $\int_n^{n+1} h \to 0$ for all $h \in L^\infty$ (one only gets $|\int_n^{n+1} h| \le \|h\|_\infty$). In fact, taking

$$h(x) = \sum_{k\in\mathbb{Z}} (-1)^k \, \mathbf{1}_{[k,k+1]}(x) \in L^\infty$$

gives

$$\int f_n h = (-1)^n,$$

which does not converge. Therefore f_n does *not* converge weakly to 0 in L^1.

Summary.

$$f_n \rightharpoonup 0 \quad \text{in } L^p(\mathbb{R}) \text{ for } 1 < p < \infty,$$

$$\text{not true in } L^1; \text{ indeed, } (f_n) \text{ has no weakly convergent subsequence in } L^1.$$

Exercise 10 — Minkowski functional (gauge) of a convex 0-neighborhood

Problem. Let X be a Banach space and suppose $A \subset X$ is convex and contains an open neighborhood of 0. Define for $x \in X$

$$\mu_A(x) := \inf\{\, t > 0 : \; t^{-1}x \in A \,\}.$$

Show that μ_A is sublinear and satisfies $\mu_A(x) \le k\|x\|$ for some $k > 0$. Moreover, if A is open then $\mu_A(y) < 1$ for every $y \in A$. (μ_A is called the *Minkowski functional* of A.)

Theory. If A is convex and absorbing (i.e. contains a neighborhood of 0), then μ_A is finite on X, positively homogeneous for $t \ge 0$, and subadditive; if A is also balanced, μ_A is a seminorm. The gauge controls the ambient norm whenever $B(0,r) \subset A$.

Solution. *(1) Positive homogeneity.* For $t > 0$,

$$\mu_A(tx) = \inf\{s > 0 : \; s^{-1}(tx) \in A\} = t \inf\{u > 0 : \; u^{-1}x \in A\} = t\, \mu_A(x).$$

(2) Subadditivity. If $\mu_A(x) < \alpha$ and $\mu_A(y) < \beta$, then $\alpha^{-1}x, \beta^{-1}y \in A$. By convexity of A,

$$(\alpha + \beta)^{-1}(x + y) = \tfrac{\alpha}{\alpha+\beta}\, (\alpha^{-1}x) + \tfrac{\beta}{\alpha+\beta}\, (\beta^{-1}y) \in A,$$

hence $\mu_A(x + y) \le \alpha + \beta$. Letting $\alpha \downarrow \mu_A(x)$ and $\beta \downarrow \mu_A(y)$ yields subadditivity.

(3) Linear bound. Since A contains an open neighborhood of 0, there is $r > 0$ with $B(0, r) \subset A$. For $x \ne 0$,

$$(\|x\|/r)^{-1}x = \tfrac{r}{\|x\|}x \in B(0, r) \subset A,$$

so $\mu_A(x) \le \|x\|/r =: k\|x\|$ (and $\mu_A(0) = 0$).

(4) Openness gives $\mu_A < 1$ on A. If A is open and $y \in A$, then for some $\delta > 0$, $(1 + \delta)^{-1}y \in A$, so $\mu_A(y) \le 1 + \delta' < 1$ for a suitable δ', hence $\mu_A(y) < 1$.

Examples. If $A = B(0,1)$ then $\mu_A(x) = \|x\|$. If $A = \{x : \|Tx\| < 1\}$ for a bounded operator T, then $\mu_A(x) = \|Tx\|$.

Exercise 11 — Prescribing values on a finite independent set

Problem. Let $\{x_1, \ldots, x_n\}$ be linearly independent in a Banach space X, and let $a_1, \ldots, a_n \in \mathbb{C}$. Show that there exists $\Lambda \in X'$ such that $\Lambda(x_i) = a_i$ for $i = 1, \ldots, n$.

Theory. Define a linear functional on the finite-dimensional subspace $E = \operatorname{span}\{x_i\}$ and extend it to X by the Hahn–Banach theorem.

Solution. Define $\Lambda_0 : E \to \mathbb{C}$ by $\Lambda_0(\sum_{i=1}^n c_i x_i) = \sum_{i=1}^n c_i a_i$. This is well-defined (independence) and continuous (finite dimension). By Hahn–Banach there exists $\Lambda \in X'$ extending Λ_0, hence $\Lambda(x_i) = a_i$ for all i.

Example. If X is a Hilbert space and (x_i) is orthonormal, then $\Lambda(x) = \sum_i a_i \langle x, x_i \rangle$ works and $\|\Lambda\| = (\sum |a_i|^2)^{1/2}$.

Exercise 12 — Mazur's separation theorem

Problem. Let M be a vector subspace of a Banach space X, and let $K \subset X$ be open, convex, and disjoint from M. Show there exists a codimension-one subspace $N \subset X$ with $M \subset N$ and $N \cap K = \varnothing$.

Theory. (Mazur's theorem.) If K is open and convex and C is closed convex with $K \cap C = \varnothing$, there exists $\Lambda \in X' \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that $\Re\Lambda(x) \le \alpha < \inf_{k \in K} \Re\Lambda(k)$ for all $x \in C$. Taking $C = M$ (closed) yields a hyperplane $N = \ker \Lambda$ separating M from K.

Solution. Apply the strong separation to K and M to obtain $\Lambda \in X' \setminus \{0\}$ with $\Re\Lambda(m) \le \alpha < \inf_{k \in K} \Re\Lambda(k)$. Replacing Λ by $\Lambda - \alpha\Phi$ for a fixed $\Phi \equiv 1$ on scalars (or by translation) we may assume $\Re\Lambda(m) = 0$ for $m \in M$ and $\Re\Lambda(k) > 0$ for $k \in K$. Then $N := \ker \Lambda$ contains M and $N \cap K = \varnothing$.

Exercise 13 — Closed subspaces of reflexive spaces are reflexive

Problem. Let X be a reflexive Banach space, and let $Y \subset X$ be a closed subspace. Show that Y is reflexive.

Theory and Solution. In a reflexive Banach space, the closed unit ball is weakly compact. Since $B_Y = B_X \cap Y$ is closed, bounded and convex, it is weakly compact in the relative weak topology of Y. By Kakutani's theorem (or James' criterion), Y is reflexive. Equivalently, the canonical map $J_Y : Y \to Y''$ is surjective because every bounded sequence in Y admits a weakly convergent subsequence; its limit represents the image under J_Y.

Examples. Any closed subspace of $L^p(\Omega)$ $(1 < p < \infty)$, e.g. $W_0^{1,p}(\Omega)$, is reflexive; any closed subspace of ℓ^2 is reflexive.

Reflexive Banach spaces: definitions, criteria, and examples

Bidual and reflexivity. Let X be a Banach space, X' its dual, and $X'' = (X')'$ its bidual. The canonical map $J_X : X \to X''$ is given by

$$J_X(x)(\varphi) = \varphi(x) \qquad (\varphi \in X').$$

It is linear and isometric. We say that X is *reflexive* if J_X is surjective (equivalently $X \cong X''$ isometrically).

Equivalent characterizations.

- **Kakutani.** X is reflexive $\iff$ the closed unit ball B_X is weakly compact.
- **Eberlein–Šmulian.** In Banach spaces, weak compactness $\iff$ weak sequential compactness. Hence X is reflexive $\iff$ every bounded sequence in X has a weakly convergent subsequence.
- **James.** X is reflexive $\iff$ every $\varphi \in X'$ attains its norm on B_X.
- **Dual stability.** X is reflexive $\iff$ X' is reflexive.
- **Uniform convexity (Milman–Pettis).** If X is uniformly convex, then X is reflexive.

Stability under subspaces and quotients. If X is reflexive and $Y \subset X$ is a closed subspace, then Y is reflexive. Moreover, the quotient X/Y is also reflexive.

Canonical examples.

- **Reflexive.** Every finite dimensional normed space; every Hilbert space; $L^p(\Omega)$ for $1 < p < \infty$; ℓ^p for $1 < p < \infty$; Sobolev spaces $W^{k,p}(\Omega)$ for $1 < p < \infty$, and closed subspaces like $W_0^{1,p}(\Omega)$.
- **Non-reflexive.** $L^1(\Omega)$ and $L^\infty(\Omega)$; ℓ^1, ℓ^∞, and c_0; $C(K)$ with $\|\cdot\|_\infty$ for infinite compact K.

Typical closed subspaces and their reflexivity.

- If H is a Hilbert space, any closed subspace $M \subset H$ is reflexive (it is itself a Hilbert space with the induced inner product).
- In $L^p(\Omega)$, $1 < p < \infty$, examples include:
  - $W_0^{1,p}(\Omega)$ (closure of C_c^∞ in $W^{1,p}$),
  - kernels of bounded operators (e.g. $\{f : \int_\Omega f = 0\}$),
  - ranges/orthogonal complements when $p = 2$.

  All are reflexive since they are closed subspaces of a reflexive space.
- In a non-reflexive space (e.g. L^1), closed subspaces can be *either* reflexive (finite-dimensional) *or* non-reflexive (e.g. a closed subspace isomorphic to ℓ^1).

Useful consequences. If X is reflexive, every bounded sequence admits a weakly convergent subsequence. Minimization of weakly lower semicontinuous convex functionals on closed, convex, bounded sets is therefore well-posed (direct method of the calculus of variations).

Quick tests for reflexivity.

- Finite-dimensional $\Rightarrow$ reflexive.
- Uniformly convex (e.g. Hilbert, L^p with $1 < p < \infty$) $\Rightarrow$ reflexive.
- Unit ball weakly compact / every bounded sequence has a weakly convergent subsequence $\Rightarrow$ reflexive.
- Presence of subspaces isomorphic to ℓ^1 or c_0 often signals non-reflexivity.

Dual, bidual, and the canonical map. Let X be a Banach space, X' its dual and $X'' = (X')'$ its bidual. The canonical map $J_X : X \to X''$ is

$$J_X(x)(\varphi) = \varphi(x) \qquad (\varphi \in X').$$

Then J_X is linear and isometric. We call X *reflexive* if J_X is surjective (equivalently $X \cong X''$ isometrically).

When is J_X surjective? (Characterizations of reflexivity). The following are equivalent:

- (Kakutani) The closed unit ball B_X is weakly compact.
- (Eberlein–Šmulian) Every bounded sequence in X admits a weakly convergent subsequence.
- (James) Every $\varphi \in X'$ attains its norm on B_X.

Moreover, if X is uniformly convex (e.g. L^p, $1 < p < \infty$), then X is reflexive (Milman–Pettis).

Canonical examples: duals, biduals, and reflexivity

$L^p(\Omega, \mu)$, $1 < p < \infty$

- **Dual.** $(L^p)' \cong L^{p'}$ via $g \mapsto \left[f \mapsto \int f g \, d\mu\right]$, with $1/p + 1/p' = 1$.
- **Bidual.** $(L^p)'' \cong (L^{p'})' \cong L^p$.
- **Why onto?** The canonical J_{L^p} is an isometric isomorphism because L^p is uniformly convex (Milman–Pettis) and the duality pairing is full. **Hence reflexive.**

$L^1(\mu)$

- **Dual.** $(L^1)' \cong L^\infty$ (pointwise multiplier functionals).
- **Bidual.** $(L^1)'' \cong (L^\infty)'$, but $(L^\infty)' = \mathrm{ba}(\mu)$ (bounded finitely additive signed measures, Yosida–Hewitt), which strictly contains L^1. The canonical embedding $L^1 \hookrightarrow (L^\infty)'$ hits only the countably additive part.
- **Conclusion.** J_{L^1} is not onto (except in essentially finite cases), so L^1 **is non-reflexive**.

$L^\infty(\mu)$

- **Dual.** $(L^\infty)' = \mathrm{ba}(\mu)$ (strictly larger than L^1).

- **Bidual.** $(L^\infty)''$ is huge; the canonical map is not surjective. **Thus non-reflexive.**

ℓ^p, $1 < p < \infty$

- **Dual.** $(\ell^p)' \cong \ell^{p'}$ via $\phi_y(x) = \sum_{k=1}^\infty x_k \overline{y_k}$.
- **Bidual.** $(\ell^p)'' \cong \ell^p$.
- **Conclusion.** Uniformly convex $\Rightarrow$ J_{ℓ^p} onto $\Rightarrow$ **reflexive**.

ℓ^1

- **Dual.** $(\ell^1)' \cong \ell^\infty$.
- **Bidual.** $(\ell^1)'' \cong (\ell^\infty)'$ (far larger than ℓ^1).
- **Conclusion.** J_{ℓ^1} not onto $\Rightarrow$ **non-reflexive**.

c_0 (sequences $\to 0$)

- **Dual.** $c_0' = \ell^1$.
- **Bidual.** $c_0'' = (\ell^1)' = \ell^\infty$. The canonical embedding $c_0 \hookrightarrow \ell^\infty$ is the inclusion, which is not onto.
- **Conclusion.** c_0 is **non-reflexive**.

$C(K)$ for compact Hausdorff K

- **Dual.** $C(K)' \cong M(K)$ (finite regular Borel measures, Riesz–Markov).
- **Bidual.** $C(K)'' \cong M(K)'$; the canonical map is onto iff $C(K)$ is finite-dimensional.
- **Conclusion.** $C(K)$ is **reflexive iff K is finite** (then $C(K) \cong \mathbb{K}^n$).

Stability facts. If X is reflexive and $Y \subset X$ is closed, then Y and X/Y are reflexive. Hence closed subspaces of L^p, $1 < p < \infty$ (e.g. $W_0^{1,p}(\Omega)$, mean-zero subspaces, kernels of bounded maps) are reflexive.

Mazur's results: convex combinations and separation

Mazur's Lemma (weak $\Rightarrow$ strong via convex combinations)

Theorem. Let (x_n) be a sequence in a Banach space X with $x_n \rightharpoonup x$. Then there exist finite convex combinations of the tails

$$y_k = \sum_{n \geq k} \alpha_n^{(k)} x_n, \qquad \alpha_n^{(k)} \geq 0,\ \sum_{n \geq k} \alpha_n^{(k)} = 1,$$

such that $y_k \to x$ in norm.

Equivalent form. For any convex $C \subset X$,

$$\overline{C}^{\text{weak}} = \overline{C}^{\|\cdot\|}.$$

Sketch of proof. If no convex averages converge strongly, the sets $A_k = \overline{\text{co}}\{x_n : n \geq k\}$ stay a positive distance from x. By Hahn–Banach, separate x from A_k with $\varphi_k \in X'$, contradicting $\varphi_k(x_n) \to \varphi_k(x)$ (weak convergence). Hence some convex averages converge in norm.

Examples.

- $X = L^p(\Omega)$, $1 \leq p < \infty$: if $f_n \rightharpoonup f$, then $\exists$ convex averages g_k with $\|g_k - f\|_p \to 0$.
- $X = H$ Hilbert: if $x_n \rightharpoonup x$, then Cesàro means of a subsequence converge to x in norm.

Mazur's Separation Theorem (strong separation of open convex sets)

Theorem. Let X be a Banach space, $K \subset X$ open and convex, $C \subset X$ closed and convex, and $K \cap C = \varnothing$. Then there exist $\Lambda \in X' \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that

$$\Re\Lambda(c) \leq \alpha \quad \text{for all } c \in C, \qquad \inf_{k \in K} \Re\Lambda(k) > \alpha.$$

In particular, if $C = M$ is a closed subspace disjoint from K, then $N := \ker\Lambda$ is a closed hyperplane (codimension one) containing M and disjoint from K.

Examples.

- $X = \mathbb{R}^2$, $K = B((0,1),1)$, $M = \{(t,0) : t \in \mathbb{R}\}$. With $\Lambda(x,y) = y$ we have $\Lambda(M) = \{0\}$ and $\Lambda(K) = (0,2)$.
- $X = \ell^2$, $M = \text{span}\{e_1\}$, $K = B(e_2, \frac{1}{2})$. The functional $\Lambda(x) = \langle x, e_1 \rangle$ and a suitable translation yield a separating hyperplane $N = \ker\Lambda$.

Consequences and where it applies.

- *Closure of convex sets:* Mazur's lemma $\Rightarrow$ weak and strong closures of convex sets coincide.
- *Optimization:* Direct method—weak limits of minimizing sequences can be averaged to obtain strong convergence.
- *Spaces:* All Banach spaces (e.g. Hilbert, L^p with $1 \le p \le \infty$, Sobolev spaces) admit both results.

ℓ^p sequence spaces. For $1 \le p < \infty$,

$$\ell^p := \Big\{x = (x_k)_{k\ge 1} : \sum_{k=1}^{\infty} |x_k|^p < \infty\Big\}, \qquad \|x\|_{\ell^p} := \Big(\sum_{k=1}^{\infty} |x_k|^p\Big)^{1/p}.$$

For $p = \infty$,

$$\ell^\infty := \Big\{x = (x_k) : \sup_k |x_k| < \infty\Big\}, \qquad \|x\|_{\ell^\infty} := \sup_k |x_k|.$$

Then $(\ell^p, \|\cdot\|_{\ell^p})$ is a Banach space for all $1 \le p \le \infty$. In particular,

$$\ell^2 \text{ is a Hilbert space with } \langle x, y\rangle := \sum_{k=1}^{\infty} x_k \overline{y_k}.$$

For $1 < p < \infty$, the dual space $(\ell^p)'$ is isometrically isomorphic to $\ell^{p'}$, where $1/p + 1/p' = 1$, via $y \mapsto [x \mapsto \sum_k x_k \overline{y_k}]$.

$L^p(\Omega, \mathcal{F}, \mu)$ function spaces. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. For $1 \le p < \infty$,

$$L^p(\mu) := \Big\{[f] : f \text{ measurable}, \int_\Omega |f|^p \, d\mu < \infty\Big\}, \qquad \|f\|_{L^p} := \Big(\int_\Omega |f|^p \, d\mu\Big)^{1/p}.$$

Elements are equivalence classes modulo equality μ-a.e. For $p = \infty$,

$$L^\infty(\mu) := \Big\{[f] : \operatorname*{ess\,sup}_\Omega |f| < \infty\Big\}, \qquad \|f\|_{L^\infty} := \operatorname*{ess\,sup}_\Omega |f|.$$

Then $L^p(\mu)$ is a Banach space for $1 \le p \le \infty$. For $1 < p < \infty$:

$$\begin{aligned} \text{(Hölder)} \quad & \int_\Omega |fg| \, d\mu \le \|f\|_{L^p} \, \|g\|_{L^{p'}}, \\ \text{(Minkowski)} \quad & \|f + g\|_{L^p} \le \|f\|_{L^p} + \|g\|_{L^p}, \\ \text{(Duality)} \quad & (L^p(\mu))' \cong L^{p'}(\mu), \quad \tfrac{1}{p} + \tfrac{1}{p'} = 1, \end{aligned}$$

via $g \mapsto \Big[f \mapsto \int_\Omega f \, g \, d\mu\Big]$.

Example 1 (strong separation in $\mathbb{R}^2$)

Setup. $X = \mathbb{R}^2$,

$$K = B\Big((0,1),1\Big) = \{(x,y) : x^2 + (y-1)^2 < 1\}, \qquad M = \{(t,0) : t \in \mathbb{R}\}.$$

(i) Assumptions. K is open and convex, M is a closed subspace, and $K \cap M = \varnothing$ (because $x^2 + (0-1)^2 = x^2 + 1 < 1$ is impossible).

(ii) Functional. Define $\Lambda : \mathbb{R}^2 \to \mathbb{R}$ by $\Lambda(x,y) = y$.

(iii) Values on M and K. For $(t,0) \in M$, $\Lambda(t,0) = 0$, hence $\Lambda(M) = \{0\}$. If $(x,y) \in K$, then $x^2 + (y-1)^2 < 1$, so $|y-1| < 1$, i.e. $0 < y < 2$. Thus $\Lambda(K) = (0,2)$.

(iv) Separation and hyperplane. With $\alpha = 0$ we have $\Re\Lambda(m) \le 0 < \inf_{k \in K} \Re\Lambda(k)$. Let $N = \ker\Lambda = \{(x,y) : y = 0\} = M$. Then $M \subset N$ and $N \cap K = \varnothing$, yielding a strict separation by a codimension-one subspace.

Example 2 (strong separation in ℓ^2)

Setup. $X = \ell^2$, let $e_1 = (1,0,0,\dots)$ and $e_2 = (0,1,0,\dots)$.

$$M = \operatorname{span}\{e_1\} = \{te_1 : t \in \mathbb{R}\}, \qquad K = B\Big(e_2, \tfrac{1}{2}\Big) = \{x \in \ell^2 : \|x - e_2\|_2 < \tfrac{1}{2}\}.$$

(i) Disjointness. Since $\operatorname{dist}(e_2, M) = \|e_2\|_2 = 1$, we have $K \cap M = \varnothing$.

(ii) Functional vanishing on M. Define $\Lambda(x) = \langle x, e_2 \rangle$ (continuous linear functional via Riesz). Then $\Lambda|_M \equiv 0$ because $e_2 \perp e_1$.

(iii) Lower bound on K. For $x \in K$, write $x = e_2 + (x - e_2)$ and compute

$$\Lambda(x) = \langle x, e_2 \rangle = \langle e_2, e_2 \rangle + \langle x - e_2, e_2 \rangle = 1 + \langle x - e_2, e_2 \rangle.$$

By Cauchy–Schwarz, $|\langle x - e_2, e_2 \rangle| \le \|x - e_2\|_2 \|e_2\|_2 < \frac{1}{2}$, hence $\Lambda(x) > \frac{1}{2}$. Therefore $\inf_{x \in K} \Lambda(x) \ge \frac{1}{2} > 0$ while $\Lambda(M) = \{0\}$.

(iv) Separation and hyperplane. With $\alpha = 0$ we have $\Re\Lambda(m) \le 0 < \inf_{k \in K} \Re\Lambda(k)$. Let $N = \ker\Lambda = \{x \in \ell^2 : \langle x, e_2 \rangle = 0\} = e_2^{\perp}$. Then $M \subset N$ and $N \cap K = \varnothing$, so N gives the required codimension-one separation.